



WFP EVALUATION

Community-based vs. data-driven targeting: Experimental evidence from the DRC

Democratic Republic of the Congo

Impact Evaluation Report



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Executive Summary

1. The Democratic Republic of the Congo (DRC) has been facing one of the world's most severe and protracted humanitarian crises. The World Food Programme (WFP) has struggled to sustain its humanitarian response in the country as needs have risen while funding has declined. In 2021, about 25.6 million people were estimated to be food insecure, but WFP could only provide food assistance to 6.3 million: 4.6 million through in-kind food assistance and 1.4 million via cash-based transfers. Since then, the crisis has only continued to worsen, yet the scale of assistance was reduced to 6.1 million in 2022 due to funding shortages and further declined to 5.3 million in 2023 and 5.2 million in 2024. In this resource-constrained context, typified by frequent outbreaks of violence, disease, and natural disasters, humanitarian actors face increasing pressure to deliver aid effectively and equitably.
2. Targeting is a cornerstone of humanitarian and development programming. It refers to the process of identifying and selecting households or communities that are most in need of assistance, ensuring that scarce resources are directed towards them. In humanitarian contexts – where needs far outstrip available resources – targeting helps enhance equity and maintain legitimacy. When done properly, targeting has the potential to improve cost-efficiency and accountability, maximizing the impact of scarce resources.¹
3. However, humanitarian operations must balance speed and precision, while also contending with limited data and volatile security and logistical conditions. These constraints not only affect the technical accuracy of targeting but also the implementers' ability to rigorously assess and learn from past experiences. As Jeong and Trako (2022) observe, *"while targeting is at the forefront of the minds of practitioners in humanitarian assistance, there is little to no quasi- or experimental literature evaluating different targeting methods in humanitarian settings."*²
4. This impact evaluation examines how two targeting approaches – one predominantly data-driven and the other community-based – perform when selecting households. The study was designed and implemented by the WFP's Office of Evaluation in partnership with the World Bank's Development Impact Department between 2023 to 2024, with support from the WFP Democratic Republic of Congo (DRC) Country Office (CO). The United States Agency for International Development's (USAID's) Bureau for Humanitarian Assistance (BHA) provided funding for the impact evaluation.
5. In the study, data-driven targeting using predicted food consumption was supplemented by insights from community focus groups (PMT Plus) to identify households in need. By contrast, under community-based targeting (CBT), WFP formed local committees to identify criteria that are associated with the most vulnerable in their respective communities.

¹ Banerjee, A., Niehaus, P., & Suri, T. (2019). Universal Basic Income in the Developing World. *Annual Review of Economics*, 11(Volume 11, 2019), 959–983. <https://doi.org/10.1146/annurev-economics-080218-030229>

² Jeong, D., & Trako, I. (2022). *Cash and In-Kind Transfers in Humanitarian Settings: A Review of Evidence and Knowledge Gaps*. The World Bank. <https://doi.org/10.1596/1813-9450-10026>

6. The evaluation aims to narrow the gap in the targeting literature by providing rigorous evidence from a humanitarian context on the impacts of applying different targeting methods. It seeks to address six evaluation questions:
 - EQ1. Coverage: How do targeting methods differ in terms of beneficiary coverage?
 - EQ2. Precision: Which targeting method is more precise in selecting households with the lowest food security?
 - EQ3. Profiles: How do targeting methods differ in the types of households they select, and do these differences lead to systematic exclusion or over-targeting of specific population subgroups?
 - EQ4. Welfare: Do the differences in targeting outcomes translate into differences in welfare for the target populations?
 - EQ5. Community acceptance and cohesion: Which targeting method leads to higher community satisfaction, trust, and social cohesion?
 - EQ6. Cost-effectiveness: Which targeting method is more cost-effective?
7. This evaluation uses a stratified cluster randomized controlled trial (RCT) in which the unit of randomization is the bloc, an informal subdivision of a village. The study sample comprises 84 blocs covering a total of 11,732 households.
8. The analysis relies on an original census of the entire population conducted by the CO in the first quarter of 2023, plus two rounds of follow-up data from a stratified random sample of about 47 households per bloc (3,998 households in total). The first follow-up survey was conducted immediately after the final transfer, while the second follow-up was done eight months later.
9. **Coverage:** The CO did not set a beneficiary quota within targeted areas, so the caseload was determined purely by need resulting from the targeting exercise. Under PMT Plus, about 80 percent of households were predicted to have poor food consumption and were selected for assistance, while CBT deemed only 69 percent of households as vulnerable. This implies that the community-based method was more restrictive in who was deemed most in need of food assistance.
10. **Precision:** In terms of targeting precision – defined as the ability to include in the assistance programme all and only food-insecure households, as measured in the baseline census – PMT Plus performs slightly better than CBT, as it leaves out fewer food-insecure households (i.e., has a lower exclusion error). However, this is entirely explained by its higher coverage rate: when coverage is held constant, exclusion errors under the two methods become virtually identical. Alternative precision metrics also show no significant differences between the two approaches.
11. **Profiles:** The two approaches select somewhat different types of households. CBT tends to prioritize households more connected to agriculture and seasonal vulnerabilities (living from agriculture, low production), while PMT Plus places relatively more weight on characteristics such as illiteracy of the household head, the presence of displaced individuals or orphaned children. When comparing beneficiary selection to actual vulnerability level (as recorded in the baseline census) by subgroup, both methods tend to under-target most groups, reflecting the fact that coverage, on average, falls below food insecurity prevalence. Nevertheless, three groups stand out as being differentially targeted across methods: households headed by a person with a disability are significantly more

likely to be targeted under CBT than under PMT Plus, while PMT Plus more strongly targets households headed by an illiterate person or a widow.

12. **Welfare:** In terms of average wellbeing outcomes across the target population (including beneficiaries and non-beneficiaries), there are no statistically significant differences between PMT Plus and CBT. By endline, the average Food Consumption Score (FCS)³ was about 15 in both arms. CBT areas show a modest, but statistically significant improvement in women's agency during the second follow-up (about 0.08 standard deviations, $p < 0.10$), while differences in risky coping strategies and life satisfaction are not significant. These patterns remain similar when accounting for different coverage rates.
13. **Community perceptions:** Perceptions about the programme are broadly positive and similar across arms. Satisfaction with targeting, perceived transparency, and complaint rates are similar. Perceived corruption is marginally lower in CBT areas, but the speed of targeting is perceived as being slower under CBT. Social cohesion levels are similar overall, but there is some increase in perceived insecurity at home and within the bloc in CBT areas, likely reflecting some distributional tensions when coverage is lower.
14. **Cost-effectiveness:** From a cost perspective, setup and targeting implementation costs are very similar between the two arms (USD 61,881 for PMT Plus and USD 62,570 for CBT). Total programme costs, however, differ substantially due to the differences in coverage: transfers disbursed under PMT Plus total USD 1.19 million versus USD 1.07 million for CBT. The difference of USD 121,091 reflects that PMT Plus provided assistance to 534 more households than CBT. Importantly, as discussed above, lower coverage under CBT did not translate into worse food security at the community level.
15. **Considerations:** This evaluation shows that a **simpler, community-driven approach (CBT) can achieve comparable precision and welfare outcomes at a lower cost than a predominantly data-driven method (PMT Plus)**. The success of CBT points to the importance of incorporating local information in targeting the most vulnerable. At the same time, humanitarian actors should rethink traditional household-level targeting by identifying "optimal saturation points" at the community level instead. By focusing on the assistance coverage actually needed by a community rather than its individual households, programmes can leverage natural community resource-sharing, which would allow agencies to expand aid to more locations instead of just supporting a few. However, increased community participation also requires strong and transparent communication and grievance mechanisms to maintain trust and social cohesion. Ultimately, the key targeting choice is not simply between data-driven and community-based methods. How targeting rules incorporate local information and, especially, how much coverage they generate may matter at least as much as the underlying selection method.

³ The FCS is a composite indicator developed by WFP to assess household food security. It combines dietary diversity, food frequency, and the relative nutritional importance of different food groups. The threshold used in DRC is the one adjusted for high sugar consumption contexts. Households are classified as having poor food security if they have a score below/equal to 28, borderline food security if they have a score above 28 and below/equal to 42, and acceptable food security if they have a score above 42.

1. Introduction

1.1. Background

16. In response to global humanitarian needs vastly outpacing available funding, organizations like the World Food Programme (WFP) need to determine how to prioritize households and target assistance. [WFP utilizes a variety of targeting and prioritization approaches](#)⁴ – ranging from data-driven vulnerability assessment models to community-based and geographic methods – to identify those most in need and to deliver assistance efficiently. While these methods are widely used, their social and economic trade-offs remain under-explored, particularly in humanitarian settings.^{5,6}
17. Humanitarian contexts present unique challenges in the application of targeting. The urgency of response typically limits opportunities for rigorous assessments, while secondary data is often unavailable, outdated, or incomplete due to the prevalence of conflict and population displacement.^{7,8} Thus, targeting decisions are often taken under limited information, which can lead to inefficient allocations and unintended social consequences, including perceptions of unfairness, social tension, and punitive behaviours towards recipients.⁹ In food-insecure and conflict-affected contexts, perceptions of arbitrary or inequitable targeting can undermine trust in aid efforts and weaken their overall effectiveness.
18. In the Democratic Republic of the Congo (DRC), these challenges are particularly acute. The country continues to face one of the world's most severe and protracted food crises, driven by a complex interplay of persistent conflict, mass displacement, economic shocks, and weather extremes.¹⁰ In this context, the WFP country office (CO) conducts regular targeting exercises to reach the most vulnerable households and to ensure optimal use of its limited resources. The CO primarily relies on proxy means testing (PMT) for targeting, which is a data-driven approach that uses statistical models based on proxy indicators to assess household vulnerability and determine eligibility for assistance.
19. PMT is one of the most widely adopted targeting approaches in humanitarian and development settings because of its perceived objectivity. However, it also has significant limitations. It entails high operational costs, including the need for census-style data

⁴ World Food Programme. (2021b). *Targeting and prioritization: Operational guidance note*.

<https://docs.wfp.org/api/documents/WFP-0000122035/download/>

⁵ Ibid.

⁶ Jeong, D., & Trako, I. (2022). *Cash and In-Kind Transfers in Humanitarian Settings: A Review of Evidence and Knowledge Gaps*. The World Bank. <https://doi.org/10.1596/1813-9450-10026>

⁷ Follett, L., & Henderson, H. (2023). A hybrid approach to targeting social assistance. *Journal of Development Economics*, 160, 103002. <https://doi.org/10.1016/j.jdeveco.2022.103002>

⁸ Pham, P. N., Johnston, L. G., Keegan, K., Wei, C., & Vinck, P. (2023). Innovative Strategies for Remotely Sampling Hard-to-Reach Populations: Assessing Phone Versus Internet Respondent-Driven Sampling Approaches Among Venezuelan Refugees and Migrants in Colombia. *American Journal of Epidemiology*, 192(10), 1613–1623. <https://doi.org/10.1093/aje/kwad116>

⁹ Della Guardia, A., Lake, M., & Schnitzer, P. (2022). Selective inclusion in cash transfer programs: Unintended consequences for social cohesion. *World Development*, 157, 105922. <https://doi.org/10.1016/j.worlddev.2022.105922>

¹⁰ Food Security Information Network (FSIN) & Global Network Against Food Crises (GNAFC). (2025). *Global Report on Food Crisis 2025* (p. 34.1 MB, 254 p.) [Pdf]. FSIN/GNAC. <https://doi.org/10.71958/WFP130664>

collection covering the entire intended population, the construction and maintenance of training datasets, and substantial analytical expertise and resources to estimate models and interpret their results. Moreover, like any statistical model, PMT predictions are subject to error, influenced by the quality and timeliness of the underlying data.^{11,12,13} In addition, the modelling process is technically complex and difficult to explain clearly to recipients, which can make the approach appear opaque and reduce perceptions of transparency and fairness.^{14,15}

20. These concerns are reflected by WFP's findings. WFP evaluations highlighted PMT-related errors and widespread misunderstanding of the approach among target populations¹⁶ and called for more "more transparent"¹⁷ approaches. These issues have raised broader concerns about the fairness of current assistance distribution and the risk of fuelling tensions between vulnerable groups. As WFP's work relies significantly on targeting, it has established a dedicated impact evaluation workstream, under the Office of Evaluation (OEV), that aims to generate rigorous evidence on optimal targeting methods in humanitarian settings.
21. Given the specific challenges observed in DRC and the broader organizational priority on targeting, the CO requested OEV, in partnership with the World Bank's Development Impact Department, to conduct a rigorous impact evaluation comparing alternative targeting approaches.

1.2. Country context

22. The DRC is the largest country in sub-Saharan Africa and is one of the most resource-rich nations in the world. However, it remains one of the poorest countries globally. As of 2023, it ranked 171 out of 193 countries and territories on the Human Development Index (HDI), and nearly three-quarters of its population lives in poverty.^{18,19} The country has also been facing one of the world's most severe and protracted humanitarian crises, and WFP has struggled to sustain its humanitarian response as needs have risen while funding has

¹¹ Follett, L., & Henderson, H. (2023). A hybrid approach to targeting social assistance. *Journal of Development Economics*, 160, 103002. <https://doi.org/10.1016/j.jdeveco.2022.103002>

¹² Banerjee, A., Hanna, R., Olken, B. A., & Lisker, D. S. (2024). Social Protection in the Developing World. *Journal of Economic Literature*, 62(4), 1349–1421. <https://doi.org/10.1257/jel.20241646>

¹³ Sharif, I. A. (2012). Can Proxy Means Testing Improve the Targeting Performance of Social Safety Nets in Bangladesh? *The Bangladesh Development Studies*, 35(2), 1–43.

¹⁴ Cameron, L., & Shah, M. (2014). Can Mistargeting Destroy Social Capital and Stimulate Crime? Evidence from a Cash Transfer Program in Indonesia. *Economic Development and Cultural Change*, 62(2), 381–415. <https://doi.org/10.1086/674102>

¹⁵ Hanna, R., & Olken, B. A. (2018). Universal Basic Incomes versus Targeted Transfers: Anti-Poverty Programs in Developing Countries. *Journal of Economic Perspectives*, 32(4), 201–226. <https://doi.org/10.1257/jep.32.4.201>

¹⁶ World Food Programme. (2023b). *Evaluation of the Policy on WFP's Role in Peacebuilding in Transition Settings* (Centralized Evaluation Report OEV/2021/001). <https://docs.wfp.org/api/documents/WFP-0000146160/download/>

¹⁷ Steets, J. & WFP-Office of Evaluation. (2026). *Strategic Evaluation of WFP's Approaches to Targeting and Prioritization for Food and Nutrition Assistance* (p. 19.7 MB, 7 v.) [Pdf]. World Food Programme (WFP). <https://doi.org/10.71958/WFP131314>

¹⁸ United Nations Development Programme. (2025). *Human Development Reports: Democratic Republic of the Congo*. <https://hdr.undp.org/data-center/specific-country-data#/countries/COD>

¹⁹ World Bank Group. (2024). *Democratic Republic of Congo: Poverty and Equity Brief – October 2024*. <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099700301032538532>

declined. In 2021, about 25.6 million people were estimated to be food insecure,²⁰ but WFP was only able to provide food assistance to 6.3 million:²¹ 4.6 million through in-kind food assistance and 1.4 million via cash-based transfers.²² The crisis has only continued to worsen. Yet, the scale of assistance was reduced to 6.1 million in 2022 due to funding shortages and further declined to 5.3 million in 2023 and 5.2 million in 2024.^{23,24} An estimated 25 million people across the country were deemed food insecure in 2025, while almost 2 million more individuals are projected to be food insecure in 2026.²⁵

23. At the epicentre of the emergency are DRC's eastern provinces – North Kivu, South Kivu, Ituri, and Tanganyika – which constitute 33 percent of the country's food-insecure population even though they only account for about 20 percent of the total population.²⁶ While the Kivus are often the focus of international attention, Tanganyika province represents a 'neglected crisis' characterized by protracted inter-communal tensions and extreme geographical isolation. The province suffers from some of the highest rates of food insecurity in the country due to a combination of limited market access, dilapidated infrastructure, and the recent impact of climate-induced shocks. In this resource-constrained context, humanitarian actors face increasing pressure to deliver aid effectively and equitably.

2. Programme and Targeting Implementation

2.1. Humanitarian General Food Assistance Programme

24. This impact evaluation builds on the WFP Kalemie Area Office's general food assistance (GFA) programme for the Tanganyika province (see Figure 1), which aims to support vulnerable households that are facing severe food insecurity with cash assistance.
25. The administrative structure of DRC comprises provinces at the first level, followed by territories/communes, and then by groupements or quartiers. Villages constitute the next

²⁰ Integrated Food Security Phase Classification. (2021). *Democratic Republic of the Congo: Food Security & Nutrition Snapshot, September 2021-August 2022*. <https://www.ipcinfo.org/ipcinfo-website/resources/resources-details/en/c/1155368/>

²¹ World Food Programme. (2022). *Democratic Republic of the Congo Annual Country Report 2021*. <https://docs.wfp.org/api/documents/WFP-0000138022/download/>

²² Ibid.

²³ World Food Programme. (2023a). *Democratic Republic of the Congo Annual Country Report 2022*. <https://docs.wfp.org/api/documents/WFP-0000147935/download/>

²⁴ World Food Programme. (2024). *Democratic Republic of the Congo Annual Country Report 2023*. <https://docs.wfp.org/api/documents/WFP-0000157699/download/>

²⁵ Integrated Food Security Phase Classification. (2025). *Democratic Republic of the Congo: Acute Food Insecurity Situation for September - December 2025 and Projection for January - June 2026*. <https://www.ipcinfo.org/ipc-country-analysis/details-map/en/c/1159775/>

²⁶ Integrated Food Security Phase Classification. (2025). *Democratic Republic of the Congo: Acute Food Insecurity Situation for September - December 2025 and Projection for January - June 2026*. <https://www.ipcinfo.org/ipc-country-analysis/details-map/en/c/1159775/>

level in rural areas, while blocs are informal subdivisions below villages that are commonly used for operational planning and programme implementation but that do not correspond to an official administrative tier.²⁷

Figure 1: Location of Tanganyika Province



26. The evaluation focuses on the GFA cycle for May to September 2023, which included 30 villages selected for this impact evaluation from four *groupements*.²⁸ Transfers for this period were made in four tranches. For the first three payments, between May and August 2023, transfers amounted to 30,000 francs per household member (about USD 13 in nominal terms²⁹ or USD 31 in 2023 PPP dollars³⁰). However, the CO adjusted the last payment in September 2023 as the estimate of the minimum expenditure basket (MEB) – which is the basis for the transfer amount – increased to 36,000 francs per household member (about USD 15 in nominal terms or 38 PPP dollars).
27. The amount of cash assistance each household received was based on the size of the household, with the goal of supporting households over a period of six months. Total transfers over the period amounted to about 126,000 francs per household member (equivalent to USD 54 or 132 2023 PPP dollars), which covers about 57 percent of the total expenditure required to meet an average household's needs (based on the MEB) over the target period.³¹
28. Transfers were delivered in cash at designated distribution points to households registered as WFP beneficiaries. Administrative records show that assistance was delivered, as planned, in four monthly instalments distributed from May to September 2023. Appendix Table A1.1 summarizes the amount disbursed in each transfer, while Figures A1.1-A1.5 describe the distribution of transfers by eligibility status, targeting arm, the number of transfers received, and data source.

²⁷ For more details, see DRC's *Organic Law n°08/016 of 7 October 2008*:

<https://www.leganet.cd/Legislation/Droit%20Public/Administration.ter/L.08.16.17.10.2008.htm>

²⁸ The selection process of the villages for the GFA and the evaluation is detailed in subsection 3.3 which focuses on the evaluation design.

²⁹ Exchange rate CDF per USD 2093.2: <https://www.xe.com/currencytables/?from=CDF&date=2023-05-01#table-section>

³⁰ Purchasing power parity (PPP) conversion factor for the DRC in 2023 is 957.73 Congolese francs per international dollar (World Bank, World Development Indicators, available at:

<https://data.worldbank.org/indicator/PA.NUS.PPP?locations=CD>)

³¹ The ratios were calculated by dividing the cumulative transfer of 126,000 Congolese francs by six times the monthly average per capita expenditure reported in the EFSA 2023 (221,182 CDF); however, actual household transfer values are based on actual household size, and not the overall average.

2.2. Targeting approaches

29. WFP's targeting protocol involves two steps. First, it uses data from the Integrated Food Security Phase Classification (IPC) to identify geographic areas at risk of severe hunger crises. Following geographic prioritization, within villages and *blocs*, WFP implementers conduct more granular targeting at the household level to identify beneficiaries. The evaluation focuses on this second step and compares two targeting approaches: PMT-based targeting (CO's status quo) and community-based targeting – a new alternative approach.

Status quo method: PMT “Plus”

30. To identify beneficiary households, the Kalemie Area Office has primarily used a PMT model supplemented with information from focus groups in the target area. This approach, called PMT “Plus”, has two core elements: 1) a data-driven model to predict household food insecurity (PMT), and 2) “lock-in” criteria selected through focus groups (the “plus” element).
- **PMT:** Following standard WFP practice, the PMT model was developed to identify food-insecure households, defined as households with a Food Consumption Score (FCS) below or equal to the [threshold of 28](#).³² The model was trained on previous baseline surveys of WFP Post-Distribution Monitoring (PDM) activities, which were conducted in the area of interest but excluded villages that were part of the evaluation. These datasets contained both the outcome variable (FCS) as well as a set of potential predictor variables. The PMT model was then applied to the census data that was collected by the CO on all potential beneficiary households between December 2022 and March 2023. Using demographic, socioeconomic, and asset-related indicators from the census, the model produced a predicted FCS for each household. Households classified as having “poor” food security (predicted FCS below or equal 28) were deemed eligible for assistance.
 - **“Plus” Element:** Additional inclusion criteria were identified through gender-separated focus group discussions (FGDs) held in three randomly selected blocs or communities in the study area. The formation of focus groups followed two strict guidelines: 1) that they occur separately between men and women, and 2) that they should not include the chief or anyone from the chief's immediate family. Groups were encouraged to have between 8 to 10 participants and to have a mix of representation across ethnicity, age, and disability, but this was not enforced. In total, six FGDs were conducted. Participants were asked to identify household characteristics that indicate a high vulnerability to food insecurity in their community.³³ Consultations across the three blocs identified twenty distinct vulnerability criteria. Eight criteria (40 percent) were common across all three blocs and were applied across all target communities. They capture commonly recognised vulnerability dimensions,³⁴ while ensuring that no community had a

³² The FCS is a composite indicator developed by WFP to assess household food security. Households are classified as having poor food security if they have a score below/equal to 28, borderline food security if they have a score above 28 and below/equal to 42, and acceptable food security if they have a score above 42.

³³ After presenting the question to participants, facilitators guided an open discussion in which participants shared and debated their views. Facilitators did not prompt or suggest examples to avoid influencing the responses. The focus groups reached agreement through collective deliberation and, once consensus was achieved, facilitators recorded the agreed-upon criteria in the data collection tablet. No scoring or ranking was undertaken as the purpose was to capture locally defined “lock-in” characteristics rather than to assign relative importance.

³⁴ These dimensions – although specific to the target population – appear in line with broader standard definitions of “persons at risk”, considered to face heightened vulnerability in emergencies, as identified in international protection

disproportionate influence over their own targeting. Any household meeting at least one of the criteria was deemed eligible for the program, regardless of their predicted FCS. Overall, the “Plus” component added 29 percent extra households to beneficiary selection. These “lock-in” criteria were:

- household headed by a woman without any other male household member engaged in income-generating activities;
 - household headed by an elderly person (aged 60 years or above);
 - household headed by a person with a disability;
 - household headed by a person with a chronic illness;
 - household with orphaned children;
 - household with a pregnant or lactating woman;
 - households headed by a widower; and
 - displaced households.
31. While PMT Plus has been the standard targeting approach in DRC, it also presents major challenges. It requires rich census-level household data, which is costly and time-intensive to collect, particularly in emergency settings. In addition, the approach has been criticized for lacking transparency and for fuelling tensions. According to the CO, there are widespread misconceptions on how targeting decisions are made with this approach, which have led to community tensions.
32. Findings from a peacebuilding evaluation similarly indicate that close to 1 in 5 community members view WFP assistance as being distributed in an unfair manner.³⁵ These patterns suggest that, despite the “Plus” element – whereby additional inclusion criteria are identified through FGDs – the approach has struggled to ensure broad acceptance at the community level.
33. To address these concerns, the CO requested OEV to assess a possible alternative that prioritizes specific local needs and that empowers the community.

Alternative method: Community-based Targeting (CBT)

34. The alternative community-based approach relies on local committees to define beneficiary selection criteria based on their specific context. Modelled after UNICEF’s practices, this method emphasizes community ownership and inclusivity. WFP’s cooperating partners (CPs) convened a meeting in each bloc to establish a targeting committee. Each committee was composed of 12 residents, six women and six men. Membership had to reflect diversity across age, ethnicity, and disability status, and included youth and elderly representatives. Chiefs and their immediate families were explicitly excluded. A total of 42 community committees were formed, and each CBT committee was tasked with defining criteria for its community’s specific context.
35. Facilitators prompted committees with this question: *“In this community, what are some characteristics of a household most in need of food assistance?”* Committees were then

guidance. See, for instance, the [UNHCR Emergency Handbook \(2024\)](#) and the [IASC Operational Guidelines on the Protection of Internally Displaced Persons \(2011\)](#).

³⁵ World Food Programme. (2023b). *Evaluation of the Policy on WFP’s Role in Peacebuilding in Transition Settings* (Centralized Evaluation Report OEV/2021/001). <https://docs.wfp.org/api/documents/WFP-0000146160/download/>

required to identify at least five criteria. A household that met at least one of the criteria was deemed eligible for assistance. This allowed for a locally defined understanding of vulnerability and shifted decision-making power towards the community itself.

36. On average, CBT committees identified 9 criteria, with the number ranging from 5 to 13. In total, committees identified 42 distinct criteria. The top criteria, both mentioned 91 percent of the time, were: “households headed by a woman without any other male household member engaged in income-generating activities” and “households headed by an elderly person”.

Key differences from community consultations

37. Although both CBT and PMT Plus involve community participation, their processes differ significantly. Unlike PMT Plus, which applies uniform lock-in criteria across all communities from a limited number of FGDs, CBT uses localized selection criteria identified by community targeting committees. This difference is reflected in the selection criteria prioritized by each method. Although the top-five CBT criteria also qualified as PMT Plus lock-in criteria, it is notable that none of the lock-in criteria was selected by 100 percent of the CBT committees. This indicates that the sources of vulnerability are highly localized. Appendix Table A1.2 presents the top vulnerability criteria selected by CBT committees in descending order, alongside the eight lock-in categories used by PMT Plus.³⁶
38. By requiring stronger community involvement, CBT has the potential to reduce tensions and increase acceptance of targeting decisions, thereby offering a more conflict-sensitive approach to targeting assistance.^{37,38} However, by delegating the targeting process to the community there may also be a higher risk of discrimination against minorities or elite capture, which might lead to an increase (rather than a decrease) in tensions and discontent. Which effect dominates and which targeting method leads to better outcomes is ultimately an empirical question, which this study aims to answer through its experimental design.

3. Evaluation Design and Methodology

³⁶ The table only includes criteria mentioned by at least 10 percent of the blocs in CBT. Some relevant criteria could not be coded due to limitations in the census data, including households without seeds, households with malnourished children, households unable to pay school fees, and households with albino children. These were excluded for lack of specific or reliable indicators.

³⁷ McCord, A. (2013). *Community-based targeting in the Social Protection sector* (Working Paper No. 514). ODI Global. <https://media.odi.org/documents/11666.pdf>

³⁸ Hillebrecht, M., Klöner, S., A Pacere, N., & Soares, A. (2020). Community-Based versus Statistical Targeting of Anti-Poverty Programs: Evidence from Burkina Faso. *Journal of African Economies*, 29(3), 271–305. <https://doi.org/10.1093/jafeco/ejz026>

3.1. Evaluation background

39. Targeting refers to the process of identifying and selecting households or communities that are most in need of assistance. In humanitarian contexts – where crises evolve rapidly, information is often incomplete and needs far outstrip available resources – effective targeting is essential. When done properly, targeting can improve cost-efficiency and accountability, maximizing the impact of scarce resources.³⁹
40. However, humanitarian operations must balance speed and precision, while also contending with limited data and volatile security and logistical conditions. These constraints not only affect the technical accuracy of targeting but also the implementers' ability to rigorously learn from past experiences. As Jeong and Trako (2022) observe, *"while targeting is at the forefront of the minds of practitioners in humanitarian assistance, there is little to no quasi- or experimental literature evaluating different targeting methods in humanitarian settings."*⁴⁰ The operational urgency and capacity constraints of such environments have, indeed, often precluded rigorous evaluation, leaving critical questions around trade-offs between accuracy, inclusiveness, and community acceptance open.

Limits of Simulations

41. Much of the existing targeting literature on humanitarian contexts relies on simulations. They model hypothetical outcomes based on existing data rather than observing actual changes generated by different targeting methods. While simulations are useful for comparing the precision and welfare performance of different approaches, they often depend on assumptions about household behaviour and welfare, and, thus, may not fully reflect real-world complexity. They are even less able to capture how targeting methods affect community acceptance, social cohesion, or broader socioeconomic outcomes.⁴¹
42. Recent work by Aiken et al. (2025)⁴² provides an example of the use and limitations of targeting simulations. They use simulated welfare effects in Bangladesh to compare a machine-learning model drawing on mobile phone data, a standard PMT, and a community-based ranking. Their findings suggest that the two data-driven methods yield higher household utility, largely due to lower targeting error. However, the analysis was unable to speak to how these approaches would perform in practice in fragile or humanitarian contexts, and most crucially, how communities would perceive or respond to them.

Advantages of causal evaluation methods & causal studies in humanitarian contexts

43. In contrast, causal evaluation methods such as randomized controlled trials (RCTs) – by directly measuring impacts in real-world settings – offer a more robust way to assess targeting approaches. Through these evaluation methods, researchers can compare

³⁹ Banerjee, A., Niehaus, P., & Suri, T. (2019). Universal Basic Income in the Developing World. *Annual Review of Economics*, 11(Volume 11, 2019), 959–983. <https://doi.org/10.1146/annurev-economics-080218-030229>

⁴⁰ Jeong, D., & Trako, I. (2022). *Cash and In-Kind Transfers in Humanitarian Settings: A Review of Evidence and Knowledge Gaps*. The World Bank. <https://doi.org/10.1596/1813-9450-10026>

⁴¹ Most of these welfare simulations assume a constant relative risk-aversion utility function following: Hanna, R., & Olken, B. A. (2018). Universal Basic Incomes versus Targeted Transfers: Anti-Poverty Programs in Developing Countries. *Journal of Economic Perspectives*, 32(4), 201–226. <https://doi.org/10.1257/jep.32.4.201>

⁴² Aiken, E., Ashraf, A., Blumenstock, J., Guiteras, R., & Mobarak, A. M. (2025). Scalable Targeting of Social Protection: When Do Algorithms Out-Perform Surveys and Community Knowledge? (Working Paper No. 33919). National Bureau of Economic Research. <https://doi.org/10.3386/w33919>

different targeting methods not only in terms of who is selected but also based on the impact that different targeting methods have on downstream outcomes such as food security, income, and psychosocial well-being. Still, most causal research on targeting (experimental or quasi-experimental) comes from development programmes that focus on social protection or poverty in non-humanitarian settings.^{43,44,45,46,47,48,49} Targeting models for these programmes generally rely on stable, long-term indicators such as consumption or asset ownership. By contrast, humanitarian interventions respond to shocks – e.g., conflict, displacement, or natural disasters – which are typified by transient needs. Given these differences, targeting approaches that perform well in stable environments may not translate effectively to crisis contexts. While evidence from development programmes generally find that PMT performs somewhat better than community-based methods in identifying consumption-poor households,^{50,51,52,53,54,55} it is unclear whether these results can be applied to humanitarian contexts as well.

44. The few causal studies on humanitarian programmes confirm the differences between humanitarian and development targeting. Premand and Schnitzer (2021)⁵⁶ in Niger and Stoeffler and Schnitzer (2024)⁵⁷ across six Sahel countries show that, although PMT outperforms CBT when welfare is measured by per-capita consumption, this advantage disappears when using alternative indicators of well-being such as the FCS. This suggests that in humanitarian settings, standard benchmarks of precision applied by development programmes may not adequately capture the former's different programme objectives. A

⁴³ Alatas, V., Banerjee, A., Hanna, R., Olken, B. A., & Tobias, J. (2012). Targeting the Poor: Evidence from a Field Experiment in Indonesia. *American Economic Review*, 102(4), 1206–1240. <https://doi.org/10.1257/aer.102.4.1206>

⁴⁴ Sabates-Wheeler, R., Hurrell, A., & Devereux, S. (2015). Targeting social transfer programmes: Comparing design and implementation errors across alternative mechanisms. *Journal of International Development*, 27(8), 1521–1545. <https://doi.org/10.1002/jid.3186>

⁴⁵ Karlan, D., & Thuysbaert, B. (2019). Targeting Ultra-Poor Households in Honduras and Peru. *The World Bank Economic Review*, 33(1), 63–94. <https://doi.org/10.1093/wber/lhw036>

⁴⁶ Basurto, M. P., Dupas, P., & Robinson, J. (2020). Decentralization and efficiency of subsidy targeting: Evidence from chiefs in rural Malawi. *Journal of Public Economics*, 185, 104047. <https://doi.org/10.1016/j.jpubeco.2019.07.006>

⁴⁷ Beaman, L., BenYishay, A., Magruder, J., & Mobarak, A. M. (2021). Can Network Theory-Based Targeting Increase Technology Adoption? *American Economic Review*, 111(6), 1918–1943. <https://doi.org/10.1257/aer.20200295>

⁴⁸ Premand, P., & Schnitzer, P. (2021). Efficiency, Legitimacy, and Impacts of Targeting Methods: Evidence from an Experiment in Niger. *The World Bank Economic Review*, 35(4), 892–920. <https://doi.org/10.1093/wber/lhaa019>

⁴⁹ Haushofer, J., Niehaus, P., Paramo, C., Miguel, E., & Walker, M. (2025). Targeting Impact versus Deprivation. *American Economic Review*, 115(6), 1936–1974. <https://doi.org/10.1257/aer.20221650>

⁵⁰ Alatas, V., Banerjee, A., Hanna, R., Olken, B. A., & Tobias, J. (2012). Targeting the Poor: Evidence from a Field Experiment in Indonesia. *American Economic Review*, 102(4), 1206–1240. <https://doi.org/10.1257/aer.102.4.1206>

⁵¹ Stoeffler, Q., Mills, B., & Del Ninno, C. (2016). *Reaching the Poor: Cash Transfer Program Targeting in Cameroon*. Elsevier. <https://doi.org/10.1596/24255>

⁵² Basurto, M. P., Dupas, P., & Robinson, J. (2020). Decentralization and efficiency of subsidy targeting: Evidence from chiefs in rural Malawi. *Journal of Public Economics*, 185, 104047. <https://doi.org/10.1016/j.jpubeco.2019.07.006>

⁵³ Trachtman, C., Permana, Y. H., & Sahadewo, G. A. (2026). How much do our neighbors really know? The limits of community-based targeting. *Journal of Development Economics*, 178, 103555. <https://doi.org/10.1016/j.jdevec.2025.103555>

⁵⁴ Sumarto, S., Satriawan, E., Olken, B. A., Banerjee, A., Tohari, A., Alatas, V., & Hanna, R. (2025). *Community Targeting at Scale* (Working Paper No. 33322). National Bureau of Economic Research. <https://doi.org/10.3386/w33322>

⁵⁵ Aiken, E., Ashraf, A., Blumenstock, J., Guiteras, R., & Mobarak, A. M. (2025). *Scalable Targeting of Social Protection: When Do Algorithms Out-Perform Surveys and Community Knowledge?* (Working Paper No. 33919). National Bureau of Economic Research. <https://doi.org/10.3386/w33919>

⁵⁶ Premand, P., & Schnitzer, P. (2021). Efficiency, Legitimacy, and Impacts of Targeting Methods: Evidence from an Experiment in Niger. *The World Bank Economic Review*, 35(4), 892–920. <https://doi.org/10.1093/wber/lhaa019>

⁵⁷ Schnitzer, P., & Stoeffler, Q. (2024). Targeting Social Safety Nets: Evidence from Nine Programs in the Sahel. *The Journal of Development Studies*, 60(4), 574–595. <https://doi.org/10.1080/00220388.2023.2291325>

recent WFP impact evaluation (Baliki et al. 2026)⁵⁸ also compares targeting methods for humanitarian assistance in Lebanon and evaluates downstream welfare outcomes across three PMT-based targeting methods based on different outcomes and one categorical expert-driven approach. The evaluation finds similar performance across the four methods, with somewhat worse outcomes for the categorical approach. However, it did not include CBT, and its household-level randomization prohibits analysing community effects.

Community dynamics

45. Even more under-explored are broader community dynamics, such as trust, social cohesion, and potential tensions, which may shape programme effectiveness and overall success. These factors are especially relevant because different targeting methods may generate different forms of community engagement and acceptance. For instance, community-based targeting might strengthen legitimacy by incorporating local knowledge and fostering a sense of ownership over the process. At the same time, it may also heighten risks of elite capture or discrimination. Despite their importance, these trade-offs remain largely untested in humanitarian settings, underscoring the need for rigorous evidence.^{59,60}
46. The limited existing evidence on community dynamics yields mixed results.⁶¹ Experimental evidence from Indonesia and Bangladesh indicate that community-based targeting is preferred by recipients over PMT for identifying consumption-poor households.^{62,63,64} In contrast, evidence from Niger finds no significant differences among beneficiaries, while non-beneficiaries are less likely to prefer community-based targeting relative to both PMT and FCS-based targeting, even though they perceive it to generate fewer targeting errors.⁶⁵

Optimal levels of support

47. In addition, most studies compare targeting methods under fixed beneficiary quotas, i.e., requiring that a predetermined share of households in each community receive assistance.⁶⁶ This design choice facilitates direct comparison of targeting errors across

⁵⁸ Baliki, G., Cruz, M. A. F., Dunsch, F., Garcia, J., Guariso, A., Heirman, J., & Weiffen, D. (2026). *Targeting humanitarian aid in crisis: Evidence from the reduction of WFP assistance in Lebanon*. World Food Programme.

<https://www.wfp.org/publications/targeting-humanitarian-aid-crisis-evidence-reduction-wfp-assistance-lebanon>

⁵⁹ Schnitzer, P., & Stoeffler, Q. (2024). Targeting Social Safety Nets: Evidence from Nine Programs in the Sahel. *The Journal of Development Studies*, 60(4), 574–595. <https://doi.org/10.1080/00220388.2023.2291325>

⁶⁰ Sabates-Wheeler, R., Hurrell, A., Devereux, S., & UNU-WIDER. (2014). *Targeting social transfer programmes: Comparing design and implementation errors across alternative mechanisms* (40th ed., Vol. 2014). UNU-WIDER. <https://doi.org/10.35188/UNU-WIDER/2014/761-5>

⁶¹ Schnitzer, P., & Stoeffler, Q. (2024). Targeting Social Safety Nets: Evidence from Nine Programs in the Sahel. *The Journal of Development Studies*, 60(4), 574–595. <https://doi.org/10.1080/00220388.2023.2291325>

⁶² Alatas, V., Banerjee, A., Hanna, R., Olken, B. A., & Tobias, J. (2012). Targeting the Poor: Evidence from a Field Experiment in Indonesia. *American Economic Review*, 102(4), 1206–1240. <https://doi.org/10.1257/aer.102.4.1206>

⁶³ Sumarto, S., Satriawan, E., Olken, B. A., Banerjee, A., Tohari, A., Alatas, V., & Hanna, R. (2025). *Community Targeting at Scale* (Working Paper No. 33322). National Bureau of Economic Research. <https://doi.org/10.3386/w33322>

⁶⁴ Aiken, E., Ashraf, A., Blumenstock, J., Guiteras, R., & Mobarak, A. M. (2025). *Scalable Targeting of Social Protection: When Do Algorithms Out-Perform Surveys and Community Knowledge?* (Working Paper No. 33919). National Bureau of Economic Research. <https://doi.org/10.3386/w33919>

⁶⁵ Premand, P., & Schnitzer, P. (2021). Efficiency, Legitimacy, and Impacts of Targeting Methods: Evidence from an Experiment in Niger. *The World Bank Economic Review*, 35(4), 892–920. <https://doi.org/10.1093/wber/lhaa019>

⁶⁶ The practice of using quotas is intended to avoid the issues that varying coverage imposes when calculating targeting errors, given that coverage has a mechanical relationship with targeting errors, especially inclusion errors. For example, even with a highly accurate method, if the quota is set below the prevalence of the indicator of interest, such as food insecurity, inclusion errors will mechanically increase, with the magnitude rising as coverage decreases.

methods, but it limits the ability to assess how different approaches affect coverage. In practice, many humanitarian organizations, including WFP, aim to assist all households deemed highly vulnerable within target areas. Therefore, when evaluating targeting methods in humanitarian contexts, considering their implications for coverage is important, not least because differences in coverage translate into differences in programme costs, which is a key dimension under tight budget constraints.

48. Emerging evidence on spillovers and multiplier effects also suggests that assisting only a fraction of households in a community may still generate broad welfare gains, as indirect effects can improve overall outcomes while freeing up resources to support other communities.^{67,68,69,70} However, rigorous evidence on the optimal level of support at the community level remains very limited, and no causal study to date has examined how targeting shapes both the share of households reached and their welfare outcomes.

3.2. Evaluation questions

49. This evaluation aims to advance humanitarian literature and practice by providing empirical evidence from a live crisis in the DRC. It evaluates both the technical accuracy of targeting as well as its social impacts to assess the overall legitimacy of aid delivery. Furthermore, the study analyses the trade-offs between cost, programme reach, and targeting intensity, offering actionable insights for maximizing the impact of limited resources while maintaining community acceptance.
50. The first part of the evaluation examines the performance of the targeting approaches in identifying food-insecure households and aims to answer the following evaluation questions (EQs):
 - **EQ1. Coverage:** How do targeting methods differ in terms of beneficiary coverage?
 - **EQ2. Precision:** Which targeting method is more precise in selecting households with the lowest food security?
 - **EQ3. Profiles:** How do targeting methods differ in the types of households they select, and do these differences lead to systematic exclusion or over-targeting of specific population subgroups?
51. Building on this, the evaluation considers whether differences in targeting outcomes lead to measurable differences in overall welfare. In addition, the study looks at community-level effects, including satisfaction and social cohesion. Specifically, the evaluation addresses the following additional questions:

⁶⁷ Angelucci, M., & De Giorgi, G. (2009). Indirect Effects of an Aid Program: How Do Cash Transfers Affect Ineligibles' Consumption? *American Economic Review*, 99(1), 486–508. <https://doi.org/10.1257/aer.99.1.486>

⁶⁸ Egger, D., Haushofer, J., Miguel, E., Niehaus, P., & Walker, M. (2022). General Equilibrium Effects of Cash Transfers: Experimental Evidence From Kenya. *Econometrica*, 90(6), 2603–2643. <https://doi.org/10.3982/ECTA17945>

⁶⁹ Filmer, D., Friedman, J., Kandpal, E., & Onishi, J. (2023). Cash Transfers, Food Prices, and Nutrition Impacts on Ineligible Children. *The Review of Economics and Statistics*, 105(2), 327–343. https://doi.org/10.1162/rest_a_01061

⁷⁰ Delius, A., & Sterck, O. (2024). Cash transfers and micro-enterprise performance: Theory and quasi-experimental evidence from Kenya. *Journal of Development Economics*, 167, 103232. <https://doi.org/10.1016/j.jdeveco.2023.103232>

- **EQ4. Welfare:** Do the differences in targeting outcomes translate into differences in welfare for the target populations?
 - **EQ5. Community acceptance and cohesion:** Which targeting method leads to higher community satisfaction, trust, and social cohesion?
52. Lastly, considering both the precision of the methods and their overall impact, the study assesses their overall cost-effectiveness and whether outcomes are achieved at different costs across the two approaches. This question is especially important for constrained COs like WFP DRC as they aim to ensure resources maximize support for the vulnerable.
- **EQ6. Cost-effectiveness:** Which targeting method is more cost-effective?

3.3. Evaluation design

53. The evaluation uses a stratified cluster RCT design embedded in the expansion of WFP activities in Tanganyika Province. The cluster used for randomization is the bloc, which groups households that are proximate to each other and share social ties, making them a meaningful unit for both programme targeting and delivery. The CO identified communities in Tanganyika province where GFA activities were planned for 2022/2023. In total, 165 blocs were identified (approximately 13,600 households), and of these, 96 blocs fulfilled the following eligibility criteria for inclusion in the impact evaluation sample:
- No WFP assistance had been provided in the bloc over the previous 12 months;
 - The bloc was identified by WFP as experiencing crisis-level food insecurity (IPC Phase 3 or higher); and
 - There were at least 40 households living in the bloc, which reflects the minimum sample size for a cluster.

Stratification and randomization

54. Within each *groupement*, blocs were ranked by the estimated number of households in the bloc and following this ordering, blocs were then grouped into “quadruplets”, or groups of four blocs of similar size.⁷¹ These quadruplets served as the strata of the study design. Within each quadruplet or stratum, blocs were randomly assigned to either the PMT Plus or the CBT group. This design improved balance and statistical power, while also mitigating the effects of potential bloc attrition in a fragile and insecure study environment.

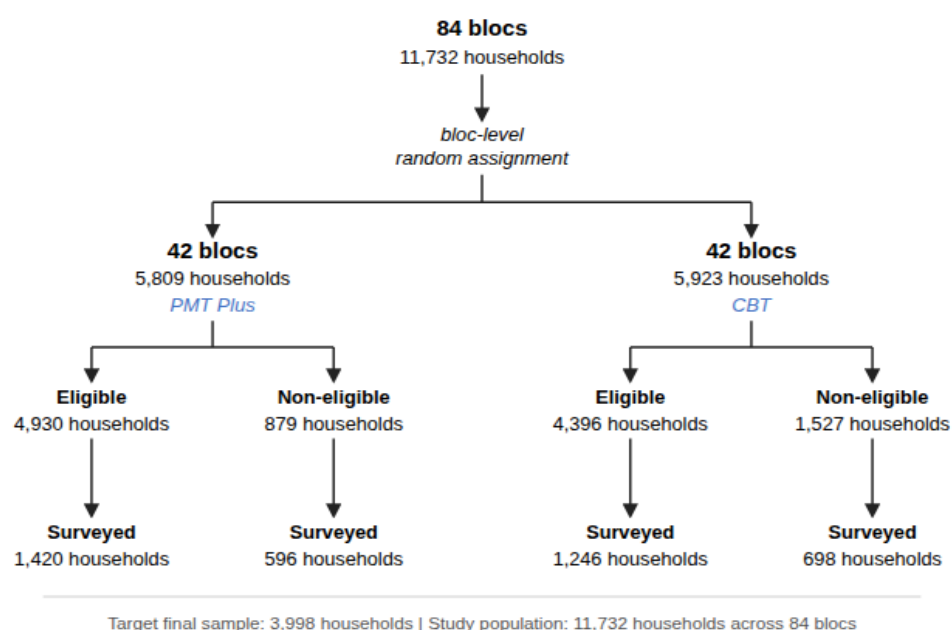
Sampling

55. The initial sampling design aimed to collect data from 40 households per bloc. However, due to security concerns, 12 of the 96 study blocs – equally distributed across the two study arms – were dropped before the study began, resulting in a final sample of 84 blocs. Appendix Tables A2.1 and A2.2 show that both the study population and the sample are well-balanced. Tables A2.3 and A2.4 in the appendix also show that the sample across arms remains well-balanced even when addressing issues related to programme implementation.

⁷¹ As the total number of blocs was not perfectly divisible by four, two blocs were allocated to the quadruplets with the smallest number of households. This resulted in two quadruplets comprising five blocs.

56. To partly compensate for the loss of statistical power, the target number of households surveyed in the remaining blocs was increased to 56, whenever possible. To maximize the power of the comparison between beneficiaries and non-beneficiaries, the study aimed for equal sample sizes: 28 beneficiaries and 28 non-beneficiaries. In practice, given the limited size of some blocs, and the widespread fragility of the context, most blocs had fewer than 28 households of either type. In such cases, all households of the type that did not meet the minimum were surveyed, and the remaining households were randomly selected from the other type, when feasible. For example, a bloc with only 15 non-beneficiary households and 50 beneficiary households would have all 15 non-beneficiary households sampled plus 41 beneficiary households (the targeted 28 plus 13 households to fill the gap).

Figure 2: Impact evaluation design

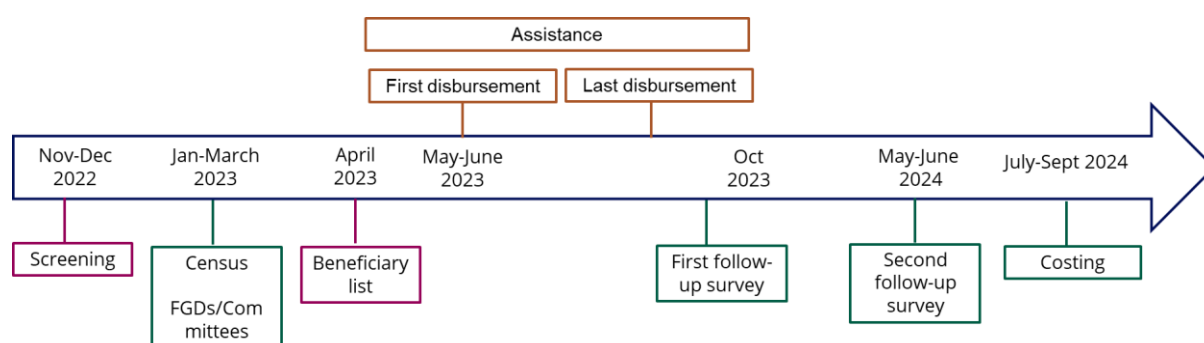


57. As a result, the final target sample comprised 3,998 households (from the study population of 11,732 households across 84 blocs). Of these, 2,694 households (67 percent) were eligible for cash transfers, while 1,304 households (33 percent) were ineligible. Figure 2 presents the final evaluation design as well as the target sample by targeting arm and beneficiary status.

3.4. Timeline

58. The timeline in Figure 3 outlines the different phases of programme and targeting implementation:

Figure 3. Implementation timeline



59. *Screening*: This process involved the CP conducting scoping visits, mapping blocs, and estimating populations within the locations identified by the CO for planned GFA.
60. *Census*: A baseline door-to-door census was conducted with the target population and established the sampling frame. Data collection lasted from January to March 2023.
61. *Community consultation and beneficiary listing*: CBT committees and PMT Plus focus groups were formed and convened. Beneficiary lists were finalised by April 2023. The evaluation did not set a beneficiary quota within targeted areas, so the caseload was determined purely by need resulting from the targeting exercise. Appeals about exclusion were assessed against the targeting rules for the relevant bloc.
62. *Programme implementation*: Assistance was delivered in four disbursements between May and September 2023. During this period, WFP conducted routine monitoring and collected administrative data, including distribution records, complaints, and appeals.
63. *Follow-up data collection rounds and costing*: The first follow-up data collection took place between September and October 2023. The second and last follow-up data collection took place between May and June 2024. A costing exercise was conducted after the completion of data collection.

4. Data and Outcomes

4.1. Data sources

64. The impact evaluation relies on three waves of survey data as well as administrative data.

Household census

65. The census is the main source of information for assessing the performance of the targeting approaches on coverage and precision. It covers the entire target population of 11,732 households and was conducted by the CO between January and March 2023. The respondent to the census was the household head and, if possible, their spouse or partner. If neither was available at the time of the interview, the adult who was the next most knowledgeable about the household's food consumption was interviewed.

66. The census collected demographic characteristics as well as other socioeconomic indicators such as asset ownership. Both represent easily observable household attributes that are commonly used in targeting in most low-income contexts, given the lack of reliable information on income or expenditure, unlike in more developed settings.⁷² For the purposes of this evaluation, the census also collected self-reported food consumption data to construct the FCS. Many organizations, including WFP, advise against using such direct outcome measures for targeting due to the risk of measurement error and strategic misreporting.^{73,74} Accordingly, the FCS is used only to compare targeting precision across the two methods since any noise or misreporting is expected to affect both arms similarly and not the targeting itself. Overall, the census provides the basis for implementing and assessing the targeting exercise across the full population while also generating baseline information for the smaller study sample.

Follow-up impact evaluation surveys on study sample

67. Two follow-up surveys were administered to the evaluation sample of 3,998 households. The study sample was drawn from the 84 blocs with census data, as the census was not implemented in the 12 blocs dropped from the study (six in each arm), given that randomization occurred prior to census data collection.
68. The surveys were addressed to the female head of household.⁷⁵ In cases where she was not available for the whole duration of data collection, she was replaced with another adult woman from the same household who was knowledgeable about household affairs. If no suitable female respondent could be identified within the household, the supervisor was authorized to select a replacement household from a pre-sorted list that matched the same beneficiary status as the original household. The questionnaires collected household-level data to compare the impact of the targeting approaches on welfare, community acceptance, and social cohesion. The two follow-up surveys were similar in structure and length, but the second included a few additional questions on social cohesion, trust, and tension that were not expected to be observed immediately after the targeting was implemented.
69. Data collection for the first follow-up was done between September and October 2023, immediately after the final transfer was disbursed. During this period, severe security disruptions affected approximately 800 households (20 percent of the planned sample of 3,998 households). Replacements were successfully identified and interviewed for 262 of these cases, resulting in a final achieved sample of 3,460 households (which corresponds to an average of about 41 households per bloc). This equals an attrition rate of 13.5 percent after replacements. Of the interviewed, 1,753 households were PMT Plus blocs and 1,707 were CBT blocs. Among the PMT Plus interviewees, 69 percent (1,215 households) were

⁷² Banerjee, A., Hanna, R., Olken, B. A., & Lisker, D. S. (2024). Social Protection in the Developing World. *Journal of Economic Literature*, 62(4), 1349–1421. <https://doi.org/10.1257/jel.20241646>

⁷³ Martinelli, C., & Parker, S. W. (2009). Deception and Misreporting in a Social Program. *Journal of the European Economic Association*, 7(4), 886–908. <https://www.jstor.org/stable/40282792>

⁷⁴ Burger, R., & Owens, T. (2010). Promoting Transparency in the NGO Sector: Examining the Availability and Reliability of Self-Reported Data. *World Development*, 38(9), 1263–1277. <https://doi.org/10.1016/j.worlddev.2009.12.018>

⁷⁵ As the household head tends to be male, the female household head refers to the person who makes household decisions besides the male head. There are exceptions like widowers.

beneficiaries, while beneficiaries comprised 64 percent (1,087 households) of the CBT interviewees.

70. The second follow-up was collected between May and June 2024, roughly eight months after the programme ended. At that time conditions had stabilized, and overall attrition was only 1 percent after replacements. The sample totalled 3,960 households (47 households per bloc on average). Out of these, 2,016 households were in PMT Plus blocs and 1,944 in CBT blocs. Of these, 70.4 percent of PMT Plus (1,420 households) and 64.1 percent of CBT (1,246 households) received the GFA.
71. In both rounds, there were no systematic differences in attrition between PMT Plus and CBT blocs. Appendix Table A2.5 summarizes attrition by data collection round, while Table A2.6 shows that the sample, even when excluding blocs affected by the security disruptions, is still balanced between the two arms.

Administrative data

72. In addition to survey data, the analysis also makes use of administrative data collected by the CO. In particular, the analysis relies on two sources:
 - records of households that received the transfer to assess adherence to the study protocol; and
 - records of complaints received by the CO to assess beneficiaries' acceptance of the targeting decision.

4.2. Main outcomes

73. Table 1 below outlines the key outcomes included in the evaluation as well as their related data sources. More details on these outcomes are provided in Appendix Table A3.1 describing the indicators and Table A3.2 explaining the construction of outcomes based on composite indices.

Table 1: Outcomes of interest

Dimension	Indicator	Census	Follow-up 1	Follow-up 2	Admin Data
Precision	Coverage rate	X			
	Inclusion error rate	X			
	Exclusion error rate	X			
Welfare	Food Consumption Score (FCS)		X	X	
	Reduced Coping Strategy Index (rCSI)		X	X	
	Livelihood Coping Strategy Index (LCSI)		X	X	
	Life Satisfaction Index		X	X	
	Women's Agency Index		X	X	
	Complaints				X
	Self-reported satisfaction		X	X	

Acceptance & Social Cohesion	Perceptions of the targeting process			X	
	Social Cohesion Index			X	
	Trust Index			X	
	Collective Action Index			X	
	Perceived safety (home)			X	
	Perceived safety (community)			X	

5. Main Findings

74. The discussion of findings is divided into six subsections, each addressing a specific evaluation question, by focusing on a different comparison across the two targeting methods, namely:
- Coverage
 - Precision in the identification of food insecure households
 - Profiles of selected (and excluded) households
 - Welfare outcomes
 - Acceptance and social cohesion
 - Cost-effectiveness
75. The analyses for the first three sub-sections are based on census data, while welfare, satisfaction and social cohesion are studied using the two survey follow-ups. The assessment of cost-effectiveness uses both survey data and administrative cost data. Appendix A4 provides details on the estimation strategy behind each evaluation question.⁷⁶

5.1. Coverage: How do targeting methods differ in terms of beneficiary coverage?

Summary of findings: The study population displayed high levels of vulnerability, with 86 percent of households reporting an FCS below or equal to 28 (the standard adjusted threshold for food insecurity) in the census data. Cash assistance coverage rates were not predetermined and were allowed to vary across blocs according to need as identified by each targeting method.

⁷⁶ There are potential threats to the validity of the study design: (1) the reduction of the sample from 96 to 84 blocs poses a risk of imbalance across study arms; (2) imperfect programme implementation resulted in some households receiving fewer or more transfers than originally intended; (3) three IDP-only blocs exhibit almost perfect coverage, mechanically limiting variation in targeting performance. To assess the implications of these issues, additional robustness checks are conducted. The results of these are presented in Appendix A5 for coverage, Appendix A6 for precision, Appendix A7 for welfare, and Appendix A8 for community acceptance and social cohesion.

On average, PMT Plus selected 80 percent of the households, while CBT selected 69 percent. The large 11-percentage-point difference is statistically significant at the 1-percent level.

76. The first evaluation question focuses on the number of households selected under each targeting approach. The CO did not impose any predetermined quota for targeting and every household identified as in need received assistance, making the *coverage rate* the first outcome for assessing targeting performance. *Coverage rate* is defined as the proportion of households in each bloc that was selected to receive assistance.
77. In the context of WFP GFA, food insecurity is the primary criterion for determining vulnerability. As mentioned previously, one of the most widely used food insecurity indicators across WFP operations is the FCS. The score ranges from 0 to 112, and households with a score equal to or below the standard threshold of 28 (adjusted for high sugar and oil consumption) are classified as food insecure. According to FCS data from the census, an average of 86 percent of households across the study blocs are classified as food insecure.⁷⁷ Both targeting methods identified a smaller share of households for assistance than the actual proportion of food insecure. This might reflect inflated self-reported FCS in the census, or under-performance of the two targeting methods.
78. Importantly, when comparing the two methods, large differences in their implied coverage rate are observed. In CBT blocs, the average coverage rate was 69 percent, compared to 80 percent in PMT Plus blocs. The 11-percentage-point difference is large and statistically significant ($p < 0.01$). Figure 4 presents coverage rates by targeting method. Panel A shows the average coverage rate across blocs, while Panel B displays the full distribution of coverage rates. The figure indicates that coverage rates under CBT are lower on average (Panel A) and exhibit greater variability across blocs compared with PMT Plus (Panel B).⁷⁸
79. The observed differences in coverage between PMT Plus and CBT might appear surprising at first. In the absence of any clear limitation, one might expect that communities would expand the vulnerability definition (and therefore programme coverage). The results show that the opposite is true and that involving communities in setting vulnerability criteria more actively actually leads to the identification of fewer beneficiaries, on average, than the more standard data-driven approach.
80. On its own, whether the lower coverage under CBT can be interpreted positively or negatively is ambiguous. In contexts such as DRC, where the needs are far larger than resources available, under a fixed budget, a lower coverage rate per community would allow assistance to be extended to a greater number of communities. At the same time, there is a risk that the lower coverage in CBT blocs was achieved at the expense of excluding

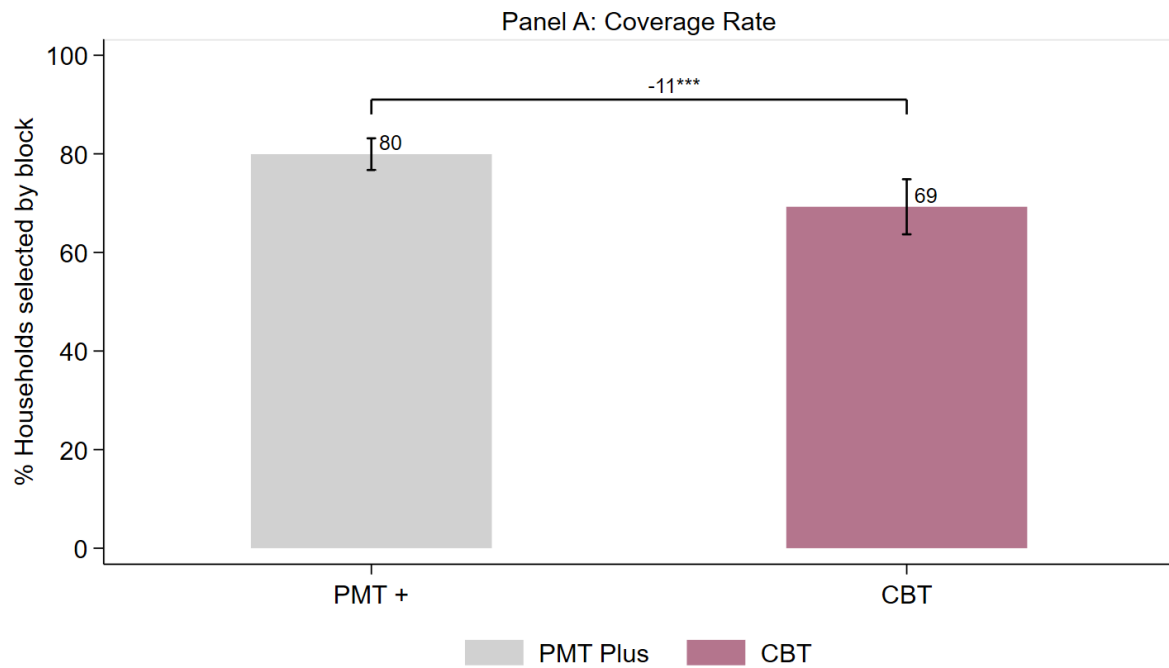
⁷⁷ Although the FCS data from the census could have been used as a benchmark, targeting followed standard operational procedures: in PMT Plus areas, eligibility was determined using predicted FCS derived from proxy indicators rather than observed (self-reported) FCS. This reflects both the prevailing practice of the CO and broader WFP guidance discouraging the use of direct outcome measures for beneficiary selection. The approach therefore mirrors real-world policy conditions, where targeting relies on observable household characteristics and proxy variables rather than realized welfare outcomes.

⁷⁸ Appendix Table A5.1 presents regression results from the main regression specification as well as the results of a robustness check excluding the three IDP blocs with almost-perfect coverage for coverage. The results from both analyses are similar.

households that would have deserved assistance but were (purposely or inadvertently) left out by the committees.

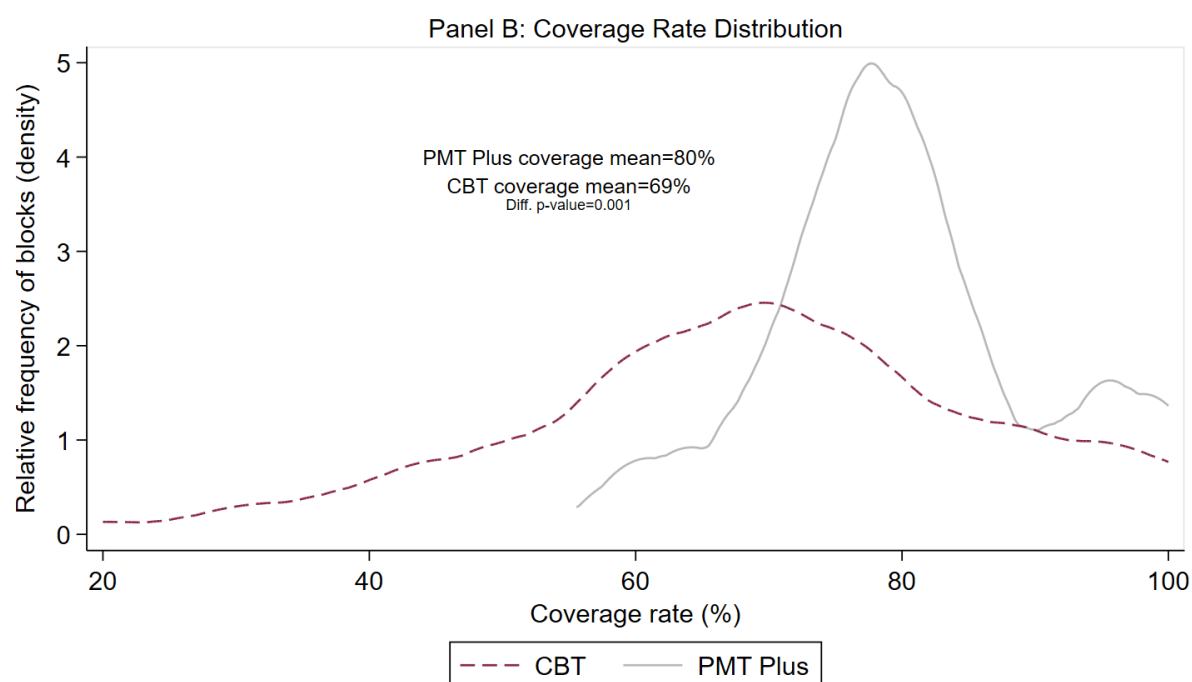
81. In order to shed light on these concerns, the next step in the analysis assesses how precise the two targeting methods were in selecting vulnerable households in need of assistance. The analysis of coverage also links to the analysis of welfare impacts further below.

Figure 4: Coverage



Note: The figure presents the average coverage rates (adjusted means) under CBT and PMT Plus. Coverage is defined as the share of households selected under each targeting approach. CBT coverage corresponds to the share of households selected under CBT criteria, while PMT Plus coverage corresponds to the share of households which were selected either through a proxy means test based on FCS

(predicted FCS ≤ 28) or through criteria set by community focus groups. The y-axis reports coverage rates. The reported difference in mean coverage rates between PMT Plus (80 percent) and CBT (69 percent) is statistically significant at the 1-percent level ($p < 0.01$).



Note: The figure presents the distribution of coverage rates under CBT and PMT Plus. The x-axis reports coverage rates, and the y-axis reports the relative frequency, or density, of blocs at each coverage level, with higher values indicating coverage rates that occur more frequently across blocs.

5.2. Precision: Which targeting method is more precise in selecting households with the lowest food security?

Summary of findings: Targeting errors, defined with reference to food security measures recorded in the census, are relatively low under both approaches, although neither method achieves perfect targeting. The two methods perform similarly in terms of inclusion errors. PMT Plus results in lower exclusion errors than CBT, but this difference is entirely driven by its higher coverage rate. When coverage is held constant in a simulation, the difference disappears, and neither food security inequality nor overall vulnerability levels meaningfully affect the relative precision of the two methods.

82. As described in the previous sections, assistance decisions in the PMT Plus arm were based on *predicted* rather than directly observed food security at baseline. This approach reflects standard WFP practice as, in most operational settings, actual FCS measures are not available for all households in the target locations and self-reported indicators may be noisy or subject to reporting bias. This study, however, collected baseline FCS data for evaluation purposes, which allows for the comparison of households' observed food security status with predicted eligibility based on the targeting methods.

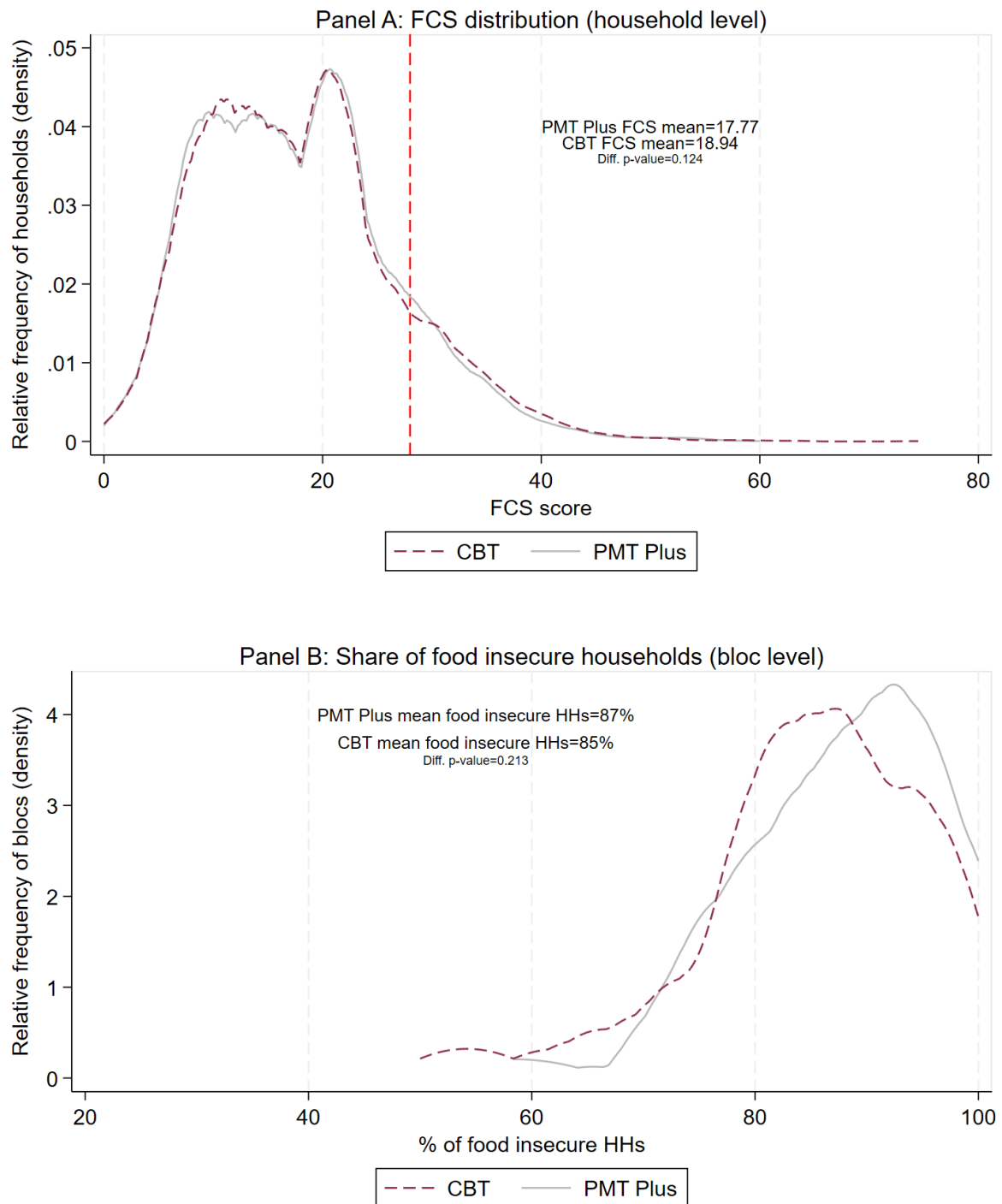
83. Panel A of Figure 5 shows the distribution of raw FCS scores for the PMT Plus and CBT arms. As expected, given random assignment, the two distributions are nearly identical and statistically indistinguishable. The vertical line indicates the threshold for defining food-insecure households (FCS = 28). The distributions reflect the widespread vulnerability of the study population, with most of the households falling below the FCS threshold (to the left of the red vertical line).⁷⁹
84. Panel B illustrates the same point from a slightly different perspective. It plots the distribution of food insecurity by showing the share of households that are classified as food-insecure (FCS ≤ 28) across blocs by treatment arm. On average, 86 percent of households in each bloc fall below the food-security threshold. Similar to Panel A, the distributions of the two targeting methods cannot be statistically distinguished.
85. As discussed in the previous section, both targeting approaches identified fewer beneficiaries than would have been selected using the score cut-off of 28 from self-reported FCS.

Inclusion and exclusion errors

86. Assessing targeting precision requires defining a “ground truth.” Although self-reported FCS measures may be affected by strategic misreporting by households hoping to receive assistance, they remain the best available proxy in this setting. Importantly, any such misreporting is likely to affect the overall level of targeting errors but not the comparison between the two targeting methods, which is the main focus of this study.

⁷⁹ For more information on food security indicators, see also: Manikas, I., Ali, B. M., & Sundarakani, B. (2023). A systematic literature review of indicators measuring food security. *Agriculture & Food Security*, 12(1), 10. <https://doi.org/10.1186/s40066-023-00415-7>

Figure 5: FCS distribution at baseline across study arms



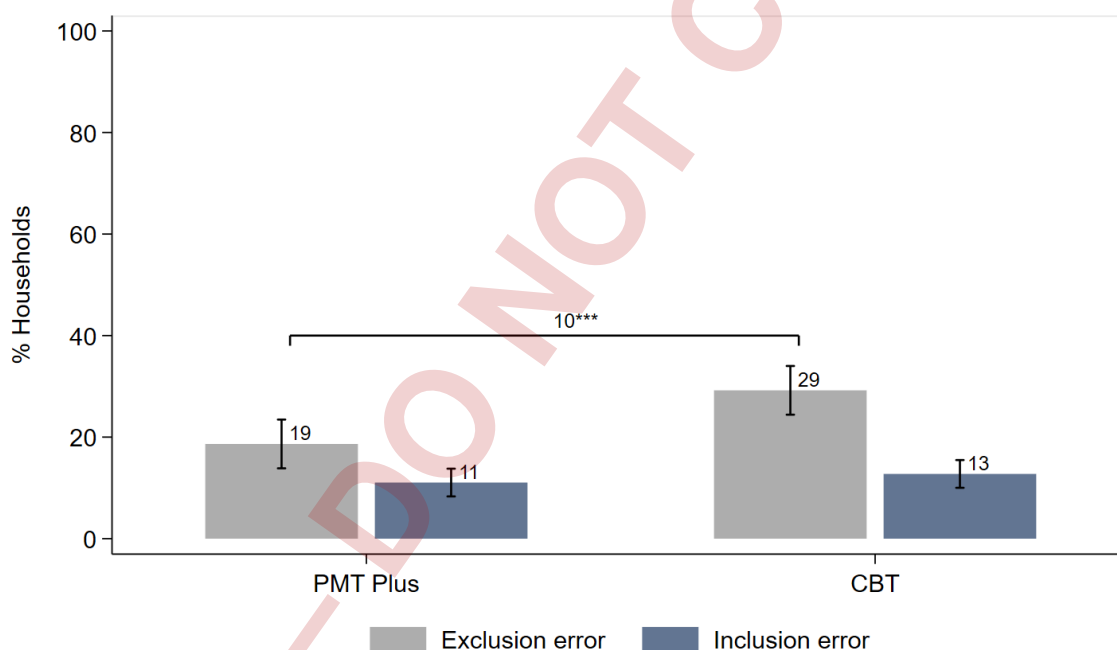
Note: In panel A, the dashed vertical red line corresponds to the adjusted threshold used to define food insecurity, i.e., FCS=28. Panel A reports household level densities. In Panel B, the percentage of food insecure households refers to the share of households within each bloc whose FCS is below 28. Panel B reports bloc-level densities. *Diff. p-value* reports the p-value from a two-sided test of equality in means between PMT Plus and CBT for the outcomes shown in each panel. A p-value larger than 0.10 denotes differences that are not statistically significant.

87. Targeting precision is assessed using two standard metrics from the targeting literature: inclusion error and exclusion error, with self-reported FCS scores serving as the benchmark. These measures are based on the idea that “perfect” targeting would include all households with an FCS of 28 or below and exclude all households with an FCS above 28. On this basis, the two indicators are defined as follows:

- *Inclusion error*: the share of non-food-insecure households ($FCS > 28$) among the selected households; and
- *Exclusion error*: the share of food-insecure households ($FCS \leq 28$) among those not selected by the targeting method.

88. Figure 6 summarises the targeting errors of the two methods. PMT Plus and CBT produced similar inclusion error rates – 11 percent and 13 percent, respectively – and the difference was not statistically significant. Appendix Table A6.1 presents regression results from the main regression specification as well as the results of a robustness check excluding the three IDP blocs with almost-perfect coverage for coverage. The results from both analyses are similar.

Figure 6: Targeting precision across study arms



Note: The reported measures are averages across blocs. The inclusion error captures the share of selected households that are not food insecure. The exclusion error measures the share of food-insecure households not selected, calculated as the ratio of excluded food-insecure households to total food-insecure households. Estimates come from regressions at the bloc level, which corresponds to the level at which indicators are estimated. Regressions include quadruplet fixed effects. Standard errors are clustered at the bloc level. Vertical lines indicate 95-percent confidence intervals. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

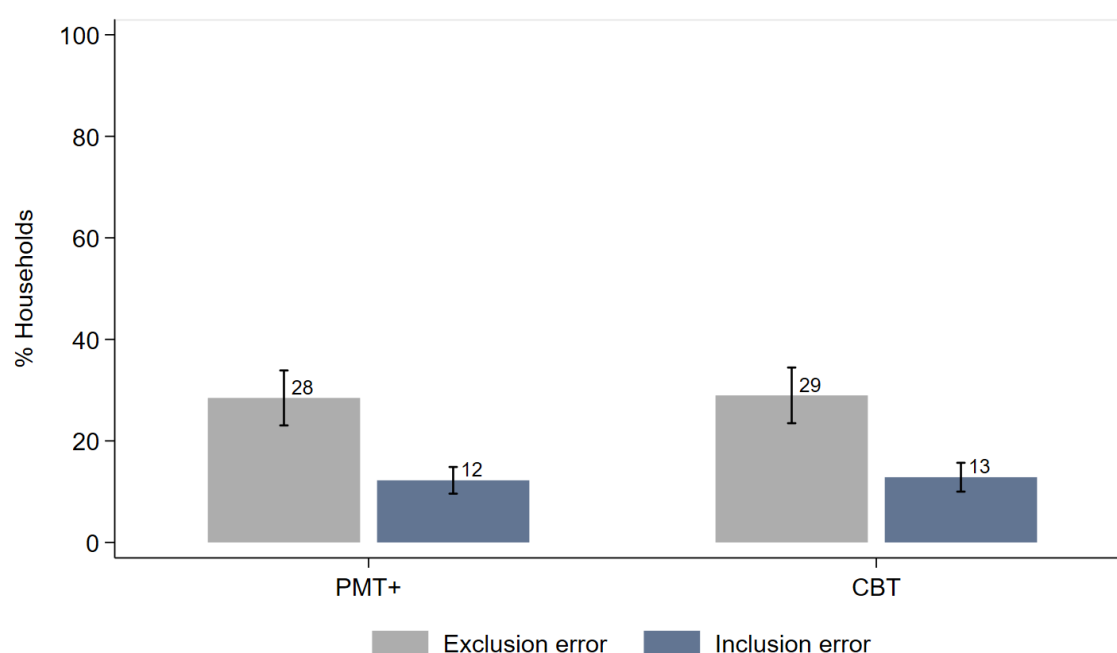
89. From a comparative perspective, despite the limitations of benchmarking to self-reported food consumption measures, both methods perform well. Targeting error rates are lower than those documented in recent experimental studies (see Appendix Table A6.2), which report median inclusion and exclusion errors of around 40 and 53 percent, respectively, although this evaluation setting displays much higher levels of food insecurity, and coverage, than most other studies.

90. This suggests that in contexts of near-universal vulnerability, the high prevalence of food insecurity naturally sets a higher floor for targeting precision, as the “pool” of eligible households is so large that the probability of selecting a non-vulnerable household is inherently minimized.
91. Significant differences between the two methods are, however, observed in the exclusion error rates: CBT led to an exclusion error of 29 percent, compared to only 19 percent for PMT Plus. This result is, however, sensitive to differences in coverage rates, as higher coverage tends to mechanically reduce exclusion errors. By contrast, there are no changes in inclusion errors, as they are less directly driven by coverage and instead depend on the composition of selected households. Inclusion error would increase with higher coverage only if the marginal households added are mostly non-vulnerable. In settings where most households are vulnerable, increases in coverage may therefore have little effect on inclusion error. The two error measures thus differ mechanically in how they relate to coverage.⁸⁰
92. To assess whether the lower exclusion error in the PMT Plus arm is due solely to higher coverage or also to the targeting mechanism itself, Figure 7 reports error rates from a simulation exercise, which models how PMT Plus would have performed if applied in CBT blocs to select the same number of households as CBT. Because targeting errors are mechanically related to coverage, comparisons across targeting approaches can be confounded by differences in the number of households selected. By holding coverage constant and applying PMT Plus selection rules within CBT blocs, the analysis enables a like-for-like comparison that separates the role of the targeting mechanism from the effect of programme size and coverage.
93. The simulation shows that, at the same level of coverage, differences in precision between the two methods disappear: exclusion errors are 29 percent and 28 percent for CBT and PMT Plus, respectively. This indicates that, under comparable coverage conditions, the targeting approaches achieve nearly identical levels of precision and targeting accuracy in identifying the most food-insecure households. When coverage constraints are relaxed, PMT Plus leads to higher coverage, which lowers its exclusion error, but without significantly impacting its inclusion error.
94. Overall, these results contrast with much of the existing literature, which generally finds that PMT outperforms alternative targeting approaches, including CBT, in terms of targeting precision. However, most of this evidence comes from social protection programs designed to identify the chronically poor, and PMT might be better suited to capturing structural poverty than to detect short-term shocks and seasonal fluctuations that are central to food insecurity.

⁸⁰ Formally, the link between error measures and coverage can be expressed as follows: let P denote the total number of actual food-insecure households. Of these, TP (True Positives) are correctly selected, while FN (False Negatives) are food-insecure households not selected. FP (False Positives) refers to food-secure households incorrectly selected. Coverage of the poor is the share of actual food-insecure households receiving assistance (TP/P). Exclusion error is the share of food-insecure households not selected (FN/P), which is mathematically equivalent to 1 minus coverage. Consequently, increasing coverage necessarily reduces exclusion error. Inclusion error, by contrast, is the share of food-secure households among the total selected ($FP / [TP + FP]$). Because this measure depends on the ratio of non-vulnerable to vulnerable households within the caseload, it is determined by the precision of the selection criteria rather than the scale of coverage alone.

95. This view is consistent with, for example, Del Ninno and Mills (2015),⁸¹ who argue that poverty-based PMT measures often fail to reflect shocks and seasonality, and with Alatas et al. (2012),⁸² who show that standard consumption-based poverty measures do not always align with community perceptions of welfare. This might explain why, in a humanitarian context where the objective is to identify households experiencing acute food insecurity rather than long-term poverty, the documented advantages of PMT are diminished. In this respect, the above findings are consistent with those of Premand and Schnitzer (2021), who find that community-based targeting performs similarly to FCS-based PMT targeting in a humanitarian setting.

Figure 7: Targeting precision across study arms simulating PMT Plus selection



Note: The reported measures are averages across blocs based on a simulation of how PMT Plus would have performed in blocs where CBT was used, assuming PMT Plus had selected the same number of households as CBT. The inclusion error captures the share of selected households that are not food insecure. The exclusion error measures the share of food-insecure households that were not selected, calculated as the ratio of excluded food-insecure households to total food-insecure households. Estimates come from regressions at the bloc level, which corresponds to the level at which indicators are estimated. Regressions include quadruplet fixed effects. Standard errors are clustered at the bloc level. Vertical lines indicate 95-percent confidence intervals. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Robustness checks and heterogeneity analyses

96. These findings are supported by a series of robustness checks which are reported in Appendix Table A6.3. First, alternative precision measures, namely the Coady Grosh Hoddinott Index (CGH) and Targeting Differentials (TD), are used to assess the progressivity of targeting and the extent to which assistance is concentrated among the more food insecure households, without relying on a single threshold. Second, to address potential bias in self-reported FCS measures, the analysis is replicated using an alternative definition of vulnerability based on a wealth index constructed from the census survey.

⁸¹ Del Ninno, C., & Mills, B. (Eds.). (2015). *Safety Nets in Africa: Effective Mechanisms to Reach the Poor and Most Vulnerable*. The World Bank. <https://doi.org/10.1596/978-1-4648-0435-9>

⁸² Alatas, V., Banerjee, A., Hanna, R., Olken, B. A., & Tobias, J. (2012). Targeting the Poor: Evidence from a Field Experiment in Indonesia. *American Economic Review*, 102(4), 1206–1240. <https://doi.org/10.1257/aer.102.4.1206>

97. In both the alternative precision measures and wealth-based proxy for vulnerability, the results do not lead to different conclusions regarding the relative performance of the two targeting approaches, whether coverage is held constant or allowed to vary.
98. Additional analyses were also conducted to examine heterogeneity in targeting performance across blocs with differing levels of food security inequality and overall vulnerability (see Appendix Tables A6.4 and A6.5). No meaningful differences are observed along these dimensions, and the results remain unchanged when controlling for coverage rates.

5.3. Profiles: How do targeting methods differ in the types of households they select, and do these differences lead to systematic exclusion or over-targeting of specific population subgroups?

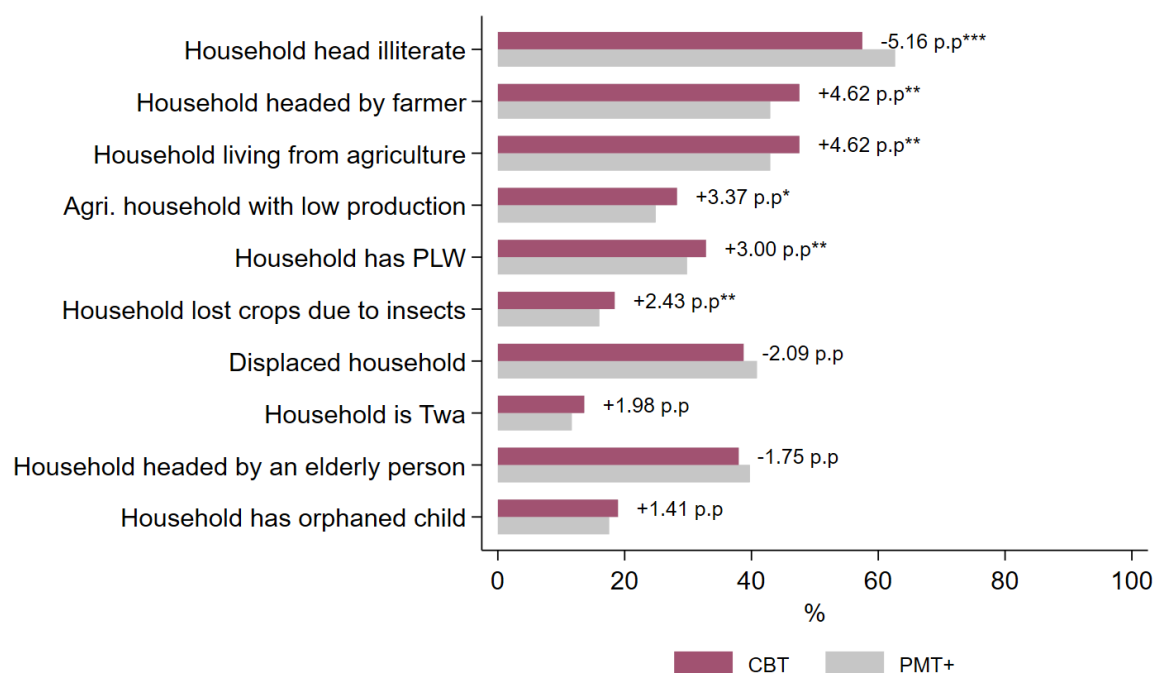
Summary of findings: PMT Plus and CBT select household profiles that overlap for many, but not all dimensions. PMT Plus puts more weight on household head demographics and displacement status, while CBT weighs agricultural and seasonal vulnerability higher. Targeting differences are concentrated in specific groups: CBT prioritizes "visible" vulnerabilities (disability, pregnancy, and agriculture) based on community perception, while PMT Plus favours socioeconomic indicators like illiteracy and widowhood. Ultimately, these patterns demonstrate that each method prioritizes distinct aspects of vulnerability.

99. For the 84 blocs, this section compares baseline characteristics of the households selected by the two methods to assess whether they prioritize different people. In order to neatly compare the actual selection decisions, abstracting from any population or coverage difference, the analysis in this section relies on comparing the actual CBT selection with the simulated (higher) PMT Plus selection within the CBT blocs.
100. Figure 8 reports the ten household characteristics with the largest differences in selection between the two approaches (as indicated by the coefficients in the figure), ordered from largest to smallest. Out of 37 baseline dimensions considered, the two targeting methods differ significantly in household selection along six of them (16 percent). Most of these dimensions reflect vulnerability criteria mentioned by many committees in CBT blocs, which instead were given lower weight in the PMT Plus methodology, including whether a household is from the Twa indigenous community, hosting displaced households, the presence of pregnant or lactating women, and crop losses due to natural shocks such as pests.
101. Overall, the patterns of selection indicate that CBT tends to give slightly more weight to agricultural and seasonal vulnerability, while PMT Plus tends to prioritize household-head demographics and displacement.
102. To assess whether either method systematically over- or under-selects specific socioeconomic groups relative to their "true" vulnerability level, the study uses a relative

representation (RR) indicator to measure the extent to which a group's share among beneficiaries selected by a targeting method differs from its share among actual food-insecure households (utilizing the FCS threshold at baseline).

103. Out of 29 baseline demographic characteristics considered, the two targeting methods differ along seven dimensions in their selection patterns. Figure 9 reports the seven characteristics with the largest differences in relative representation across targeting methods.

Figure 8: Differences in characteristics of selected households by targeting method

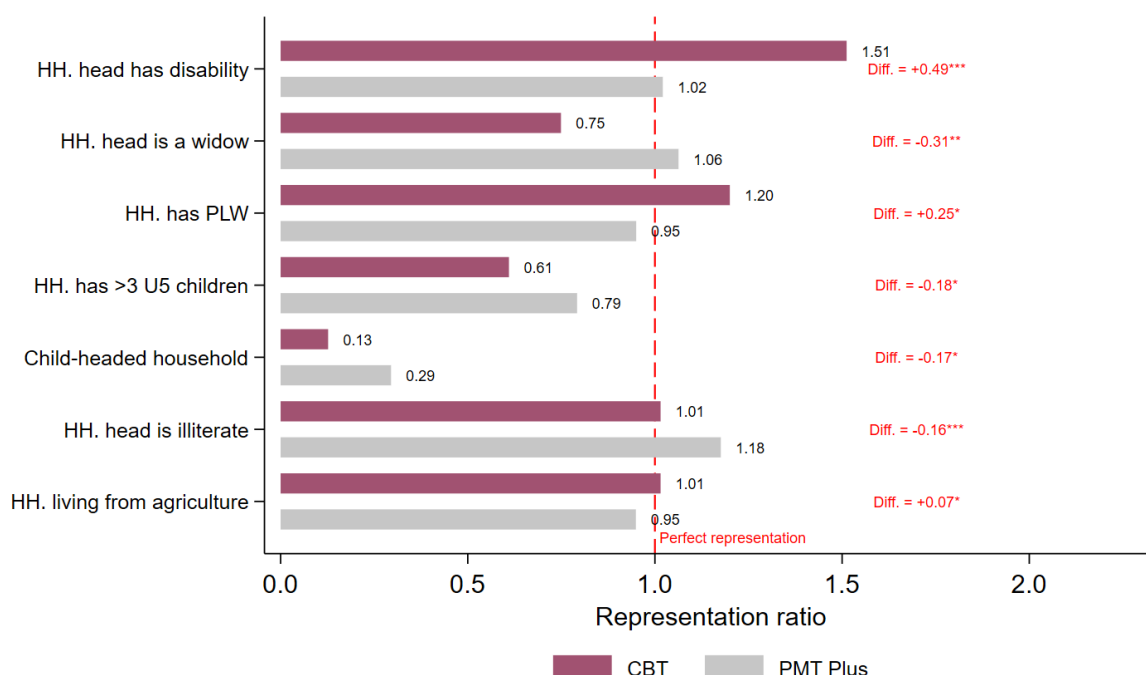


Note: The figure presents the characteristics of households identified as vulnerable by PMT Plus (simulation) and CBT (in practice) in CBT blocs. The figure presents the ten socioeconomic characteristics that show the largest differences between the two methods, while the coefficients represent the percentage point difference associated with CBT relative to PMT Plus. Statistical significance is indicated by asterisks: * p < 0.10, ** p < 0.05, *** p < 0.01.

104. Households headed by a person with a disability are substantially more over-targeted – higher inclusion error – under CBT than under PMT Plus (PMT Plus RR 1.02, CBT RR 1.51; difference +0.49; p < 0.01). CBT also more strongly over-targets households with pregnant and lactating women (PMT Plus RR 0.95, CBT RR 1.20; difference +0.25; p < 0.10) and households primarily living from agriculture (PMT Plus RR 0.95, CBT RR 1.02; difference +0.07; p < 0.10).
105. Although CBT aligns with the vulnerability profiles emphasized by community committees, these locally prioritized categories do not always correspond to the most food-insecure households, as recorded through FCS. For example, households primarily engaged in agriculture are more likely to be selected by CBT, especially when criteria such as crop loss are emphasized, as these shocks are highly visible within the community. However, households relying primarily on casual labour experience lower levels of food security (average FCS of 16), compared with households engaged in agriculture (average FCS of 19). As the census was conducted during the lean season, it is likely that agricultural households

are still better-off during other periods of the year. This potentially reflects the fact that agricultural households may have access to land or productive assets, whereas households dependent solely on casual labour are far less likely to have this asset. This suggests that CBT captures “visible” forms of vulnerability, which do not necessarily coincide with the highest levels of actual food insecurity.

Figure 9: Differences in relative representation



Note: The figure reports a relative representation (RR) index, defined as the ratio of a subgroup's share among beneficiary households to its share among food-insecure households, as measured by observed FCS. Values greater than 1 indicate overrepresentation relative to need, while values below 1 indicate underrepresentation. The red labels on the right report the difference in RR associated with CBT relative to PMT Plus. Statistical significance of differences across targeting methods is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

106. In contrast, PMT Plus over-targets households headed by an illiterate person significantly more than CBT (PMT Plus RR 1.18, CBT RR 1.02, difference -0.16 ; $p < 0.01$) as well as widow-headed households (PMT Plus RR 1.06, CBT RR 0.75, difference -0.31 ; $p < 0.05$). Although it is more likely than CBT to target households with more than three children under five (PMT Plus RR 0.79, CBT RR 0.61; difference -0.18 ; $p < 0.10$) as well as child-headed households (PMT Plus RR 0.30, CBT RR 0.13; difference -0.17 ; $p < 0.10$), it still significantly under-targets these groups when considering perfect representation.

5.4. Welfare: Do the differences in targeting outcomes translate into differences in welfare for the target populations?

Summary of findings: The lower programme coverage observed in CBT blocs (-11 percentage points) did not translate into differences in welfare outcomes across the blocs (including both beneficiaries and non-beneficiaries). Core food security indicators (FCS, use of coping strategies) show no consistent pattern favouring either targeting method. Similarly, changes in broader

welfare indicators (life satisfaction and women's agency) are limited and not stable over time. However, the GFA itself generated statistically significant improvements in food security among beneficiary households when compared with non-beneficiaries. Improved FCS persisted after programme completion, indicating that the lack of welfare differences between people allocated to the two targeting methods is unlikely due to a lack of overall programme effectiveness.

107. This analysis compares the levels of welfare in communities targeted through the two methods using data from the follow-up surveys. It focuses on the average difference across households living in blocs assigned to the two targeting methods and includes both beneficiaries and non-beneficiaries.⁸³ It first examines core food security outcomes commonly used by WFP and then considers additional indicators related to broader welfare, including subjective life satisfaction and an index of women's agency. As discussed above, any observed difference reflects two features of the targeting approaches: differences in selection criteria and differences in coverage rates.⁸⁴

Food security

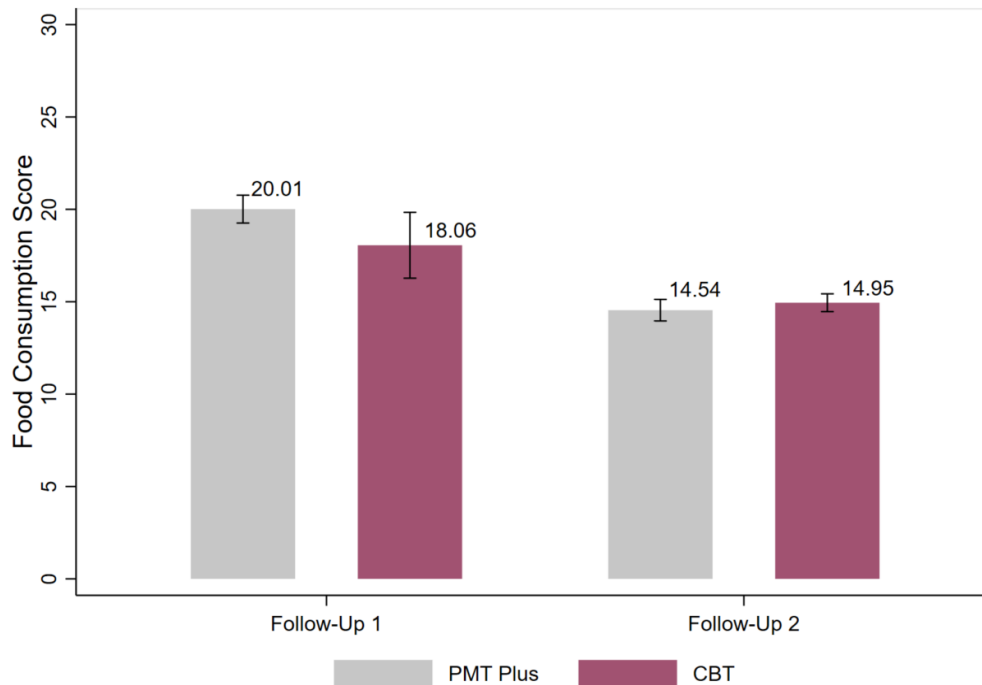
108. Figure 10 reports the estimated difference in FCS – the primary outcome of interest – between the PMT Plus and CBT study arms across the two follow-up rounds. The figure shows no statistically significant differences between the two approaches at either survey round, and the estimated differences are small in magnitude. Average FCS is slightly higher in PMT Plus blocs at the first follow-up, while it is slightly higher in CBT blocs at the second follow-up. Average FCS decreased substantially between the two data collection rounds for both groups, which is not surprising given that the second follow-up took place eight months after programme completion, when households were no longer receiving assistance.
109. The next food security dimension examined is the use of coping strategies, captured by two standard indicators: the reduced Coping Strategies Index ([rCSI](#)), which measures short-term food consumption-related behaviours, and the Livelihood Coping Strategies Index ([LCSI](#)), which reflects pressures on household livelihoods. Figure 11 shows no statistically significant differences in average rCSI between CBT and PMT Plus blocs at either follow-up, and the estimated differences are small in magnitude. As with FCS, the direction of the difference reverses across survey rounds, reinforcing the conclusion that, on average, neither method delivers better food security outcomes than the other. In both groups, rCSI scores increase by the second follow-up, indicating a deterioration in food security

⁸³ A complication for this analysis is that the study sample is not representative of the overall bloc population, as it was stratified by eligibility status, to ensure the inclusion of both targeted and non-targeted households. This limitation can be addressed by applying sampling weights, allowing results to be interpreted at the population level. Weights were obtained by multiplying design weights with non-response weights. The design weights reflect the inverse of the probability of selection, calculated from the ratio of households selected to the total number of households in each bloc and eligibility category. The non-response weights reflect the probability of response, based on the ratio of responding households to the total number selected in each bloc. Random non-response was assumed, which is unlikely to bias the estimates given the very low attrition rate (1 percent) at endline.

⁸⁴ Appendix Tables A5.3 to A5.5 provide robustness checks for welfare outcomes that address the threats to study validity.

conditions, likely reflecting the end of assistance. The absence of significant differences in the aggregate index may, however, mask changes in its individual components.

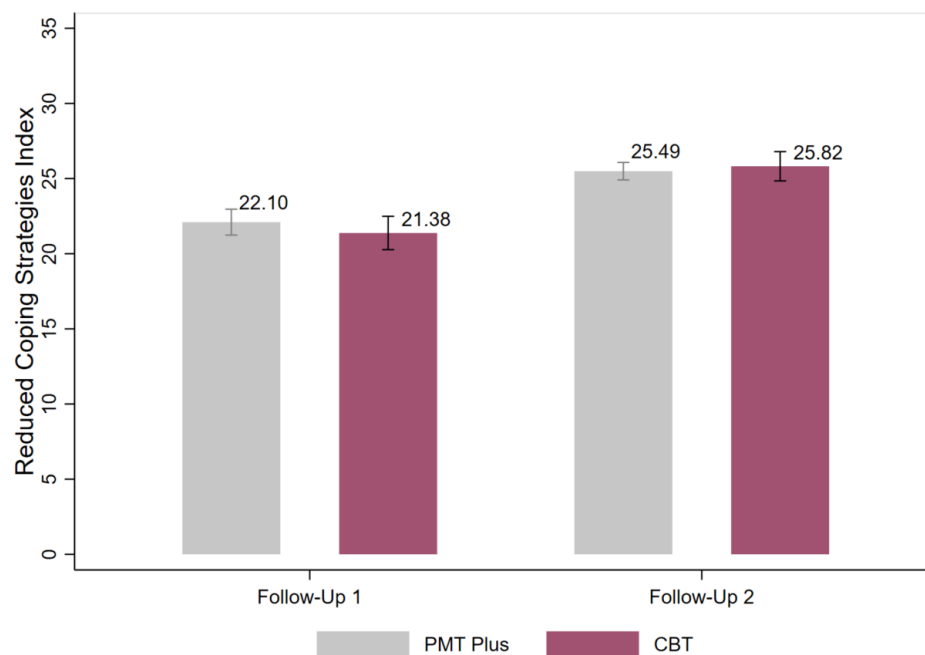
Figure 10: Impact of CBT on FCS



Note: The figure presents the adjusted mean of the FCS, which ranges from 0 to 112. Adjusted means refer to estimated outcome values obtained from regressions that control for additional variables. Regressions include control variables selected using post-double selection Lasso and quadruplet fixed effects. Analytical weights are applied to account for sampling design and non-response. Standard errors are clustered at the bloc (treatment) level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

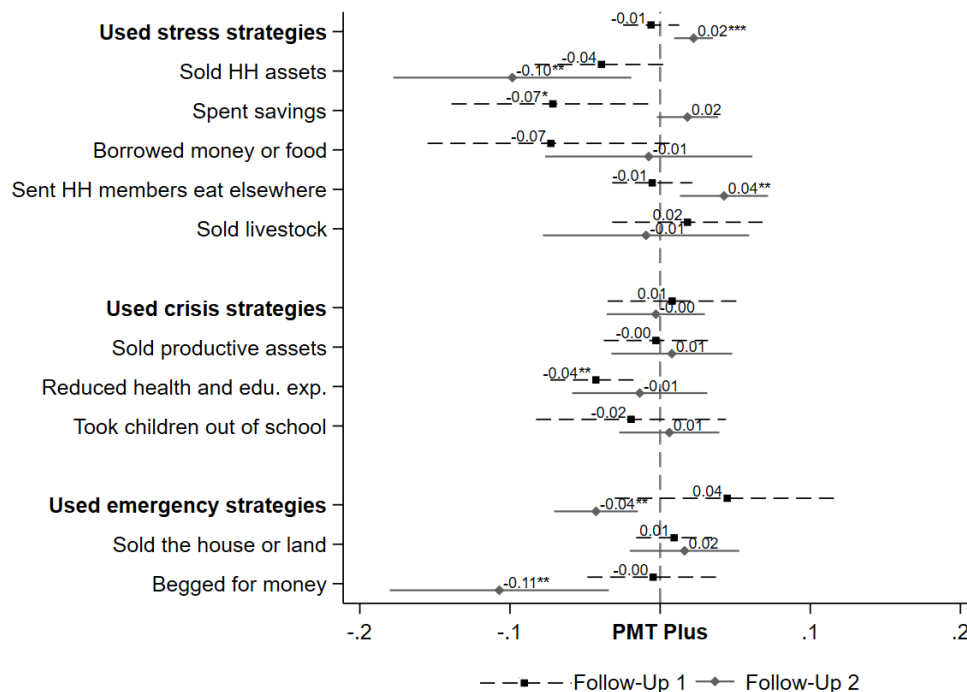
110. Appendix Figure A7.1 reports the impacts on the individual rCSI components at both follow-ups. At the first follow-up, households in CBT blocs were significantly less likely to report reducing meal size (0.42 fewer days, $p < 0.05$) or reducing the number of meals (0.41 fewer days, $p < 0.05$). By the second follow-up, households in CBT blocs were more likely to report borrowing food (0.18 more days, $p < 0.10$). No other components display significant differences.
111. Similar to the results for FCS and rCSI, no consistent pattern emerges that would favour one targeting method over the other when considering LCSi. The findings do not indicate any systematic difference between CBT and PMT Plus in the use of coping strategies. Although a few statistically significant differences appear at specific points in time, they vary in direction across follow-up rounds and across types of coping strategies (see Figure 12).
112. Appendix Table A7.1 provides the estimations of the impact of CBT on the primary outcomes. It shows results for both the main specification and for regressions controlling for the coverage rate. The results are virtually identical.

Figure 11: Impact of CBT on the reduced Coping Strategies Index (rCSI)



Note: The figure presents adjusted mean values of the reduced Coping Strategies Index (rCSI), which ranges from 0 to 56. Higher scores indicate greater reliance on negative coping strategies. Regressions include control variables selected using post-double selection Lasso and quadruplet fixed effects, with analytical weights applied to account for sampling and non-response. Standard errors are clustered at the bloc (treatment) level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 12: Impact of CBT on Livelihood Coping Strategies



Note: The figure shows the estimated impact of being assigned to CBT (ITT) blocs for both beneficiaries and non-beneficiaries on the likelihood of adopting livelihood-related coping strategies. Each point reflects the change in probability for households in CBT blocs, with values to the right indicating increased use and values to the left reduced use relative to those in PMT Plus blocs. Regressions include control variables selected using post-double selection Lasso quadruplet fixed effects, with analytical weights applied to account for sampling and non-response. Standard errors are clustered at the bloc (treatment) level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Secondary welfare outcomes

113. Appendix Table A7.2 reports the results for the life satisfaction index⁸⁵, which aggregates three questions capturing how respondents perceive their overall position in life, how respected they feel in the community, and how widely their opinions are held.
114. Households in CBT blocs report, on average, slightly higher satisfaction regarding their perceived standing in the community during the first follow-up (0.28 points on the 0 to 10 ladder, $p < 0.10$), although this represents a small increase and does not persist into the second follow-up. Across the three components, there are no consistent or meaningful differences in overall life satisfaction attributable to being in a CBT bloc as opposed to a PMT Plus bloc.
115. Additionally, the analysis considers women's agency⁸⁶ as participation in the local committees set up in CBT blocs may have impacted women's voice and influence in their communities.⁸⁷ Appendix Table A7.3 summarises the results.
116. At the first follow-up, the two targeting methods perform similarly, with no substantive differences in the index or its components. In the second follow-up, however, CBT households exhibit a small, but significant increase in the index (0.08 standard deviations, $p < 0.10$), driven by improved autonomy in social interactions, such as the ability to visit or meet others. Nonetheless, these changes remain limited in scope and do not suggest systematic differences.
117. Appendix Table A7.4 provides robustness checks for the welfare outcomes and addresses issues related to bloc attrition, imperfect programme adherence, and perfect coverage. The results remain largely unchanged.

Heterogeneity analysis

118. Potential heterogeneities are also explored to assess whether targeting and its impacts on household welfare vary with bloc size (Table A7.5), coverage levels (Table A7.6), and the prevalence of food insecurity (Table A7.7). The heterogeneity analysis subsection in Appendix A7 discusses the results in detail.
119. Overall, there is no consistent evidence of heterogeneity across these dimensions. While some differences arise for specific outcomes – such as women's agency in smaller blocs and increased reliance on coping strategies in high coverage or more food-insecure settings – these patterns are not robust across outcomes or specifications. Notably, higher reliance on

⁸⁵ The life satisfaction and women's agency indices were constructed by combining their component variables using weights based on the inverse of the covariance matrix following the method proposed in Anderson, M. L. (2008). Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects. *Journal of the American Statistical Association*, 103(484), 1481–1495.
<https://doi.org/10.1198/016214508000000841>

⁸⁶ Measured following Christian, P., Dinarte, L., Dunsch, F., Heirman, J., Jeong, D., Kelley, E., Kondylis, F., Lane, G., Loeser, J. A., Uckat, H., Balantrapu, T., Fiorina, M.-A., McCollum, K., & Spindler, E. (n.d.). *Impact Evaluation of Cash-Based Transfers on Food Security and Gender Equality in El Salvador*. World Food Programme. Retrieved <https://www.calpnetwork.org/wp-content/uploads/2023/12/Impact-Evaluation-of-Cash-Based-Transfers-on-Food-Security-and-Gender-Equality-in-El-Salvador-December-2023.pdf>

⁸⁷ Markham, S. (2013). *Women as Agents of Change: Having Voice in Society and Influencing Policy*. World Bank, Washington, DC. <https://doi.org/10.1596/21031>

coping strategies is in some cases accompanied by higher life satisfaction, suggesting that these indicators may capture different dimensions of welfare.

Impacts on beneficiaries and non-beneficiaries

120. A similar comparison is conducted between beneficiary and non-beneficiary households across the two targeting methods. This analysis should be interpreted with caution, as selection into assistance is non-random and instead driven by households' level of vulnerability. Nevertheless, the results may offer suggestive evidence on the relationship between the targeting method and outcomes between the two groups.
121. Among beneficiaries (Appendix Table A7.8), there are no significant gaps in FCS between CBT and PMT Plus at either follow-up. Small differences appear on coping that are scattered and not consistent across follow-ups. For non-beneficiaries (Appendix Table A7.9), the only consistent changes are higher food-related coping at the second follow-up and higher women's agency at the first follow-up, both of which hold even after controlling for the coverage rate.

Impacts of cash assistance on food security

122. It is striking that no statistically significant differences in welfare outcomes emerge between the entire groups subject to the two targeting methods, despite coverage in CBT blocs being substantially lower – by about 11 percentage points on average. This finding raises an important question: does the assistance programme itself generate meaningful improvements in household welfare? If the programme had little impact, the lack of differences across targeting methods would not be surprising.
123. To investigate the impacts of the cash assistance itself, the study includes a panel analysis of FCS from the three rounds of survey data collection, which provides information before, during, and after GFA implementation. Figure A7.2 shows the results of this analysis, which point towards the programme's effectiveness in improving in food security among beneficiary households. Positive results persist after programme completion, indicating that the programme was effective at the household level. Consequently, the lack of significant welfare differences between the two targeting methods at the community level is unlikely to be due to a lack of programme impact at the household level.

5.5. Community acceptance and cohesion: Which targeting method leads to higher community satisfaction, trust, and social cohesion?

Summary of findings: Although CBT blocs had significantly lower coverage than PMT Plus blocs, overall satisfaction with the targeting process was similar between the two methods. Administrative complaints were rare and evenly distributed between arms, and most households in both groups viewed the targeting process as transparent. Understanding of the targeting process was slightly higher in CBT blocs, and households were marginally less likely to perceive corruption in the process but were also less likely to consider the process as being timely. In terms of broader community dynamics, the two targeting approaches produced

similar average levels of trust and social cohesion. However, households in CBT blocs reported somewhat lower participation in community groups and collective action activities, as well as slightly higher perceptions of insecurity within their communities.

Overall, the results suggest that, despite the different coverage, the two targeting methods can achieve comparable levels of community satisfaction and perceived transparency, even though some differences emerged in community participation and perceived safety. This underscores the importance of process design and effective communication in shaping community trust and acceptance.

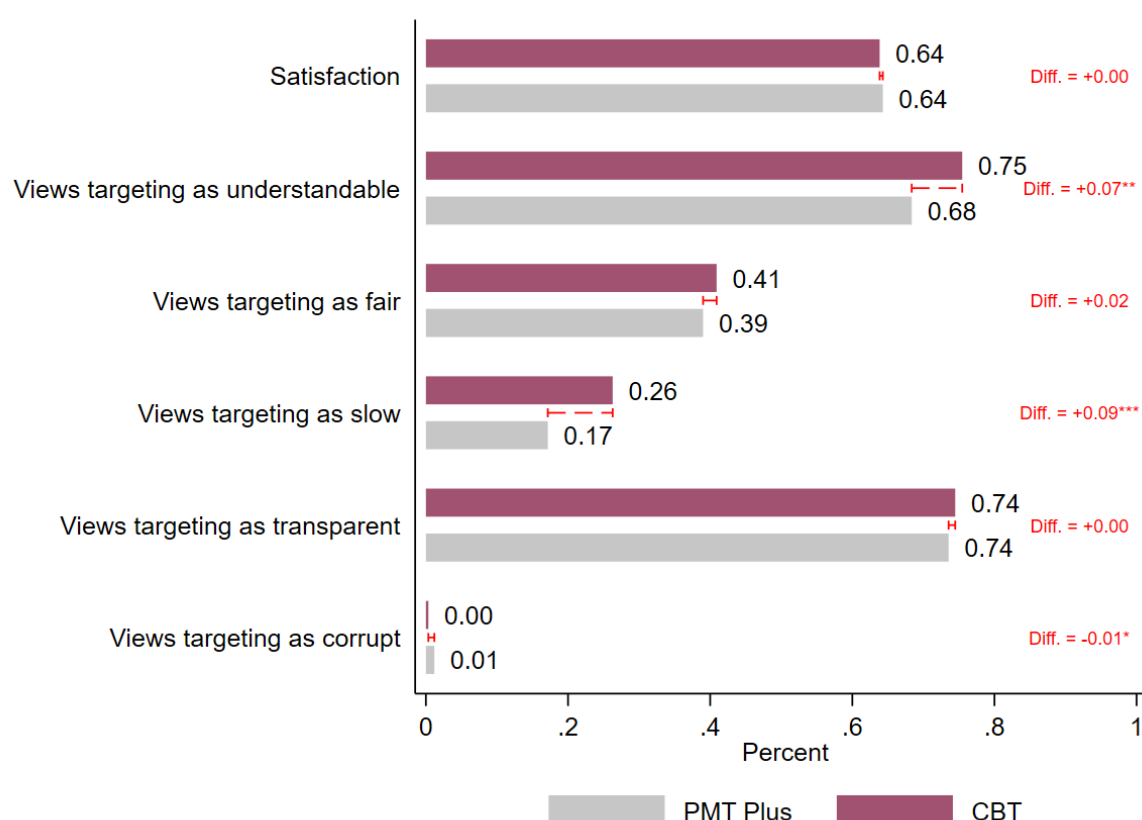
Satisfaction

124. Community satisfaction with the targeting process was assessed through both administrative records and survey questions. Administrative data was collected via the standard WFP feedback system that was equally accessible to all communities selected for assistance. Community sensitisation on communication channels was carried out through announcements in local languages, engagement with community leaders, radio broadcasts, and dissemination of information via village committees. The channels included: (i) a dedicated telephone helpline, (ii) secure submission boxes, and (iii) community-based committees.
125. A total of 651 cases were recorded from the study blocs. Most cases concerned simple administrative/logistical issues, e.g., the absence of the principal household member who is authorized to collect assistance, issues with tokens or account identifiers, and general assistance requests. Only 148 cases were issues about targeting errors, or the exclusion/inclusion of households, and these were split evenly between PMT Plus (75) and CBT (73).
126. Around 26 percent of blocs in the PMT Plus arm and 24 percent in the CBT arm registered at least one complaint; this difference was not statistically significant. The average number of complaints per bloc was also similar between the two targeting arms, at approximately 1.9 for PMT Plus and 1.8 for CBT (see Appendix Figure A8.1).
127. The endline survey also collected information on households' awareness and satisfaction with the targeting process, which are presented in Figure 13. Understanding of the targeting process was high for both arms, with over 70 percent of respondents correctly describing the targeting process to enumerators. However, understanding among households in CBT blocs was 7 percentage points higher compared to PMT Plus blocs ($p < 0.05$).
128. It is also notable that satisfaction with the targeting process was similar across the two arms, despite the significantly lower coverage observed under CBT. When disaggregated by beneficiary status, 69 percent of beneficiaries reported being satisfied with the process, compared with 45 percent of non-beneficiaries. This suggests that satisfaction is not solely driven by whether households were selected to receive assistance.⁸⁸

⁸⁸ Appendix Tables A5.6 to A5.8 provide robustness checks for community acceptance and cohesion outcomes that address the threats to study validity.

129. The high levels of self-reported satisfaction are consistent with evidence from similar field studies. One concern is that subjective satisfaction measures may be affected by response bias, as respondents often feel inclined to provide favourable assessments rather than express critical views about fairness or quality.⁸⁹ To capture a more nuanced picture, at the second follow-up, respondents were also asked about specific attributes of the targeting process, such as perceived transparency, corruption, and speed.

Figure 13: Impact of CBT on self-reported satisfaction



Note: The figure presents the percentage of households reporting satisfaction with different aspects of the targeting process, disaggregated by targeting arm. The reported estimates represent adjusted means. Lines in red indicate the difference

between the two methods, while the coefficient differences are shown on the right. Regressions include control variables selected using post-double selection Lasso and quadruplet fixed effects, with analytical weights applied to account for sampling and non-response. Standard errors are clustered at the bloc (treatment) level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

130. Perceived transparency was generally high: 75 percent of households across both approaches considered the targeting process transparent, with no significant difference

⁸⁹ The most notable case is that of patient satisfaction, where such biases have been widely documented in survey-based assessments of service delivery. In patient satisfaction surveys, respondents often provide favourable evaluations that may not reflect genuine satisfaction but rather limited expectations or a lack of alternatives, particularly in low-income settings where individuals may value any available service. Respondents may also agree with interviewers out of politeness or a desire to appear cooperative (social desirability or acquiescence bias), or they may choose the first response option as a cognitively easier shortcut (satisficing) as shown in Dunsch, F., Evans, D. K., Macis, M., & Wang, Q. (2018). Bias in patient satisfaction surveys: A threat to measuring healthcare quality. *BMJ Global Health*, 3(2), e000694. <https://doi.org/10.1136/bmjgh-2017-000694> and Krosnick, J. A. (2000). The threat of satisficing in surveys: The shortcuts respondents take in answering questions. *Survey Methods Newsletter*, 20, 4–8.

between methods. In contrast, a small but statistically significant difference ($p < 0.05$) emerged in perceptions of corruption: households in CBT blocs were 1 percentage point less likely to view the process as corrupt compared to those in PMT Plus blocs. Perception of speed showed greater variation, with 26 percent of households in CBT blocs reporting that the process was slow, compared to 17 percent among PMT Plus households.⁹⁰

131. Appendix Table A8.1, with the full regression results, shows that the pattern in satisfaction is similar regardless of whether coverage controls are applied, while Table A8.2 provides the results of robustness checks, which show that the results are consistent even when accounting for issues related to programme implementation.
132. No significant differences were found between methods in other dimensions, including knowledge of programme objectives, understanding of the targeting process and selection criteria, or perceptions of targeting errors. Appendix Table A8.3 summarises household responses by study arm for these and other indicators of understanding and satisfaction with the targeting process.

Trust and social cohesion

133. Any targeting approach, by defining beneficiaries and non-beneficiaries, has the potential to trigger tensions or social divisions within communities.^{91,92,93} Figure 14 compares overall trust, social cohesion, and collective action measures between PMT Plus and CBT blocs.⁹⁴ No significant differences are observed in terms of average trust and social cohesion.
134. However, in CBT blocs there is a 0.19 standard deviation drop in household participation in community groups and collective action activities (such as volunteering for the community or participating in local organizations) compared to PMT Plus blocs ($p < 0.05$).
135. In CBT areas, households also report feeling less safe at home and in their bloc. Some of these effects are quite large: the share of households feeling unsafe at home or within the bloc is 61 percent for PMT Plus blocs and 69 percent for CBT, representing a 8-percentage-point difference ($p < 0.05$).
136. While it is hard to pin down the reasons behind these effects, these may be attributable to the lower coverage in CBT blocs, as the feeling of being unsafe at home or in the bloc is 4 percentage points ($p < 0.05$) higher among non-beneficiaries, at 67 percent, compared with

⁹⁰ This difference should be interpreted cautiously, as the study design standardized the time of the targeting process across methods. PMT Plus could be implemented quickly using census data and a few FGDs, while CBT required committee-based data collection in addition to the census data collection. As highlighted in previous sections, in practice, CBT could require a much lighter data collection, only in relation to the dimensions identified by the committees (potentially available through pre-existing registries).

⁹¹ Adato, M. (2000). The Impact of PROGRESA on Community Social Relationships: Final Report. <https://doi.org/10.22004/AG.ECON.16015>

⁹² Della Guardia, A., Lake, M., & Schnitzer, P. (2022). Selective inclusion in cash transfer programs: Unintended consequences for social cohesion. *World Development*, 157, 105922. <https://doi.org/10.1016/j.worlddev.2022.105922>

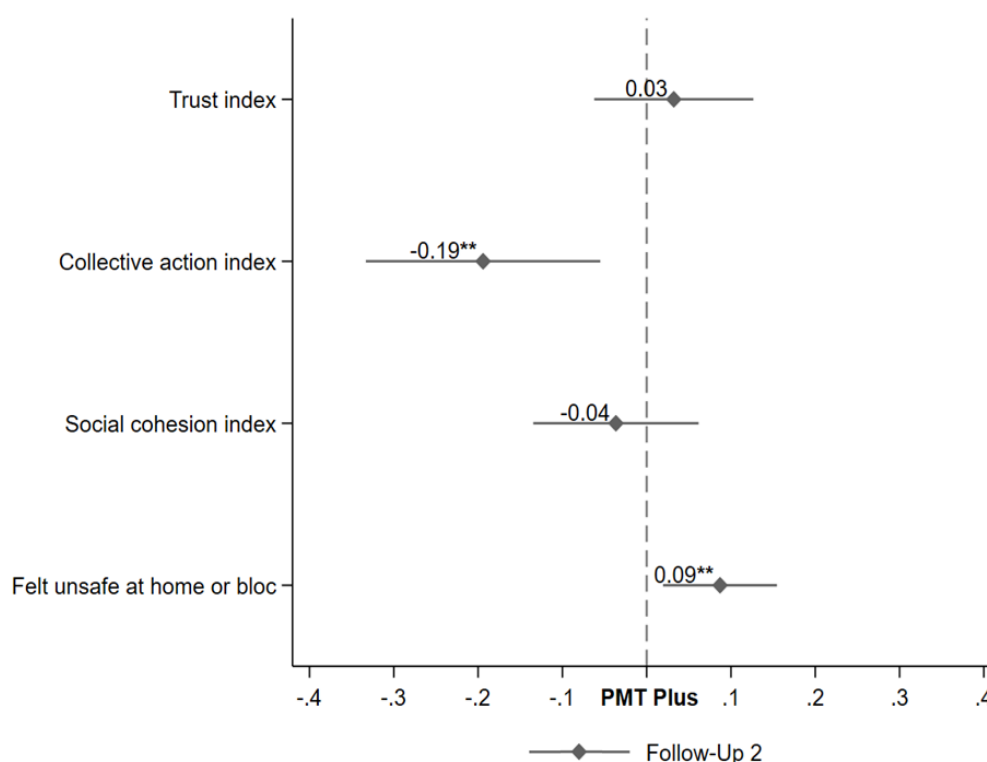
⁹³ Pavanello, S., Watson, C., Onyango-Ouma, W., & Bukuluki, P. (2016). Effects of Cash Transfers on Community Interactions: Emerging Evidence. *The Journal of Development Studies*, 52(8), 1147–1161. <https://doi.org/10.1080/00220388.2015.1134774>

⁹⁴ Details on the construction of the indices can be found in Appendix A3. The Social Cohesion, Collective Action, and Trust indices are adapted from Bossuoy, T., Goldstein, M., Karimou, B., Karlan, D., Kazianga, H., Parienté, W., Premand, P., Thomas, C. C., Udry, C., Vaillant, J., & Wright, K. A. (2022). Tackling psychosocial and capital constraints to alleviate poverty. *Nature*, 605(7909), 291–297. <https://doi.org/10.1038/s41586-022-04647-8>

63 percent among beneficiaries. Moreover, once coverage is controlled for, the estimated effect is no longer statistically significant.

137. For the full regression results on trust and social cohesion outcomes, see Appendix Table A8.4, while Table A8.5 provides the results from robustness checks. These additional validation analyses show that results are generally consistent between the different specifications.

Figure 14: Impact of CBT on social cohesion, trust, and tension



Note: The figure presents the impact of CBT on indices of trust, social cohesion, and collective action, as well as on reported perceptions of safety, relative to PMT Plus. The trust, social cohesion, and collective action measures are standardized indices, with coefficients expressed in standard deviation units, while the perceptions of safety measures are binary variables, with coefficients expressed in percentage points. Index standard deviation is computed as the standardized (z-score) mean of its components. Coefficients capture the difference between CBT and PMT Plus. Horizontal lines indicate 95 percent confidence intervals around the point estimates. Regressions include control variables selected using post-double selection Lasso and always include quadruplet fixed effects, with analytical weights applied to account for sampling and non-response. Standard errors are clustered at the bloc (treatment) level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.6. Cost-effectiveness: Which targeting method is more cost-effective?

Summary of findings: The cost analysis shows that total programme costs were lower under CBT (USD 1.07 million) than under PMT Plus (USD 1.19 million). This difference does not stem from lower targeting setup costs, which were nearly identical, but from the lower coverage realized under CBT and the resulting reduction in transfer disbursements. Given that both approaches produced comparable targeting precision and similar welfare outcomes, the lower total cost under CBT implies that similar programme objectives could be reached at reduced

expenditure, allowing programmes to extend assistance to a larger number of communities within a fixed budget.

138. This section focuses on cost-effectiveness, assessing which targeting method can deliver comparable or superior outcomes at the lowest cost, thereby enabling programmes to maximise horizontal coverage in contexts of widespread need.
139. Cost data were collected using a template developed by the World Bank's Development Impact Department. A member of the evaluation team worked with the CO – including teams from beneficiary registration, vulnerability assessment and mapping, and other relevant units – to compile information from administrative records. Files such as receipts, ledgers, and driving logs were used to verify day rates, mileage, and other operational costs.

Table 2: Actual targeting and programme costs (84 blocs, in USD)

	PMT Plus (42 blocs)	CBT (42 blocs)
Households		
Households screened/listed	5,809	5,923
Households selected	4,930	4,396
Costs		
Targeting costs	61,881	62,570
(1) Preparation	1,775	0
(2) Consultation	860	2,420
(3) Data collection	58,090	58,090
(4) Production of lists	1,156	2,060
Total transfers	1,124,517	1,002,737
Total programme costs	1,186,398	1,065,307
Targeting Cost per HH	10.65	10.56
Total Cost per HH Screened	<u>204.23</u>	<u>179.86</u>
Total Cost per HH Selected	<u>240.65</u>	<u>242.34</u>

Note: (1) Preparation: amount for analyst to develop model. (2) Consultation: includes cost of facilitators, transport costs and tablet rentals. (3) Data collection: field coordinators, enumerators, research assistants, training venue, transport for training, training material, venue costs, tablets, costs related to the survey firm, and overall administrative costs. (4) Production of lists. (5) Transfers: approx. 228 USD per selected household. Targeting costs are nominal USD as recorded. These costs refer to the 84 blocs where data collection was conducted.

140. The previous subsections showed that the approaches achieve similar precision in identifying food-insecure households and generate comparable welfare outcomes at the bloc level. Table 2 reports the costs associated with targeting and with the transfer itself, illustrating the resources required to implement the programme across an equal number of blocs (42 PMT Plus and 42 CBT).

141. In terms of targeting costs, the resources required to set up and implement the process were almost identical: USD 61,881 for PMT Plus and USD 62,570 for CBT. As a result, screening costs were also very similar, amounting to USD 10.65 per screened household in PMT Plus blocs and USD 10.56 in CBT blocs.
142. Data preparation costs are set to zero for CBT, as it does not require constructing a targeting model, unlike PMT Plus which involves analyst time. Data collection costs are assumed to be equal between arms, since enumerators were paid daily rather than per survey. As a result, costs were largely fixed at the bloc level, with similar expenditures per community regardless of the number of households surveyed.
143. When the value of transfers provided to households is accounted for, however, total programme costs amounted to USD 1,186,398 under PMT Plus, compared with USD 1,065,307 under CBT. Transfers are estimated from WFP's payment records as the average amount disbursed per selected household, USD 228 in both arms, multiplied by the number of households each method selected. The difference therefore reflects the 534 additional households selected by PMT Plus.
144. Consequently, CBT cost USD 121,091 less than PMT Plus to screen, target, and provide assistance to households across 42 blocs.

Cost simulations

145. The difference in total costs between the two methods is due to the sizeable gap in their coverage rates. Table 3 compares the total cost of targeting and delivering assistance in one bloc consisting of 140 households (average bloc size). The results show that CBT costs 12.7 percent less than PMT Plus.

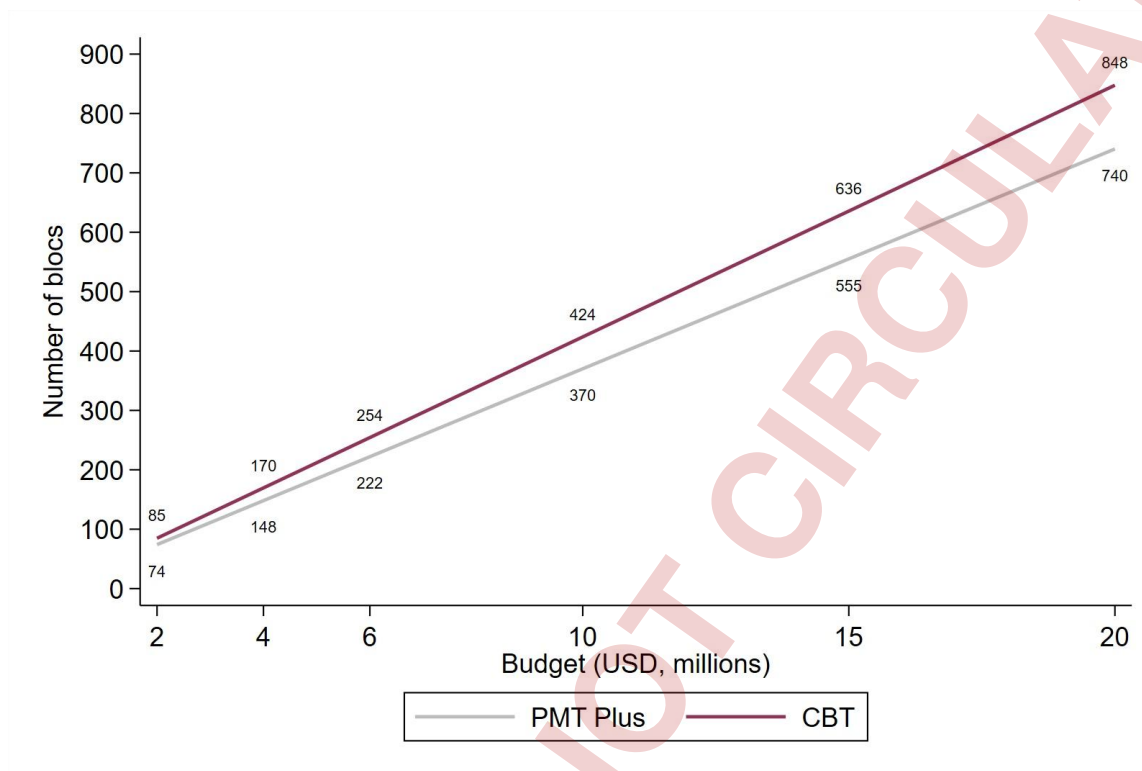
Table 3: Projected targeting and programme costs (1 bloc, in USD)

	PMT Plus	CBT
Targeting costs	1,491	1,478
No. of households screened	140	140
×		
Targeting Cost per HH	10.65	10.56
Transfer costs	25,536	22,116
No. of households screened	140	140
×		
Coverage rate	80%	69%
×		
Average transfer per HH	228	228
Total costs (targeting + transfer costs)	27,027	23,594

146. The cost implications of this difference can be further illustrated through a simple simulation. Figure 15 shows the number of blocs that could be assisted under different

budget scenarios assuming the same average bloc size and average transfer value as Table 3. Assuming similar expansion costs, the simulation indicates that CBT would allow a larger number of blocs to be covered, thereby extending geographic reach.

Figure 15: Projection of blocs assisted based on different budget scenarios



Note: The figure shows the number of blocs that can be assisted under each targeting approach as total programme budget increases. Estimates are based on the average cost of targeting and assistance for a bloc of about 140 households, equal to 27,027 USD under PMT Plus and 23,594 USD under CBT. Coverage is set at each method's observed rate: 79.9 percent under PMT Plus and 69.3 percent under CBT. The average transfer per assisted household is 228 USD. The analysis assumes a constant marginal cost of expanding coverage.

147. It is worth noting that the estimated cost of CBT may be conservative, as it includes the expense of conducting a full census. These costs could be reduced if an existing population registry or previously collected census data were used, or if registration exercises were streamlined to collect the additional characteristics most frequently cited by community committees. Under such a scenario, the census component of the targeting cost – approximately USD 1,383 per bloc – could be redirected towards expanding coverage, allowing additional blocs to be assisted within the same overall budget.
148. The findings could indicate potential "optimal saturation points" for aid coverage. While most targeting systems assess vulnerability at the household level, they often overlook community dynamics such as the internal sharing of food and resources. These results (similar food security with varying coverage and costs) suggest there may be a threshold of assistance beyond which it is more effective to extend aid to additional communities rather than attempting to reach every vulnerable individual within a single location. Future research should investigate these social interactions to determine how "saturation" influences the overall impact and efficiency of aid.

6. Conclusions and Considerations for Future Programming

149. By comparing a predominantly data-driven model (PMT Plus) and a community-based method (CBT) within WFP's GFA programme in DRC, the impact evaluation provides one of the first causal studies of household-level targeting under real-world humanitarian conditions marked by urgency and insecurity.
150. The evaluation goes beyond measuring technical accuracy and examines differences in coverage, welfare outcomes, community perceptions, and cost-efficiency – dimensions that are central to the legitimacy and effectiveness of humanitarian operations.

6.1. Conclusions

151. Overall, this evaluation demonstrates that PMT Plus and CBT perform similarly in relation to precision and food security. They, however, differ substantially in coverage and, thus, costs. Under tight budget constraints, community-based approaches can offer a cost-effective, transparent, and socially acceptable alternative to more common data-driven approaches, provided they are supported by strong facilitation, communication, and grievance mechanisms.

EQ1: How do targeting methods differ in terms of beneficiary coverage?

152. **If coverage is left free to vary based on assessed needs, community-based targeting would lead to a lower number of beneficiaries for a given population.** Coverage of the overall study sample was high (75 percent) but significantly lower under CBT (69 percent) than under PMT Plus (80 percent). The 11-percentage-point difference is large and statistically significant ($p < 0.01$). Additionally, coverage rates under CBT exhibit greater variability across blocs compared with PMT Plus. The results show that involving communities in setting vulnerability criteria more actively actually leads to the identification of fewer beneficiaries, on average, than the more standard data-driven approach.

EQ2: Which targeting method is more precise in selecting households with the lowest food security?

153. **Although the higher PMT Plus coverage leads to lower exclusion errors, when coverage is fixed, the two targeting approaches achieve similar precision in identifying vulnerable households.** PMT Plus displayed slightly lower exclusion errors, but this difference was entirely explained by its higher coverage. Targeting precision and inclusion error rates were instead statistically indistinguishable across methods (11 percent for PMT Plus and 13 percent for CBT).

154. **Error rates of both approaches evaluated are lower than those documented in recent experimental studies (see Appendix Table A6.2).** As the evaluation setting displays much higher levels of food insecurity (and coverage) than most other studies, this shows that in contexts of near-universal vulnerability, the high prevalence of food insecurity naturally sets a higher floor for targeting precision, as the “pool” of eligible households is so large that the probability of selecting a non-vulnerable household is inherently minimized.

EQ3: How do targeting methods differ in the types of households they select, and do these differences lead to systematic exclusion or over-targeting of specific population subgroups?

155. **Differences in which types of households are prioritized indicate that CBT might be better suited to capture livelihood-related forms of vulnerability, whereas PMT Plus focuses on more structural or chronic characteristics.** PMT Plus tended to select households defined by demographic vulnerability and displacement status (e.g., female-headed, illiterate head), reflecting its uniform selection rule across blocs. CBT committees, by contrast, more often identified agricultural households and households recently affected by shocks, such as crop losses or poor harvests.

EQ4: Do the differences in targeting outcomes translate into differences in welfare for the target populations?

156. **The two targeting approaches achieve broadly comparable welfare outcomes despite significantly lower coverage under CBT.** Despite reaching fewer households, CBT communities achieved welfare outcomes that are statistically indistinguishable to PMT Plus communities. Reliance on negative coping strategies and life satisfaction also showed no consistent differences.
157. **Community context is an important consideration – i.e., bloc characteristics – for relative effectiveness.** Although no major differences emerged across bloc size or coverage, the results show nuanced patterns. In smaller blocs, CBT improved women's agency relative to PMT Plus, likely reflecting the visibility and social interaction inherent in smaller communities. In high-coverage blocs, CBT was associated with higher life satisfaction but slightly worse coping-strategy scores. These differences underscore the importance of contextual factors – population size, coverage intensity, and social cohesion – in shaping programme outcomes.

EQ5: Which targeting method leads to higher community satisfaction, trust, and social cohesion?

158. **The targeting methods achieved comparable levels of perceived transparency, fairness, and satisfaction, demonstrating that effective communication and process design can sustain trust regardless of technical modality and coverage.** Although CBT blocs had lower coverage, average satisfaction with the targeting process did not differ from PMT Plus. Around three-quarters of households across both arms perceived the process as transparent, and complaint rates were low and similar across arms. Perceived corruption was marginally lower in CBT blocs, while perceptions of speed were slightly less favourable to CBT.

159. **Social cohesion remained stable, while safety concerns increased with CBT.** Levels of trust and social cohesion were similar across study arms, though perceived insecurity at home and within the bloc increased in CBT areas. This may be an indication that lower coverage heightens perceptions of insecurity. These findings highlight the importance of monitoring community dynamics, especially at lower levels of coverage.

EQ6. Cost-effectiveness: Which targeting method is more cost-effective?

160. **Lower total cost under CBT – due to its lower coverage rate – implies that similar programme objectives could be achieved at reduced expenditure, allowing programmes to extend assistance to a larger number of communities within a fixed budget.** Total programme costs were USD 1.07 million under CBT and USD 1.19 million under PMT Plus, a difference of about USD 121,091. This difference does not stem from lower targeting setup costs, which were nearly identical, but from the lower coverage achieved under CBT and the resulting reduction in transfer disbursements.

6.2. Considerations for future programming

161. This evaluation shows that **simpler, community-driven approaches can achieve comparable precision and welfare outcomes as more advanced data-driven approaches**, and at lower cost, potentially freeing resources for broader coverage or faster delivery. Humanitarian actors operating under acute funding constraints should therefore consider expanding reliance on such an approach.
162. The evaluation also highlights **potential "optimal saturation points" for aid coverage**. As most targeting methods assess vulnerability at the household level, they often overlook community dynamics such as the sharing of food and resources. These results suggest there may be a threshold of assistance beyond which it is more effective to extend aid to additional communities rather than attempting to reach every vulnerable individual within a single location. Future research should investigate these social interactions to determine how "saturation" influences the overall impact and efficiency of aid.
163. The results also show that the perceptions of fairness depend less on the targeting design than on how the process is explained and managed. However, the increase in safety concerns observed in CBT blocs suggests that lower coverage can trigger social friction and perceived insecurity. Therefore, while community participation is essential for legitimacy, it must be paired with robust "Do No Harm" protocols. This includes **proactive protection monitoring and conflict-mediation strategies to ensure that the targeting process does not inadvertently erode social cohesion** or compromise the safety of community members.
164. The success of CBT points to the importance of incorporating local information in targeting the most vulnerable. However, ultimately, the key targeting choice is not simply between data-driven and community-based methods. How targeting rules incorporate local information and, especially, how much coverage they generate may matter at least as much as the underlying selection method.

7. References

- Adato, M. (2000). *The Impact of PROGRESA on Community Social Relationships: Final Report*. <https://doi.org/10.22004/AG.ECON.16015>
- Aiken, E., Ashraf, A., Blumenstock, J., Guiteras, R., & Mobarak, A. M. (2025). *Scalable Targeting of Social Protection: When Do Algorithms Out-Perform Surveys and Community Knowledge?* (Working Paper No. 33919). National Bureau of Economic Research. <https://doi.org/10.3386/w33919>
- Alatas, V., Banerjee, A., Hanna, R., Olken, B. A., & Tobias, J. (2012). Targeting the Poor: Evidence from a Field Experiment in Indonesia. *American Economic Review*, 102(4), 1206–1240. <https://doi.org/10.1257/aer.102.4.1206>
- Altındağ, O., O’Connell, S. D., Şaşmaz, A., Balcioğlu, Z., Cadoni, P., Jerneck, M., & Foong, A. K. (2021). Targeting humanitarian aid using administrative data: Model design and validation. *Journal of Development Economics*, 148, 102564. <https://doi.org/10.1016/j.jdeveco.2020.102564>
- Anderson, M. L. (2008). Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects. *Journal of the American Statistical Association*, 103(484), 1481–1495. <https://doi.org/10.1198/016214508000000841>
- Angelucci, M., & De Giorgi, G. (2009). Indirect Effects of an Aid Program: How Do Cash Transfers Affect Ineligibles’ Consumption? *American Economic Review*, 99(1), 486–508. <https://doi.org/10.1257/aer.99.1.486>
- Angrist, J. D., & Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist’s Companion*. Princeton University Press. <https://doi.org/10.2307/j.ctvcm4j72>
- Baliki, G., Cruz, M. A. F., Dunsch, F., Garcia, J., Guariso, A., Heirman, J., & Weiffen, D. (2026). *Targeting humanitarian aid in crisis: Evidence from the reduction of WFP assistance in Lebanon*. World Food Programme. <https://www.wfp.org/publications/targeting-humanitarian-aid-crisis-evidence-reduction-wfp-assistance-lebanon>
- Banerjee, A., Hanna, R., Olken, B. A., & Lisker, D. S. (2024). Social Protection in the Developing World. *Journal of Economic Literature*, 62(4), 1349–1421. <https://doi.org/10.1257/jel.20241646>
- Banerjee, A., Niehaus, P., & Suri, T. (2019). Universal Basic Income in the Developing World. *Annual Review of Economics*, 11(Volume 11, 2019), 959–983. <https://doi.org/10.1146/annurev-economics-080218-030229>
- Basurto, M. P., Dupas, P., & Robinson, J. (2020). Decentralization and efficiency of subsidy targeting: Evidence from chiefs in rural Malawi. *Journal of Public Economics*, 185, 104047. <https://doi.org/10.1016/j.jpubeco.2019.07.006>

Beaman, L., BenYishay, A., Magruder, J., & Mobarak, A. M. (2021). Can Network Theory-Based Targeting Increase Technology Adoption? *American Economic Review*, 111(6), 1918–1943. <https://doi.org/10.1257/aer.20200295>

Bossuroy, T., Goldstein, M., Karimou, B., Karlan, D., Kazianga, H., Parienté, W., Premand, P., Thomas, C. C., Udry, C., Vaillant, J., & Wright, K. A. (2022). Tackling psychosocial and capital constraints to alleviate poverty. *Nature*, 605(7909), 291–297. <https://doi.org/10.1038/s41586-022-04647-8>

Brown, C., Ravallion, M., & van de Walle, D. (2018). A poor means test? Econometric targeting in Africa. *Journal of Development Economics*, 134, 109–124. <https://doi.org/10.1016/j.jdeveco.2018.05.004>

Burger, R., & Owens, T. (2010). Promoting Transparency in the NGO Sector: Examining the Availability and Reliability of Self-Reported Data. *World Development*, 38(9), 1263–1277. <https://doi.org/10.1016/j.worlddev.2009.12.018>

Cameron, L., & Shah, M. (2014). Can Mistargeting Destroy Social Capital and Stimulate Crime? Evidence from a Cash Transfer Program in Indonesia. *Economic Development and Cultural Change*, 62(2), 381–415. <https://doi.org/10.1086/674102>

Christian, P., Dinarte, L., Dunsch, F., Heirman, J., Jeong, D., Kelley, E., Kondylis, F., Lane, G., Loeser, J. A., Uckat, H., Balantrapu, T., Fiorina, M.-A., McCollum, K., & Spindler, E. (n.d.). *Impact Evaluation of Cash-Based Transfers on Food Security and Gender Equality in El Salvador*. World Food Programme. Retrieved <https://www.calpnetwork.org/wp-content/uploads/2023/12/Impact-Evaluation-of-Cash-Based-Transfers-on-Food-Security-and-Gender-Equality-in-El-Salvador-December-2023.pdf>

Coady, D., Grosh, M., & Hoddinott, J. (2004). *Targeting of Transfers in Developing Countries: Review of Lessons and Experience*. The World Bank. <https://doi.org/10.1596/0-8213-5769-7>

Del Ninno, C., & Mills, B. (Eds.). (2015). *Safety Nets in Africa: Effective Mechanisms to Reach the Poor and Most Vulnerable*. The World Bank. <https://doi.org/10.1596/978-1-4648-0435-9>

Delius, A., & Sterck, O. (2024). Cash transfers and micro-enterprise performance: Theory and quasi-experimental evidence from Kenya. *Journal of Development Economics*, 167, 103232. <https://doi.org/10.1016/j.jdeveco.2023.103232>

Della Guardia, A., Lake, M., & Schnitzer, P. (2022). Selective inclusion in cash transfer programs: Unintended consequences for social cohesion. *World Development*, 157, 105922. <https://doi.org/10.1016/j.worlddev.2022.105922>

Devereux, S., Masset, E., Sabates-Wheeler, R., Samson, M., Rivas, A.-M., & te Lintelo, D. (2017). The targeting effectiveness of social transfers. *Journal of Development Effectiveness*, 9(2), 162–211. <https://doi.org/10.1080/19439342.2017.1305981>

Dunsch, F., Evans, D. K., Macis, M., & Wang, Q. (2018). Bias in patient satisfaction surveys: A threat to measuring healthcare quality. *BMJ Global Health*, 3(2), e000694. <https://doi.org/10.1136/bmjgh-2017-000694>

Egger, D., Haushofer, J., Miguel, E., Niehaus, P., & Walker, M. (2022). General Equilibrium Effects of Cash Transfers: Experimental Evidence From Kenya. *Econometrica*, 90(6), 2603–2643. <https://doi.org/10.3982/ECTA17945>

Everitt, B., & Skrondal, A. (2010). *The Cambridge dictionary of statistics* (4th edition). Cambridge University Press.

Filmer, D., Friedman, J., Kandpal, E., & Onishi, J. (2023). Cash Transfers, Food Prices, and Nutrition Impacts on Ineligible Children. *The Review of Economics and Statistics*, 105(2), 327–343. https://doi.org/10.1162/rest_a_01061

Filmer, D., & Pritchett, L. H. (2001). Estimating Wealth Effects without Expenditure Data-or Tears: An Application to Educational Enrollments in States of India. *Demography*, 38(1), 115–132. <https://doi.org/10.2307/3088292>

Follett, L., & Henderson, H. (2023). A hybrid approach to targeting social assistance. *Journal of Development Economics*, 160, 103002. <https://doi.org/10.1016/j.jdeveco.2022.103002>

Food Security Information Network (FSIN) & Global Network Against Food Crises (GNAFC). (2025). *Global Report on Food Crisis 2025* (p. 34.1 MB, 254 p.) [Pdf]. FSIN/GNAC,. <https://doi.org/10.71958/WFP130664>

Galasso, E., & Ravallion, M. (2005). Decentralized targeting of an antipoverty program. *Journal of Public Economics, Cornell - ISPE Conference on Public Finance and Development*, 89(4), 705–727. <https://doi.org/10.1016/j.jpubeco.2003.01.002>

Grosh, M., Leite, P., Wai-Poi, M., & Tesliuc, E. (Eds.). (2022). *Revisiting Targeting in Social Assistance: A New Look at Old Dilemmas*. The World Bank. <https://doi.org/10.1596/978-1-4648-1814-1>

Hanna, R., & Olken, B. A. (2018). Universal Basic Incomes versus Targeted Transfers: Anti-Poverty Programs in Developing Countries. *Journal of Economic Perspectives*, 32(4), 201–226. <https://doi.org/10.1257/jep.32.4.201>

Haushofer, J., Niehaus, P., Paramo, C., Miguel, E., & Walker, M. (2025). Targeting Impact versus Deprivation. *American Economic Review*, 115(6), 1936–1974. <https://doi.org/10.1257/aer.20221650>

Hillebrecht, M., Klonner, S., A Pacere, N., & Souares, A. (2020). Community-Based versus Statistical Targeting of Anti-Poverty Programs: Evidence from Burkina Faso. *Journal of African Economies*, 29(3), 271–305. <https://doi.org/10.1093/jafeco/ejz026>

Humanitarian Funding Forecast. (2022, August 9). *Underfunded Crisis Index*. <https://humanitarianfundingforecast.org/index-underfunded-crisis/>

Integrated Food Security Phase Classification. (2021). *Democratic Republic of the Congo: Food Security & Nutrition Snapshot, September 2021-August 2022*. <https://www.ipcinfo.org/ipcinfo-website/resources/resources-details/en/c/1155368/>

Integrated Food Security Phase Classification. (2025). *Democratic Republic of the Congo: Acute Food Insecurity Situation for September - December 2025 and Projection for January - June 2026*. <https://www.ipcinfo.org/ipc-country-analysis/details-map/en/c/1159775/>

Jayachandran, S., Biradavolu, M., & Cooper, J. (2023). Using machine learning and qualitative interviews to design a five-question survey module for women's agency. *World Development*, 161, 106076. <https://doi.org/10.1016/j.worlddev.2022.106076>

Jeong, D., & Trako, I. (2022). *Cash and In-Kind Transfers in Humanitarian Settings: A Review of Evidence and Knowledge Gaps*. The World Bank. <https://doi.org/10.1596/1813-9450-10026>

Karlan, D., & Thuysbaert, B. (2019). Targeting Ultra-Poor Households in Honduras and Peru. *The World Bank Economic Review*, 33(1), 63–94. <https://doi.org/10.1093/wber/lhw036>

Krosnick, J. A. (2000). The threat of satisficing in surveys: The shortcuts respondents take in answering questions. *Survey Methods Newsletter*, 20, 4–8.

Manikas, I., Ali, B. M., & Sundarakani, B. (2023). A systematic literature review of indicators measuring food security. *Agriculture & Food Security*, 12(1), 10. <https://doi.org/10.1186/s40066-023-00415-7>

Markham, S. (2013). *Women as Agents of Change: Having Voice in Society and Influencing Policy*. World Bank, Washington, DC. <https://doi.org/10.1596/21031>

Martinelli, C., & Parker, S. W. (2009). Deception and Misreporting in a Social Program. *Journal of the European Economic Association*, 7(4), 886–908. <https://www.jstor.org/stable/40282792>

McCord, A. (2013). *Community-based targeting in the Social Protection sector* (Working Paper No. 514). ODI Global. <https://media.odi.org/documents/11666.pdf>

Pathmind. (n.d.). *A Beginner's Guide to Eigenvectors, Eigenvalues, PCA, Covariance and Entropy*. A.I. Wiki. Retrieved <https://wiki.pathmind.com/eigenvector#principal>

Pavanello, S., Watson, C., Onyango-Ouma, W., & Bukuluki, P. (2016). Effects of Cash Transfers on Community Interactions: Emerging Evidence. *The Journal of Development Studies*, 52(8), 1147–1161. <https://doi.org/10.1080/00220388.2015.1134774>

Pham, P. N., Johnston, L. G., Keegan, K., Wei, C., & Vinck, P. (2023). Innovative Strategies for Remotely Sampling Hard-to-Reach Populations: Assessing Phone Versus Internet Respondent-Driven Sampling Approaches Among Venezuelan Refugees and Migrants in Colombia. *American Journal of Epidemiology*, 192(10), 1613–1623. <https://doi.org/10.1093/aje/kwad116>

Premand, P., & Schnitzer, P. (2021). Efficiency, Legitimacy, and Impacts of Targeting Methods: Evidence from an Experiment in Niger. *The World Bank Economic Review*, 35(4), 892–920. <https://doi.org/10.1093/wber/lhaa019>

Sabates-Wheeler, R., Hurrell, A., Devereux, S., & UNU-WIDER. (2014). *Targeting social transfer programmes: Comparing design and implementation errors across alternative mechanisms* (40th ed., Vol. 2014). UNU-WIDER. <https://doi.org/10.35188/UNU-WIDER/2014/761-5>

Schnitzer, P., & Stoeffler, Q. (2024). Targeting Social Safety Nets: Evidence from Nine Programs in the Sahel. *The Journal of Development Studies*, 60(4), 574–595.
<https://doi.org/10.1080/00220388.2023.2291325>

Sharif, I. A. (2012). Can Proxy Means Testing Improve the Targeting Performance of Social Safety Nets in Bangladesh? *The Bangladesh Development Studies*, 35(2), 1–43.
<https://www.jstor.org/stable/41968817>

Steets, J. & WFP-Office of Evaluation. (2026). *Strategic Evaluation of WFP's Approaches to Targeting and Prioritization for Food and Nutrition Assistance* (p. 19.7 MB, 7 v.) [PDF]. World Food Programme (WFP),. <https://doi.org/10.71958/WFP131314>

Stoeffler, Q., Mills, B., & Del Ninno, C. (2016). *Reaching the Poor: Cash Transfer Program Targeting in Cameroon*. Elsevier. <https://doi.org/10.1596/24255>

Sumarto, S., Satriawan, E., Olken, B. A., Banerjee, A., Tohari, A., Alatas, V., & Hanna, R. (2025). *Community Targeting at Scale* (Working Paper No. 33322). National Bureau of Economic Research. <https://doi.org/10.3386/w33322>

The Brookings-Bern Project on Internal Displacement. (2011). *IASC Operational Guidelines on the Protection of Persons in Situations of Natural Disasters*.
https://www.ohchr.org/sites/default/files/Documents/Issues/IDPersons/OperationalGuidelines_I DP.pdf

The Government of the Democratic Republic of the Congo. Loi Organique N° 08/016 Du 07 Octobre 2008 Portant Composition, Organization et Fonctionnement Des Entités Territoriales Décentralisées et Leurs Rapports Avec l'Etat et Les Provinces, Pub. L. No. 08/016. Retrieved April 15, 2026, from
<https://www.leganet.cd/Legislation/Droit%20Public/Administration.ter/L.08.16.17.10.2008.htm>

The Office of the United Nations High Commissioner for Refugees (UNHCR). (2018). *UNHCR Emergency Handbook: Shelter in urban areas* (4th edition).
<https://www.humanitarianlibrary.org/sites/default/files/2018/06/Emergency%20handbook.pdf>

Trachtman, C., Permana, Y. H., & Sahadewo, G. A. (2026). How much do our neighbors really know? The limits of community-based targeting. *Journal of Development Economics*, 178, 103555.
<https://doi.org/10.1016/j.jdeveco.2025.103555>

United Nations Development Programme. (2025). *Human Development Reports: Democratic Republic of the Congo*. <https://hdr.undp.org/data-center/specific-country-data#/countries/COD>

Urmitsky, O., Hansen, C., & Chernozhukov, V. (2016). Using Double-Lasso Regression for Principled Variable Selection. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.2733374>

World Bank Group. (2024). *Democratic Republic of Congo: Poverty and Equity Brief – October 2024*.
<https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099700301032538532>

World Bank Group. (n.d.). *PPP conversion factor, GDP (LCU per international \$)*. World Bank Open Data. Retrieved April 15, 2026, from <https://data.worldbank.org>

World Food Programme. (2021b). *Targeting and prioritization: Operational guidance note*. <https://docs.wfp.org/api/documents/WFP-0000122035/download/>

World Food Programme. (2022). *Democratic Republic of the Congo Annual Country Report 2021*. <https://docs.wfp.org/api/documents/WFP-0000138022/download/>

World Food Programme. (2023a). *Democratic Republic of the Congo Annual Country Report 2022*. <https://docs.wfp.org/api/documents/WFP-0000147935/download/>

World Food Programme. (2023b). *Evaluation of the Policy on WFP's Role in Peacebuilding in Transition Settings* (Centralized Evaluation Report OEV/2021/001). <https://docs.wfp.org/api/documents/WFP-0000146160/download/>

World Food Programme. (2024). *Democratic Republic of the Congo Annual Country Report 2023*. <https://docs.wfp.org/api/documents/WFP-0000157699/download/>

World Food Programme & DRC National Nutrition Programme. (2023). *Emergency Food Security Assessment (EFSA) 2023*.

Xe. (2023, May 1). *Historical Rates Tables—CDF* | Xe. <https://www.xe.com/currencytables/?from=CDF&date=2023-05-01>

8. Acronyms

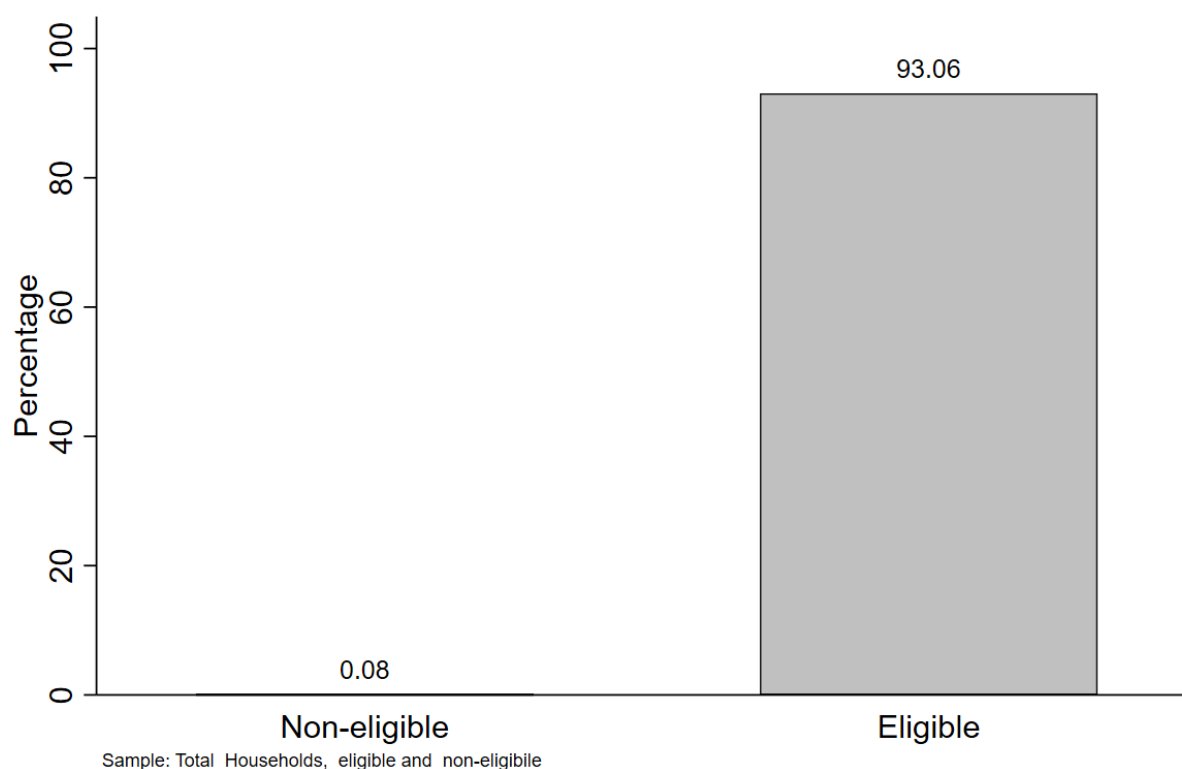
CBT	Community-based targeting
CGH	Coady Grosh Hoddinott Index
CO	Country office
CP	Cooperating partner
DiD	Difference-in-differences
DI	Development Impact (World Bank)
DRC	Democratic Republic of the Congo
EQ	Evaluation question
FCS	Food Consumption Score
FGD	Focus group discussion
GFA	General food assistance
HDI	Human Development Index
HH	Household
HoH	Head of household
IDP	Internally displaced person
IPC	Integrated Food Security Phase Classification
IRB	Institutional Review Board
LCSI	Livelihood Coping Strategies Index
MEB	Minimum expenditure basket
NGO	Non-governmental organization
OEV	Office of Evaluation
PDM	Post-distribution monitoring
PLW	Pregnant/lactating woman
PMT	Proxy means test
PPP	Purchasing power parity
RCSI	reduced Coping Strategies Index
RCT	Randomized control trial
RR	Relative representation
SD	Standard deviation
TD	Targeting Differentials
USD	United States dollar
WFP	World Food Programme

9. Appendix

A1. Additional details on programme implementation

165. Adherence to the protocol within the sample was high, with 93 percent of eligible households receiving at least one transfer (82 percent received all four), and only one non-eligible household receiving assistance.⁹⁵ Rates were similar across the two arms: for CBT, 92 percent received at least one, and 81 percent received all four; for PMT Plus, this was 94 percent and 84 percent, respectively. Appendix Figures A1.1–A1.5 describe the distribution of transfers by eligibility status, targeting arm, the number of transfers received, and data source.

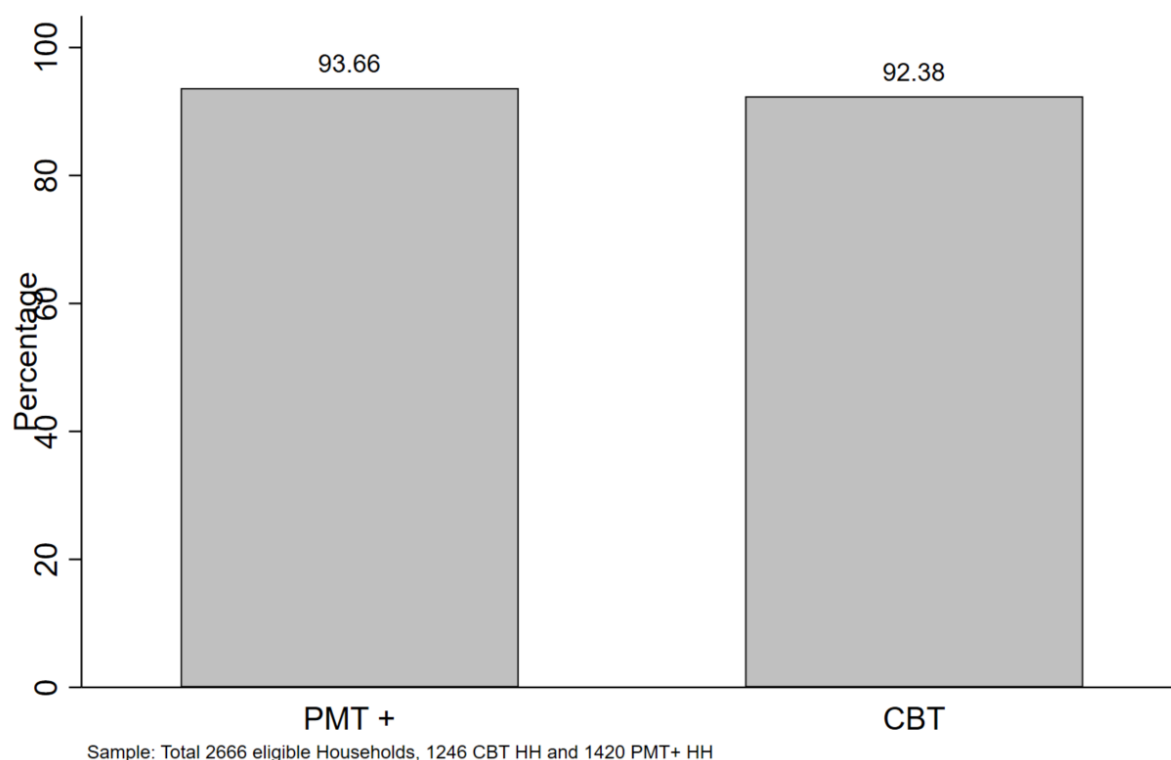
Figure A1.1: Percentage of households that received at least 1 transfer by eligibility



Note: The figure reports the share of eligible households that received at least one transfer during the programme period, separately for the PMT Plus and CBT arms. Source: WFP SCOPE administrative data.

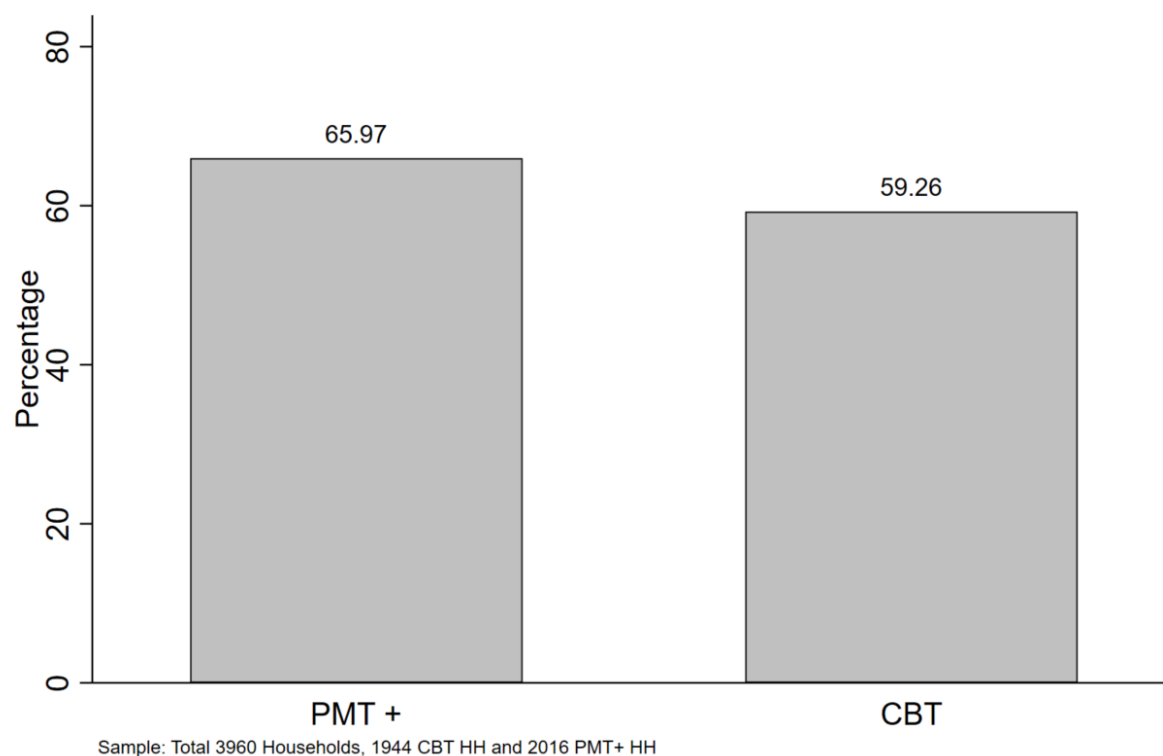
⁹⁵ Given this is a highly mobile population, these rates are within the range of normal expectations.

Figure A1.2: Percent of eligible households that received at least 1 transfer by targeting arm



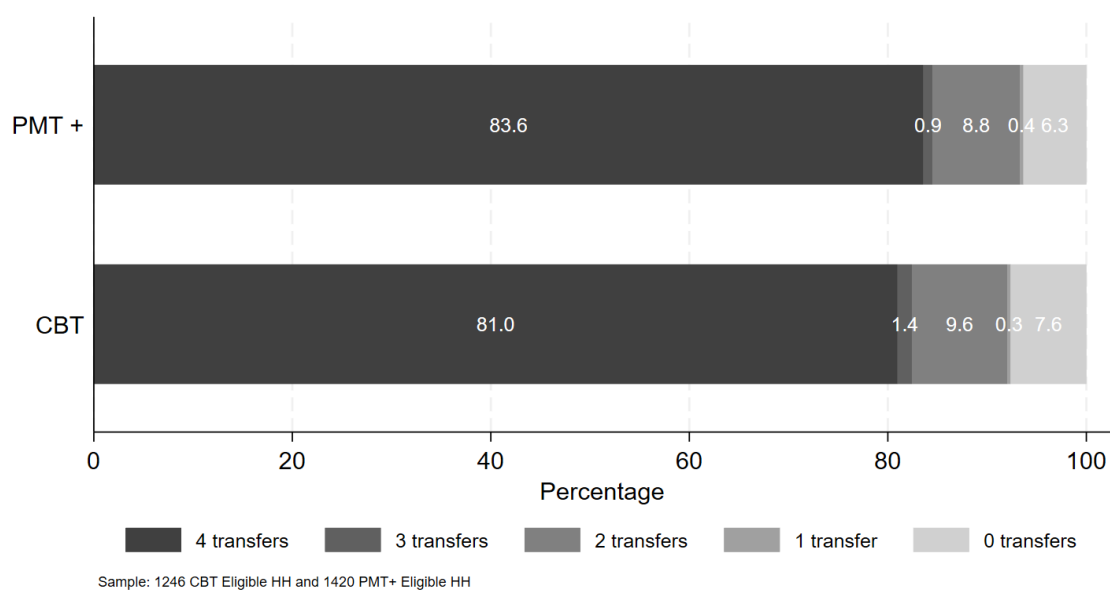
Note: The figure shows the percentage of eligible households that received at least one transfer during the programme period, separately for the PMT Plus and CBT targeting arms. Source: WFP SCOPE administrative data.

Figure A1.3: Percentage of all households that received at least 1 transfer by targeting arm (eligible and non-eligible)



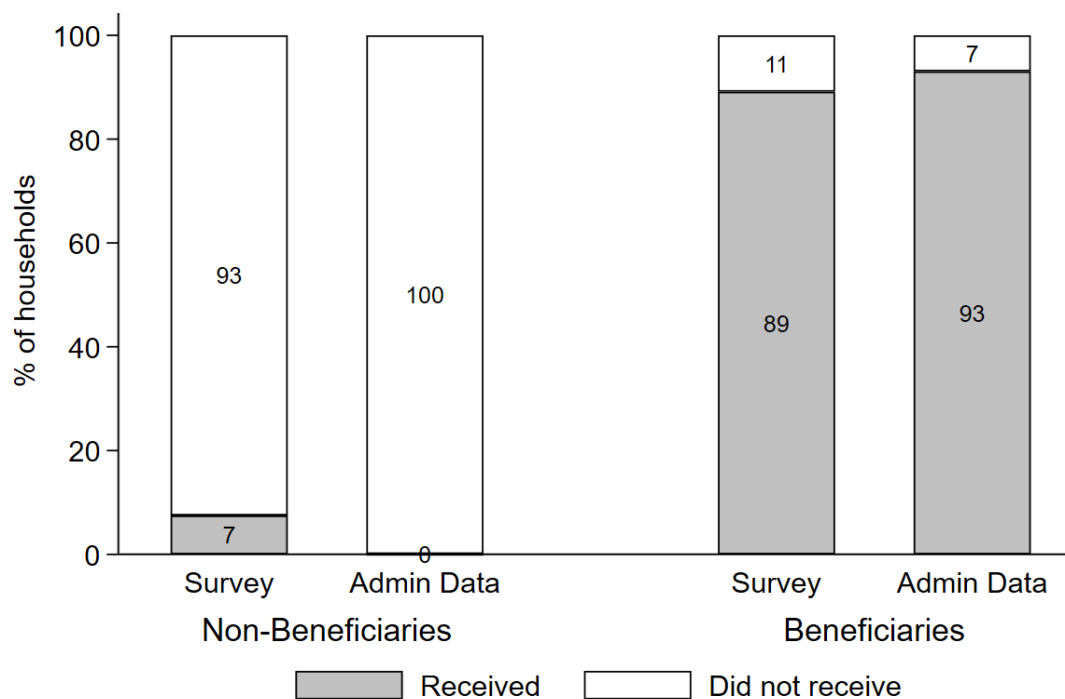
Note: The figure reports the share of households that received at least one transfer during the programme period, separately for the PMT Plus and CBT arms. The sample includes both eligible and non eligible households. Source: WFP SCOPE administrative data.

Figure A1.4: Percentage of eligible households by total number of transfers received



Note: The figure shows the distribution of eligible households by the total number of transfers received during the programme period, separately for CBT and PMT Plus arms. Source: WFP SCOPE administrative data.

Figure A1.5: Percentage of households that received assistance by data source



Note: The figure compares survey responses (self-reported) from the first follow-up survey and WFP SCOPE administrative records on whether households received assistance. Shares are reported separately for beneficiaries and non-beneficiaries.

Table A1.1: General Food Assistance transfer amounts (USD PPP 2023)

Treatment Arm	Transfer 1	Transfer 2	Transfer 3	Transfer 4
Amount per beneficiary	31	31	31	38
Total disbursements (PMT Plus households)	179,575	180,113	199,834	238,928
Total disbursements (CBT households)	156,283	155,999	176,763	211,584
% eligible households who received the transfer	83%	83%	93%	92%
% non-eligible households receiving the transfer	0%	0%	0%	0%

Table A1.2: CBT criteria with frequency above 10% and lock-in criteria

Criteria	CBT Frequency (%)	Lock-in PMT Plus
Household headed by woman without provider	90.7	Yes
Household headed by an elderly person	90.7	Yes
Household headed by a person with disability	86.1	Yes
Household has orphaned child	79.1	Yes
Household head is pregnant	62.8	Yes
Household is Twa	55.8	
Household has person with disability	51.2	
Household headed by a person with chronic illness	48.8	Yes
Household has a paltry shelter	37.2	
Household head is nursing	34.9	
Household hosting displaced	27.9	
Household has at least one elderly person	23.3	
Household has a pregnant/lactating woman (PLW)	20.9	
Woman-headed household	18.6	
Orphan headed household	16.3	
Household headed by a widower	13.9	Yes
Displaced household	11.6	Yes
Household lost crops due to natural disaster (e.g., pests)	11.6	
Child-headed household	11.6	

A2. Additional details on sampling and attrition

Sampling balance

The tables include only a subset of the variables collected at baseline that are relevant for the analysis. When considering the full set of baseline variables and conducting balance checks across the two arms, a total of 120 comparisons are generated. Of these, 9 comparisons (7.5 percent) show differences significant at the 10-percent level, 5 comparisons (4.2 percent) are significant at the 5-percent level, and 4 comparisons (3.3 percent) are significant at the 1-percent level. Balance is assessed using the 84 blocs for which baseline data were collected, as no information was gathered for 12 blocs that were dropped from the study (6 in each arm).

Table A2.1: Balance comparison - population vs. sample

	A. Full Population			B. Study Sample		
	N	Mean	SD	N	Mean	SD
Single	11732	0.03	0.16	3960	0.03	0.17
Married	11732	0.80	0.40	3960	0.79	0.41
Widow(er)	11732	0.14	0.34	3960	0.14	0.34
Others (Divorced, living together)	11732	0.04	0.19	3960	0.05	0.21
No education	11721	0.52	0.50	3956	0.49	0.50
Below secondary	11721	0.46	0.50	3956	0.49	0.50
Secondary and above	11721	0.02	0.14	3956	0.02	0.13
HH Head age	11730	41.8	14.38	3960	42.5	14.94
Woman-headed household	11732	0.18	0.38	3960	0.16	0.36
Household headed by an elderly person	11732	0.14	0.35	3960	0.16	0.37
Household headed by a person with chronic illness	11732	0.03	0.18	3960	0.03	0.16
Household headed by a person with disability	11732	0.04	0.19	3960	0.04	0.20
Household size	11732	6.00	1.90	3960	5.96	1.88
Household has more than 3 U5 children	11732	0.03	0.18	3960	0.03	0.18
Household is Twa	11732	0.11	0.32	3960	0.11	0.31
Household living from agriculture	11732	0.47	0.50	3960	0.48	0.50
hh_status==Displaced in a camp	11726	0.27	0.44	3959	0.26	0.44
hh_status==Refugee	11726	0.03	0.17	3959	0.03	0.17
hh_status==Resident	11726	0.59	0.49	3959	0.59	0.49
Displaced household	11732	0.30	0.46	3960	0.29	0.45
Average household monthly income	10443	1.16	0.42	3577	1.17	0.41
Agriculture and Livestock	11668	0.87	0.33	3939	0.89	0.31
Small businesses and non-agri	11668	0.07	0.26	3939	0.05	0.22
Others	11668	0.05	0.22	3939	0.04	0.19
Household received assistance from WFP in 2022	11732	0.29	0.45	3960	0.27	0.44
Wealth index	10555	1.47	0.46	3483	1.47	0.48

Food Consumption Score	11732	18.29	9.02	3960	18.35	9.07
FCSCat28==Poor (0-28)	11732	0.86	0.35	3960	0.85	0.36
FCSCat28==Borderline (28.5-42)	11732	0.13	0.34	3960	0.14	0.35
FCSCat28==Acceptable (42.5 and over)	11732	0.01	0.10	3960	0.01	0.10
HH didn't adopt any coping strategies	11732	0.06	0.24	3960	0.06	0.23
Stress coping strategies	11732	0.08	0.27	3960	0.09	0.28
Crisis coping strategies	11732	0.14	0.34	3960	0.17	0.37
Emergency coping strategies	11732	0.72	0.45	3960	0.69	0.46

Note: Reported values for the study sample are weighted means. Sampling weights are applied as probability weights.

Table A2.2: Balance comparison (84 blocs) – PMT Plus vs. CBT

	PMT Plus (42 blocs)			CBT (42 blocs)			P-val t- test
	N	Mean	SE	N	Mean	SE	
Single	2037	0.04	(0.01)	1961	0.03	(0.01)	0.01
Married	2037	0.76	(0.05)	1961	0.80	(0.02)	-0.04
Widow(er)	2037	0.16	(0.04)	1961	0.12	(0.01)	0.04
Others (Divorced, living together)	2037	0.04	(0.01)	1961	0.05	(0.01)	-0.00
No education	2035	0.51	(0.06)	1959	0.54	(0.04)	-0.03
Below secondary	2035	0.47	(0.05)	1959	0.44	(0.04)	0.03
Secondary and above	2035	0.01	(0.00)	1959	0.02	(0.00)	-0.00
HH Head age	2037	42.00	(1.64)	1961	42.26	(1.19)	-0.26
Woman-headed household	2037	0.15	(0.02)	1961	0.16	(0.02)	-0.01
Household headed by an elderly person	2037	0.16	(0.04)	1961	0.16	(0.02)	-0.01
Household headed by a person with chronic illness	2037	0.02	(0.00)	1961	0.03	(0.01)	-0.01
Household headed by a person with disability	2037	0.04	(0.01)	1961	0.05	(0.01)	-0.01
Household size	2037	5.80	(0.17)	1961	6.17	(0.08)	-0.37***
Household has more than 3 U5 children	2037	0.05	(0.01)	1961	0.03	(0.01)	0.02**
Household is Twa	2037	0.12	(0.04)	1961	0.10	(0.03)	0.02
Household living from agriculture	2037	0.46	(0.09)	1961	0.50	(0.04)	-0.05
Displaced in a camp	2036	0.26	(0.18)	1961	0.28	(0.13)	-0.02
Refugee	2036	0.03	(0.01)	1961	0.04	(0.02)	-0.01
Resident	2036	0.61	(0.17)	1961	0.56	(0.13)	0.05
Displaced household	2037	0.29	(0.20)	1961	0.32	(0.14)	-0.03
Average household monthly income	1811	1.16	(0.04)	1795	1.15	(0.03)	0.02

Agriculture and Livestock	2029	0.88	(0.01)	1949	0.87	(0.02)	0.01
Small businesses and non-agri	2029	0.06	(0.01)	1949	0.06	(0.01)	-0.00
Others	2029	0.04	(0.01)	1949	0.04	(0.01)	0.00
Household received assistance from WFP in 2022	2037	0.31	(0.11)	1961	0.20	(0.06)	0.10
Wealth Index (PCA)	1784	1.45	(0.11)	1736	1.46	(0.09)	-0.01
Food Consumption Score	2037	17.58	(0.46)	1961	17.62	(0.80)	-0.04
FCS Category=Poor (0-28)	2037	0.88	(0.01)	1961	0.87	(0.02)	0.01
FCS Category=Borderline (28.5-42)	2037	0.12	(0.01)	1961	0.12	(0.02)	-0.01
FCS Category=Acceptable (42.5 and over)	2037	0.01	(0.00)	1961	0.01	(0.00)	-0.01*
HH didn't adopt any coping strategies	2037	0.06	(0.01)	1961	0.05	(0.01)	0.01
Used stress coping strategies	2037	0.90	(0.01)	1961	0.91	(0.02)	-0.01
Used crisis coping strategies	2037	0.79	(0.02)	1961	0.84	(0.02)	-0.04**
Used emergency coping strategies	2037	0.67	(0.03)	1961	0.75	(0.02)	-0.07***

Note: Reported values are weighted means. Pairwise regressions include quadruplet fixed effects. Standard errors, clustered at the bloc level, are reported in parentheses. Sampling weights are applied as probability weights. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Table A2.3: Balance comparison – dropping households with more or less than 4 transfers

	PMT Plus (42 blocs)			CBT (42 blocs)			P-val t- test
	N	Mean	SE	N	Mean	SE	
Single	2037	0.04	(0.01)	1961	0.03	(0.01)	0.01
Married	2037	0.76	(0.05)	1961	0.80	(0.02)	-0.04
Widow(er)	2037	0.16	(0.04)	1961	0.12	(0.01)	0.04
Others (Divorced, living together)	2037	0.04	(0.01)	1961	0.05	(0.01)	-0.00
No education	2035	0.51	(0.06)	1959	0.54	(0.04)	-0.03
Below secondary	2035	0.47	(0.05)	1959	0.44	(0.04)	0.03
Secondary and above	2035	0.01	(0.00)	1959	0.02	(0.00)	-0.00
HH Head age	2037	42.12	(1.71)	1961	42.38	(1.23)	-0.25
Woman-headed household	2037	0.15	(0.02)	1961	0.16	(0.02)	-0.01
Household headed by an elderly person	2037	0.16	(0.04)	1961	0.16	(0.02)	-0.01
Household headed by a person with chronic illness	2037	0.02	(0.00)	1961	0.03	(0.01)	-0.01
Household headed by a person with disability	2037	0.04	(0.01)	1961	0.05	(0.01)	-0.01
Household size	2037	5.81	(0.17)	1961	6.18	(0.09)	-0.37***
Household has more than 3 U5 children	2037	0.05	(0.01)	1961	0.03	(0.01)	0.02**
Household is Twa	2037	0.12	(0.04)	1961	0.10	(0.03)	0.02
Household living from agriculture	2037	0.46	(0.09)	1961	0.50	(0.04)	-0.05
Displaced in a camp	2036	0.26	(0.18)	1961	0.28	(0.13)	-0.02
Refugee	2036	0.03	(0.01)	1961	0.04	(0.02)	-0.01
Resident	2036	0.61	(0.17)	1961	0.56	(0.13)	0.05
Displaced household	2037	0.29	(0.20)	1961	0.32	(0.14)	-0.03
Average household monthly income	1811	1.16	(0.04)	1795	1.15	(0.03)	0.02

Agriculture and Livestock	2029	0.88	(0.01)	1949	0.87	(0.02)	0.01
Small businesses and non-agri	2029	0.06	(0.01)	1949	0.06	(0.01)	-0.00
Others	2029	0.04	(0.01)	1949	0.04	(0.01)	0.00
Household received assistance from WFP in 2022	2037	0.31	(0.11)	1961	0.20	(0.06)	0.10
Wealth Index (PCA)	1784	1.45	(0.11)	1736	1.46	(0.09)	-0.01
Food Consumption Score	2037	17.58	(0.46)	1961	17.62	(0.80)	-0.04
FCS Category=Poor (0-28)	2037	0.88	(0.01)	1961	0.87	(0.02)	0.01
FCS Category=Borderline (28.5-42)	2037	0.12	(0.01)	1961	0.12	(0.02)	-0.01
FCS Category=Acceptable (42.5 and over)	2037	0.01	(0.00)	1961	0.01	(0.00)	-0.01*
HH didn't adopt any coping strategies	2037	0.06	(0.01)	1961	0.05	(0.01)	0.01
Used stress coping strategies	2037	0.90	(0.01)	1961	0.91	(0.02)	-0.01
Used crisis coping strategies	2037	0.79	(0.02)	1961	0.84	(0.02)	-0.04**
Used emergency coping strategies	2037	0.67	(0.03)	1961	0.75	(0.02)	-0.07***

Note: Reported values are weighted means. Pairwise regressions include quadruplet fixed effects. Standard errors, clustered at the bloc level, are reported in parentheses. Sampling weights are applied as probability weights. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Table A2.4: Balance comparison – dropping IDP blocs with perfect coverage

	PMT Plus (41 blocs)			CBT (40 blocs)			P-val t- test
	N	Mean	SE	N	Mean	SE	
Single	1981	0.03	(0.00)	1849	0.03	(0.01)	0.00
Married	1981	0.82	(0.02)	1849	0.82	(0.02)	-0.00
Widow(er)	1981	0.04	(0.01)	1849	0.03	(0.01)	0.01*
Others (Divorced, living together)	1981	0.11	(0.02)	1849	0.12	(0.02)	-0.01
No education	1979	0.58	(0.03)	1847	0.56	(0.05)	0.02
Below secondary	1979	0.40	(0.03)	1847	0.42	(0.05)	-0.02
Secondary and above	1979	0.01	(0.00)	1847	0.02	(0.00)	-0.01
HH Head age	1981	39.95	(0.98)	1849	41.65	(1.69)	-1.70**
Woman-headed household	1981	0.15	(0.03)	1849	0.15	(0.02)	-0.01
Household headed by an elderly person	1981	0.10	(0.01)	1849	0.14	(0.03)	-0.03**
Household headed by a person with chronic illness	1981	0.02	(0.01)	1849	0.04	(0.01)	-0.01**
Household headed by a person with disability	1981	0.04	(0.01)	1849	0.04	(0.01)	-0.00
Household size	1981	6.03	(0.13)	1849	6.25	(0.11)	-0.22*
Household has more than 3 U5 children	1981	0.05	(0.01)	1849	0.04	(0.01)	0.01*
Household is Twa	1981	0.17	(0.03)	1849	0.14	(0.03)	0.03
Household living from agriculture	1981	0.57	(0.02)	1849	0.56	(0.02)	0.01
Displaced in a camp	1980	0.01	(0.00)	1849	0.03	(0.02)	-0.02*
Refugee	1980	0.01	(0.00)	1849	0.02	(0.01)	-0.01
Resident	1980	0.85	(0.04)	1849	0.80	(0.05)	0.05
Displaced household	1981	0.02	(0.01)	1849	0.05	(0.02)	-0.03
Average household monthly income	1761	1.22	(0.01)	1705	1.19	(0.02)	0.03

Agriculture and Livestock	1973	0.88	(0.01)	1837	0.90	(0.02)	-0.02
Small businesses and non-agri	1973	0.06	(0.01)	1837	0.06	(0.01)	0.01
Others	1973	0.04	(0.01)	1837	0.04	(0.01)	0.00
Household received assistance from WFP in 2022	1981	0.17	(0.04)	1849	0.19	(0.04)	-0.02
Wealth Index (PCA)	1735	1.60	(0.03)	1631	1.63	(0.03)	-0.03*
Food Consumption Score	1981	18.01	(0.46)	1849	18.63	(0.47)	-0.62
FCS Category=Poor (0-28)	1981	0.89	(0.01)	1849	0.84	(0.01)	0.04***
FCS Category=Borderline (28.5-42)	1981	0.10	(0.01)	1849	0.15	(0.01)	-0.04***
FCS Category=Acceptable (42.5 and over)	1981	0.01	(0.00)	1849	0.01	(0.00)	-0.00
HH didn't adopt any coping strategies	1981	0.07	(0.01)	1849	0.04	(0.02)	0.03**
Used stress coping strategies	1981	0.88	(0.01)	1849	0.93	(0.02)	-0.04***
Used crisis coping strategies	1981	0.78	(0.02)	1849	0.82	(0.02)	-0.04**
Used emergency coping strategies	1981	0.72	(0.02)	1849	0.75	(0.02)	-0.04**

Note: Reported values are weighted means. Pairwise regressions include quadruplet fixed effects. Standard errors, clustered at the bloc level, are reported in parentheses. Sampling weights are applied as probability weights. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

Attrition

166. At the first follow-up, about 20 percent of the selected households could not be reached. This was in part due to security issues in certain blocs, initial refusals by local leaders to authorise interviews, and seasonal factors such as the *Bulumba* hunting season, which drew households away from their blocs.⁹⁶ By endline, most households had returned to their homes and attrition with respect to the target sample went down to 9 percent (see Table A2.5).

Table A2.5: Attrition by data collection round

	Census	First follow-up	Second follow-up
Number of successful interviews	11,732 Sample: 3,998	3,460	3,960
Overall Attrition (percent)	-	19.96% (no replacements) 13.5% (w/ replacements)	9% (no replacements) 1% (w/ replacements)
Differential attrition (percentage points)	-	3.56 p.p. (no replacements) 1 p.p. (w/replacements)	0 p.p. (no replacements) 0.1 p.p. (w/replacements)
Differential attrition signif. test (p-value)	-	0.560 (no replacements) 0.488 (w/replacements)	0.876 (no replacements) 0.582 (w/replacements)

167. Attrition was balanced across PMT Plus and CBT groups. When a household could not be found, or the household refused to participate, a replacement household was sought. This happened in 7.5 and 8.2 percent of the cases for first and second follow-up respectively, without any systematic difference between PMT Plus and CBT blocs.
168. The randomization design, which stratified blocs in quadruplets, allows the exclusion of the quadruplets affected by the security issues while maintaining a balanced sample. After dropping these quadruplets, the resulting sample comprises 70 clusters in total, evenly divided between the two treatment arms (35 clusters each), and includes 3,423 households out of the initial 3,960. This represents a reduction in sample size of approximately 13.6 percent. Table A2.6 shows that the two arms remain well-balanced and comparable to the

⁹⁶ Fieldwork in September 2023 was repeatedly disrupted by access and security challenges. In the blocs of Likasi, Likasi IDP, and Kyempele, chiefs initially refused authorisation for interviews, citing exclusion from targeting. In Kisalaba and Mulange, insecurity was a constant concern: incidents involving militia groups such as *Tête Rouge*, clashes between the Kilima militia and the self-defence group *La Rondo*, and the killing of civilians triggered repeated displacement and fear among residents. Beyond conflict, seasonal movements such as the *Bulumba* hunting season affected households in Twa Pilipili, Kabila Yuma, Nkikanza, Kiboko, Kinshasa, Mutenta, and Kantogne, while in several blocs respondents were reluctant to participate in a survey not linked to direct assistance. Awareness-raising and tracking allowed some resumption of activities, but volatility in affected blocs persisted throughout the month.

balance observed for the sample of 84 blocs, even after the exclusion of the quadruplets containing the dropped blocs.

Table A2.6: Balance comparison – dropping blocs with security issues

	PMT Plus (35 blocs)			CBT (35 blocs)			P-val t- test
	N	Mean	SE	N	Mean	SE	
Single	1748	0.03	(0.01)	1704	0.03	(0.01)	0.01
Married	1748	0.75	(0.05)	1704	0.80	(0.02)	-0.04
Widow(er)	1748	0.04	(0.01)	1704	0.05	(0.01)	0.04
Others (Divorced, living together)	1748	0.17	(0.04)	1704	0.13	(0.01)	-0.00
No education	1746	0.50	(0.06)	1702	0.53	(0.04)	-0.03
Below secondary	1746	0.48	(0.06)	1702	0.45	(0.04)	0.03
Secondary and above	1746	0.01	(0.00)	1702	0.02	(0.00)	-0.00
HH Head age	1748	42.70	(1.68)	1704	42.97	(1.19)	-0.26
Woman-headed household	1748	0.16	(0.02)	1704	0.17	(0.02)	-0.01
Household headed by an elderly person	1748	0.17	(0.04)	1704	0.17	(0.02)	-0.01
Household headed by a person with chronic illness	1748	0.02	(0.00)	1704	0.03	(0.01)	-0.01
Household headed by a person with disability	1748	0.04	(0.01)	1704	0.05	(0.01)	-0.01
Household size	1748	5.82	(0.19)	1704	6.24	(0.08)	-0.37***
Household has more than 3 U5 children	1748	0.05	(0.01)	1704	0.03	(0.01)	0.02**
Household is Twa	1748	0.12	(0.04)	1704	0.09	(0.03)	0.02
Household living from agriculture	1748	0.46	(0.09)	1704	0.51	(0.04)	-0.05
Displaced in a camp	1747	0.28	(0.19)	1704	0.30	(0.13)	-0.02
Refugee	1747	0.03	(0.01)	1704	0.04	(0.02)	-0.01
Resident	1747	0.59	(0.18)	1704	0.54	(0.13)	0.05
Displaced household	1748	0.31	(0.21)	1704	0.34	(0.14)	-0.03

Average household monthly income	1548	1.16	(0.04)	1573	1.15	(0.03)	0.02
Agriculture and Livestock	1741	0.89	(0.01)	1692	0.88	(0.02)	0.01
Small businesses and non-agri	1741	0.06	(0.00)	1692	0.06	(0.01)	-0.00
Others	1741	0.04	(0.01)	1692	0.04	(0.01)	0.00
Household received assistance from WFP in 2022	1748	0.32	(0.11)	1704	0.21	(0.06)	0.10
Wealth Index (PCA)	1536	1.45	(0.12)	1480	1.45	(0.09)	-0.01
Food Consumption Score	1748	17.66	(0.51)	1704	17.73	(0.80)	-0.04
FCS Category=Poor (0-28)	1748	0.88	(0.01)	1704	0.86	(0.02)	0.01
FCS Category=Borderline (28.5-42)	1748	0.12	(0.01)	1704	0.13	(0.02)	-0.01
FCS Category=Acceptable (42.5 and over)	1748	0.01	(0.00)	1704	0.01	(0.00)	-0.01*
HH didn't adopt any coping strategies	1748	0.05	(0.01)	1704	0.04	(0.01)	0.01
Used stress coping strategies	1748	0.90	(0.01)	1704	0.92	(0.02)	-0.01
Used crisis coping strategies	1748	0.80	(0.02)	1704	0.85	(0.02)	-0.04**
Used emergency coping strategies	1748	0.67	(0.04)	1704	0.75	(0.02)	-0.07***

Note: Reported values are weighted means. Pairwise regressions include quadruplet fixed effects. Standard errors, clustered at the bloc level, are reported in parentheses. Sampling weights are applied as probability weights. Statistical significance is denoted by *** $p < 0.01$, ** $p < 0.05$, and * $p < 0.10$.

A3. Additional details on outcomes

Table A3.1: Additional details on outcomes of interest

Dimension	Indicator	Description
Performance	Coverage rate	The coverage rate indicates the proportion of households selected for assistance within each bloc. Calculated as Ben_{hh}/Tot_{hh} where the numerator, Ben_{hh} , represents the number households selected for assistance divided by the total number of households in the respective bloc. The coverage rate for each targeting arm is calculated as the average coverage rate across all blocs within that arm. <i>Source: census</i>
	Inclusion error rate	Number of non-food insecure households ($FCS > 28$) selected for assistance, divided by the total number of selected households. <i>Source: census</i>
	Exclusion error rate	Number of food insecure households ($FCS \leq 28$) not selected for assistance, divided by the total number of food insecure households. <i>Source: census</i>
Welfare	Food Consumption Score (FCS)	FCS is a widely used measure of food security. It combines information on dietary diversity and frequency of consumption across eight food groups over the past seven days. The score is calculated by multiplying the number of days each food group was consumed by a standard weight. The total score ranges from 0 to 112. <i>Source: survey data.</i> <i>Source: follow-up survey</i>
	Reduced Coping Strategy Index (rCSI)	The index measures the frequency and severity of behaviours that households adopt when facing food shortages. These include skipping meals, reducing portion sizes, or relying on less preferred foods. Each behaviour is assigned a severity weight (1 to 4), and the number of days each behaviour was used in the past seven days is multiplied by its respective weight. The weighted frequencies are then summed to produce the total rCSI score, which typically ranges from 0 to 56, with higher scores indicating greater food insecurity. <i>Source: follow-up survey</i>
	Livelihood Coping Strategy Index (LCSI)	The LCSI is used by humanitarian organizations to assess the severity of livelihood stress caused by food insecurity or economic shocks. ⁹⁷ It captures longer-term coping strategies that can undermine future household resilience. The index is composed of four stress, three crisis, and three emergency strategy questions. A binary indicator was created for each

⁹⁷ The standard LCSI module uses “the past 30 days” as the timeframe for the questions; however, for the study, a significantly longer timeframe to match the study’s timeline. For round 1, the timeframe was “since the start of the year” which coincided with the beginning of the new food assistance cycle and the re-targeting of households. For round 2, the timeframe was “since July 2024” which was the endpoint of round 1 household surveys.

		category—stress, crisis, and emergency—based on whether a household reported using at least one type of strategy within that category. <i>Source: survey data.</i>
	Life Satisfaction Index	The index uses the Cantril Ladder approach to aggregate three 0-to-10 scale questions on subjective well-being. It captures how respondents perceive their quality of life, respect within the community, and how widely their opinion is valued. Individual questions are standardized and then combined into a single index using weights based on the inverse of their covariance matrix. ⁹⁸ <i>Source: follow-up survey</i>
	Women's Agency Index	Standardized index based on Jayachandran et al. ⁹⁹ , captures key aspects of women's decision-making power and mobility. It includes five equally weighted binary indicators on whether a woman can express her opinion on major purchases, go to the market or visit others without permission, decide on her children's health, and use public transport freely. Individual questions are standardized and then combined into a single index using weights based on the inverse of their covariance matrix. <i>Source: follow-up survey</i>
Satisfaction & Social Cohesion	Complaints	Formal grievances submitted by excluded households. These are recorded through the Complaints and Feedback Mechanism of the CO <i>Source: WFP's admin data.</i>
	Self-reported satisfaction	Binary variable based on two Likert-scale questions on satisfaction with the overall targeting process and the formation of the local targeting mechanism. It takes the value of one if the respondent reports being very satisfied or satisfied to at least one of the two questions. <i>Source: follow-up survey</i>
	Perceptions of the targeting process	Measured by asking respondents to choose up to three descriptive words from a predefined list to characterise how the process was conducted in their community. This provides insight into whether the process was seen as fair, transparent, slow, or otherwise. <i>Source: follow-up survey</i>
	Social Cohesion Index	Adapted from Bossuroy et al. (2022) ¹⁰⁰ , this standardized index captures interpersonal support and community connectedness.

⁹⁸ Anderson, M. L. (2008). Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects. *Journal of the American Statistical Association*, 103(484), 1481–1495. <https://doi.org/10.1198/016214508000000841>

⁹⁹ Jayachandran, S., Biradavolu, M., & Cooper, J. (2023). Using machine learning and qualitative interviews to design a five-question survey module for women's agency. *World Development*, 161, 106076. <https://doi.org/10.1016/j.worlddev.2022.106076>

¹⁰⁰ Bossuroy, T., Goldstein, M., Karimou, B., Karlan, D., Kazianga, H., Parienté, W., Premand, P., Thomas, C. C., Udry, C., Vaillant, J., & Wright, K. A. (2022). Tackling psychosocial and capital constraints to alleviate poverty. *Nature*, 605(7909), 291–297. <https://doi.org/10.1038/s41586-022-04647-8>

		It is based on four equally weighted items: whether the respondent feels they can rely on neighbours for help in times of difficulty, how many people out of ten they feel they can count on, whether there has been any tension between community members in the past six months, and how close the respondent feels to their community. Individual responses were first standardized to have mean zero and standard deviation one within each component of the index. These standardized values were then averaged without weights to form a composite index, which was itself re-standardized. <i>Source: follow-up survey</i>
	Trust Index	Includes Likert-scale questions measuring respondents' trust in a range of groups and institutions, including family, neighbours, religious organizations, NGOs, local and national government, migrants, and other ethnic groups. Individual responses were first standardized to have mean zero and standard deviation one within each component of the index. These standardized values were then averaged without weights to form a composite index, which was itself re-standardized. <i>Source: follow-up survey</i>
	Collective Action Index	Captures household engagement in community groups and local collective activities over the past six months. It is constructed from four equally weighted components: (i) number of groups or associations a household member belongs to, (ii) number of leadership or responsibility roles held within those groups, (iii) number of days household members volunteered in community activities, and (iv) amount of money donated to bloc projects. Individual responses were first standardized to have mean zero and standard deviation one within each component of the index. These standardized values were then averaged without weights to form a composite index, which was itself re-standardized. <i>Source: follow-up survey</i>
	Perceived safety (home)	Binary indicator of feeling unsafe at home in past 6 months. The indicator takes the value of one if the respondent reported feeling unsafe at least occasionally. <i>Source: follow-up survey</i>
	Perceived safety (community)	Binary indicator of feeling unsafe in community in past 6 months. The indicator takes the value of one if the respondent reported feeling unsafe at least occasionally. <i>Source: follow-up survey</i>

Table A3.2: Details on the construction of indices

Indicator	Components/ Reference	Sub-components	Possible answers	Weights	Score range
Food Consumption Score (FCS)	Weighted frequency of food groups consumed over past 7 days	Cereals, roots, and tubers	0 - 7 days	2	0-112
		Pulses	0 - 7 days	3	
		Vegetables	0 - 7 days	1	
		Fruits	0 - 7 days	1	
		Meat and fish	0 - 7 days	4	
		Milk	0 - 7 days	4	
		Sugar	0 - 7 days	0.5	
		Oil	0 - 7 days	0.5	
Reduced Coping Strategy Index (rCSI)	Frequency of 5 coping strategies used in past 7 days	Rely on less preferred or less expensive foods	0 - 7 days	1	0-56
		Borrow food or rely on help from a friend or relative	0 - 7 days	2	
		Limit portion size at mealtimes	0 - 7 days	1	
		Restrict consumption by adults so children can eat	0 - 7 days	3	
		Reduce number of meals eaten in a day	0 - 7 days	1	
Life Satisfaction Index	Cantril Ladder approach	Please imagine a ladder with steps numbered from 0 at the bottom to 10 at the top. The top of the ladder represents the best	10 Highest to 0 Lowest		

		possible life for you and the bottom of the ladder represents the worst possible life for you. Which step of the ladder best represents the way you personally feel you stand at this time?			
		Now the staircase represents how respected you are in the community. At the top are the most respected people, such as the village chief and religious leaders, and at the bottom are the least respected people, such as the outcasts. Can you point out which step of the stairs you are on?			
		Now the staircase represents how widely your opinion is held in the community. At the top are the people whose opinions are most widely held in the community, and at the bottom are the people whose opinions are least held. Can you tell me which step of the stairs you are on these days?			
Women's Agency	Jayachandran et al. ¹⁰¹	Is your opinion heard when an expensive item like a bicycle or cow is purchased?	1=Never to 6=Yes, always		
		Do you need permission from other household members to buy clothing for yourself?	1=Never to 6=Yes, always		

¹⁰¹ Jayachandran, S., Biradavolu, M., & Cooper, J. (2023). Using machine learning and qualitative interviews to design a five-question survey module for women's agency. *World Development*, 161, Article 106076. <https://doi.org/10.1016/j.worlddev.2022.106076>

		Are you allowed to buy things in the market without asking your partner?	1=Never to 6=Yes, always		
		Are you permitted to visit women in other neighborhoods to talk with them?	1=Never to 6=Yes, always		
		Who do you consult with for decisions regarding your children's health care?	1=Never to 6=Yes, always		
Trust Index	Generalised trust	Generally speaking, would you say that most people can be trusted or that you must be very careful in dealing with people?	1 = Most people can be trusted; 2 = You must be very careful with people; -99 = Don't know; -999 = No answer	n.a.	0-30
	Perception of trust on institutions	On a scale of 1 (not at all confident) to 5 (full confidence): How much confidence do you feel you have in your extended family?	1 = Not at all confident to 4 = Full confident; -98 = Don't know	n.a.	
		On a scale of 1 (not at all confident) to 5 (full confidence): How much confidence do you feel you have in your immediate neighbors?	1 = Not at all confident to 4 = Full confident; -98 = Don't know	n.a.	
		On a scale of 1 (not at all confident) to 5 (full confidence): How much confidence do you feel you have in the people of your bloc?	1 = Not at all confident to 4 = Full confident; -98 = Don't know	n.a.	

		On a scale of 1 (not at all confident) to 5 (full confidence): How much confidence do you feel you have in the chief of your bloc/village?	1 = Not at all confident to 4 = Full confident; -98 = Don't know	n.a.	
		On a scale of 1 (not at all confident) to 5 (full confidence): How much confidence do you feel you have in the government?	1 = Not at all confident to 4 = Full confident; -98 = Don't know	n.a.	
		On a scale of 1 (not at all confident) to 5 (full confidence): How much confidence do you feel you have in the foreigners?	1 = Not at all confident to 4 = Full confident; -98 = Don't know	n.a.	
		On a scale of 1 (not at all confident) to 5 (full confidence): How much confidence do you feel you have in NGOs?	1 = Not at all confident to 4 = Full confident; -98 = Don't know	n.a.	
		On a scale of 1 (not at all confident) to 5 (full confidence): How much confidence do you feel you have in the people from another ethnic group / tribe?	1 = Not at all confident to 4 = Full confident; -98 = Don't know		
Social Cohesion Index	Bossuroy et al. ¹⁰²	You can count on the help of people of your bloc when you have difficulties. Do you agree?	1 = Not at all to 4 = Absolutely;		
		Out of 10 people in the bloc, how many are honest and trustworthy people?	Integer		

¹⁰² Bossuroy, T., Goldstein, M., Karimou, B., Karlan, D., Kazianga, H., Parienté, W., Premand, P., Thomas, C. C., Udry, C., Vaillant, J., & Wright, K. A. (2022). Tackling Psychosocial and Capital Constraints to Alleviate Poverty. *Nature*, 605(7909), 291–297. <https://doi.org/10.1038/s41586-022-04647-8>

		In the last 6 months, has there been tension between members of your bloc?	1 = Yes, a lot to 4 = Not at all		
		How close do you feel to your bloc community?	1 = Independent 4 = Integrated		
Collective Action Index	Bossuroy et al.	How many groups or associations in your bloc (such as youth groups or tontines) have you or any household member been a member of over the last 6 months?	Integer	n.a.	
		How many positions of responsibility you or another family member held in any of these group over the last 6 months?	Integer	n.a.	
		Over the past 6 months, was there any project of any type (e.g. educational, infrastructure, cleaning etc) realized within your bloc, where bloc members could contribute either with donations and/or by volunteering?	1 = Yes; 2 = NO; -98= I don't know		
		How much money, if any, have you or another family member given in donations to bloc projects in the last 6 months?	Integer		
		How many days, if any, have you or other household members volunteered in bloc activities over the last 6 months?	Integer		

		Do you or your household members work with the community of your bloc to achieve common goals?	1 = Not at all to 4 = Absolutely;		
Tensions	Bossuroy et al.	Over the past 6 months, how many times, if any, have you or a family member felt unsafe in your bloc?	1 = Never to 5 = Always; -9 = I don't know	n.a.	
		Over the past 6 months, how many times, if any, have you or a family member feared being a victim of a crime in your own home?	1 = Never to 5 = Always; -9 = I don't know		

A4. Estimation strategy

169. This appendix provides a more detailed description of the specific analysis used to address each research question, under the original design, which was pre-registered in the American Economic Association's RCT Registry (AEARCTR-0010350).

EQ1: How do targeting methods differ in terms of beneficiary coverage?

170. To answer **EQ1**, across targeting methods, the full census data is used. The estimating equation is the following:

$$CVR_b = \alpha CBT_b + \eta_q + w_b \quad (1)$$

171. where CVR_b represents the coverage rate in bloc b . CBT_b is an indicator variable that takes value 1 if bloc b was assigned to the CBT targeting and to zero if it was instead assigned to PMT Plus targeting. Coefficient α captures the average impact of adopting CBT targeting on the coverage rate, with respect to the coverage identified through PMT Plus targeting. η_q represents quadruplet fixed effects (as randomization into the two arms was stratified by that), and w_b is the error term. The unit of analysis is at the bloc level and measures are constructed using the full census data. Results always compute and report robust standard errors.

EQ2. Precision: Which targeting method is more precise in selecting households with the lowest food security?

172. For EQ2, targeting errors are again measured at the bloc level using full census data. Figure 6 estimates Equation 1 with inclusion and exclusion errors as the outcome variables instead of coverage. The corresponding specification can be written as follows:

$$y_b = \beta CBT_b + \eta_q + \varepsilon_b \quad (2)$$

173. where y_b denotes the targeting error outcome, either inclusion or exclusion error, for bloc b . The indicator CBT_b equals one if the bloc is assigned to the CBT targeting method and zero if it is assigned to PMT Plus. The coefficient β therefore captures the average difference in targeting performance between CBT and PMT Plus across blocs. η_q represents quadruplet fixed effects, and ε_b denotes robust standard errors.
174. Because coverage rates vary across targeting methods and mechanically affect targeting error measures, particularly exclusion errors,¹⁰³ these estimates do not isolate differences attributable solely to the targeting approach. To address this, the analysis implements a simulation exercise. Restricting the sample to CBT blocs, the PMT Plus allocation rule is replicated up to the realised CBT coverage level, generating a counterfactual PMT Plus assignment under identical coverage.
175. For each bloc, two observations are constructed: one corresponding to the observed CBT assignment and one corresponding to the simulated PMT Plus assignment. This allows

¹⁰³ Brown, C., Ravallion, M., & van de Walle, D. (2018). A poor means test? Econometric targeting in Africa. *Journal of Development Economics*, 134, 109–124. <https://doi.org/10.1016/j.jdeveco.2018.05.004>

errors to be compared within blocs, isolating differences driven by the targeting method rather than by coverage variation. The corresponding specification can be written as follows:

$$y_b = \gamma CBT_b + \eta_q + \varepsilon_b \quad (3)$$

176. where y_b denotes the targeting error outcome, either inclusion or exclusion error, for bloc b . The indicator CBT_b equals one for the observed CBT assignment and zero for the simulated PMT Plus assignment constructed at identical coverage within the same bloc. The coefficient γ captures the average difference in targeting performance between CBT and PMT Plus within blocs, holding coverage constant. η_q represents quadruplet fixed effects, and ε_b denotes robust standard errors.
177. Heterogeneities in precision are examined by expanding Equation (2) to interact the CBT indicator with two bloc level measures: inequality in food security, proxied by the coefficient of variation (CV) of the FCS, and overall vulnerability, measured as the share of households classified as food insecure (Fshare). The specification can be written as follows:

$$y_b = \lambda CBT_b + \delta(CBT_b \times XTrait_b) + \psi XTrait + \eta_q + \varepsilon_b \quad (4)$$

178. where y_b denotes the targeting error outcome (either inclusion or exclusion error) for bloc b . CBT_b is an indicator variable that takes value 1 if bloc b was assigned to the CBT targeting and to zero if it was assigned to PMT Plus. $XTrait_b$ denotes the bloc level characteristic used to capture heterogeneity (CV or Fshare). The coefficient λ captures the average difference in targeting performance between CBT and PMT Plus when the trait equals zero, while γ is the coefficient of interest and measures how this difference varies with the level of the trait. The parameter δ captures the direct association between the bloc level trait and the targeting error outcome. η_q represents quadruplet fixed effects, and ε_b is the error term. Standard errors are robust.
179. To isolate differences in targeting performance from variation in coverage rates, this analysis controls for coverage. Equation 4 is expanded as follows:

$$y_b = \Phi CBT_b + \delta(CBT_b \times XTrait_b) + \psi XTrait + \kappa CVR_b + \eta_q + \varepsilon_b \quad (5)$$

180. where CVR_b denotes the programme coverage rate in bloc b . The coefficient Φ captures the difference in targeting performance between CBT and PMT Plus conditional on coverage, while δ measures how this difference varies with the bloc level characteristic, holding coverage constant. The parameter ψ captures the direct association between the bloc level characteristic and the targeting error outcome, and κ captures the association between coverage and the targeting error outcome. η_q represents quadruplet fixed effects, and ε_b is the error term. Standard errors are robust.

EQ3. Profiles: How do targeting methods differ in the types of households they select, and do these differences lead to systematic exclusion or over-targeting of specific population subgroups?

181. To examine EQ3, which assesses whether the two targeting approaches select different household profiles, the analysis first simulates PMT Plus selection within CBT blocs as in EQ2. T tests are then used to compare baseline characteristics of selected households across methods, and the ten characteristics with the largest statistically significant differences are retained for presentation.

182. To further assess whether either approach systematically over or under selects specific socioeconomic groups relative to their actual vulnerability, a relative representation indicator, denoted RR_{gb} , is constructed.
183. Let g index a socioeconomic subgroup, such as households primarily engaged in agriculture, widow headed households, or households headed by a person with a disability. Let B_b denote the total number of beneficiaries in bloc b , and FI_b the total number of food insecure households in bloc b . For subgroup g in bloc b , let $Selected_{gb}$, denote the number of selected households belonging to subgroup g , and $Vulnerable_{gb}$, the number of food insecure households in subgroup g . The relative representation index is then defined as the ratio of the subgroup's share among beneficiaries to its share among food insecure households:

$$RR_{gb} = (Selected_{gb}/B_b) / (Vulnerable_{gb}/FI_b) \quad (6)$$

EQ4. Welfare: Do the differences in targeting outcomes translate into differences in welfare for the target populations?

EQ5. Community acceptance and cohesion: Which targeting method leads to higher community satisfaction, trust, and social cohesion?

184. Household-level outcomes (EQ4 and EQ5) are estimated using data from the first and second follow-up rounds. Both specifications are estimated using the following equation:

$$y_{ib} = \lambda CBT_b + \theta Z'_i + \eta_q + \Omega_{ib} \quad (7)$$

185. where now y_{ib} represents an outcome of interest (e.g. FCS, rCSI, satisfaction, social cohesion) for household i residing in bloc b . The coefficient λ captures the (intention-to-treat) effect of living in a bloc assigned to CBT rather than to PMT Plus. Z_{ib} is a vector of control variables taken from the baseline census and selected through a double-lasso estimation¹⁰⁴, which always includes the baseline values of the outcome y as non-penalized variable. η_q are again the quadruplet fixed effects, and Ω_{ib} is the error term. In the household-level analysis, standard errors are clustered at the bloc level, which corresponds to the unit of randomization.
186. As described in Section 3, the survey sampled equal numbers of eligible and non-eligible households. Given the high share of eligible households in the study areas, this design over represents non-eligible households in the sample. The analysis therefore applies sampling weights to recover population representative estimates.
187. Because programme coverage differs across targeting methods, the analysis also estimates a specification that controls for bloc-level coverage:

$$y_{ib} = \tau CBT_b + \kappa CVR_b + \theta Z'_i + \eta_q + \Omega_{ib} \quad (8)$$

188. where CVR_b denotes the programme coverage rate in bloc b . The coefficient τ captures the difference in outcomes between CBT and PMT Plus conditional on coverage, while κ captures the association between bloc-level coverage and household outcomes.

¹⁰⁴ Urmitsky, O., Hansen, C., & Chernozhukov, V. (2016). Using Double-Lasso Regression for Principled Variable Selection. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.2733374>

189. The analysis also examines the heterogeneity of impact across various initial bloc-level and household-level dimensions. This is implemented by augmenting Equation (7) with an interaction between treatment assignment and the characteristic of interest, while including the main effect of that characteristic in the regression. The estimating equation is:

$$y_{ib} = \rho CBT_b + \Pi(CBT_b \times H_b) + \Gamma H_b + \theta Z'_i + \eta_q + \Omega_{ib} \quad (9)$$

190. where H_b denotes a bloc-level dummy capturing the heterogeneity dimension considered, namely bloc size, realised coverage rate, or the share of food insecure households, and zero otherwise. The coefficient Π is the coefficient of interest and captures whether the impact of CBT relative to PMT Plus differs in blocs with higher levels of H_b . η_q are the quadruplet fixed effects, and Ω_{ib} is the error term. The analysis uses standard errors clustered at the bloc level and applies weights to recover population representative estimates.
191. Programme impact presented in section 5.4 follows a difference-in-differences (DiD) approach that can be applied to outcomes measured both before and after the intervention. Since the FCS was the only outcome collected in the baseline census, it serves as the sole focus for assessing the programme's impact on beneficiaries.
192. The estimation of the DiD follows the specification presented in the equation below, comparing changes in outcomes over time between beneficiary and non-beneficiary households, conditional on baseline eligibility.

$$Y_{it} = \alpha + \tau(Eligible_i \times Post_t) + \mu_t + \lambda_i + \psi_{it} \quad (10)$$

193. where $Eligible_i$ indicates whether household i was eligible for assistance at baseline, and $Post_t$ is a dummy equal to one in the follow-up period. τ represents the DiD estimator and captures the average treatment effect on the treated under the assumptions of parallel trends, no anticipatory effects, and independent sampling. μ_t is unit fixed effects and λ_i time fixed effects. ψ_{it} is the error term.

EQ6. Cost-effectiveness: Which targeting method is more cost-effective?

194. EQ6 evaluates the relative budgetary reach of PMT Plus and CBT under alternative budget scenarios. Given the absence of meaningful differences in welfare outcomes between the two methods, the comparison focuses on programme costs and the number of blocs that can be covered with a fixed budget.
195. For each targeting approach mmm , the number of households assisted in an average bloc is given by $N_m = H_{km}$ where H is average bloc size and km is the observed coverage rate.
196. Transfer costs are computed as $TC_m = \tau N_m'$, with τ denoting the transfer value per assisted household, while targeting costs are given by $SC_m = c_m H$, where c_m is the targeting cost per screened household.
197. Total cost per bloc is, thus, $C_m = TC_m + SC_m$. The number of blocs that can be supported under budget B is then $G_m = \frac{B}{C_m}$. This simulation captures the trade-off between within bloc coverage and geographic expansion across targeting methods.

A5. Additional details and analyses on coverage

Table A5.1: Impact of CBT on coverage rate

	PMT Plus (mean)	CBT
	(1)	(2)
Panel A: Main results		
Coverage Rate	0.794	-0.118*** (0.036)
Panel B: Dropping IDP blocs with perfect coverage		
Coverage Rate	0.799	-0.108*** (0.035)

*Note: Regressions are done at the bloc level, which is the level which indicators are estimated. Regressions include quadruplet fixed effects, with robust standard errors reported in parentheses. Column (1) reports the mean of the outcome for PMT Plus. Column (2) reports the CBT coefficient from regressions without controls. Coverage rate is defined as the share of food-insecure households in a bloc receiving transfers. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

A6. Additional details and analyses on precision

Precision error rates

Table A6.1: Impact of CBT on precision error rates

	PMT Plus (mean)	CBT	CBT
	(1)	(2)	(3)
Panel A: Main results			
Exclusion Error Rate	0.189	0.106*** (0.034)	0.009 (0.014)
Inclusion Error Rate	0.11	0.017 (0.019)	0.014 (0.021)
Panel B: Dropping IDP blocs with perfect coverage			
Exclusion Error Rate	0.193	0.116*** (0.035)	0.011 (0.010)
Inclusion Error Rate	0.109	0.020 (0.020)	0.018 (0.026)
Controls: coverage rate			Yes

*Note: Regressions are done at the bloc level, which is the level which indicators are estimated. Regressions include quadruplet fixed effects, with robust standard errors reported in parentheses. Column (1) reports the mean of the outcome for PMT Plus. Column (2) reports the CBT coefficient from regressions without controls. Coverage rate is defined as the share of food-insecure households in a bloc receiving transfers. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

Table A6.2: Precision error rates from selected studies

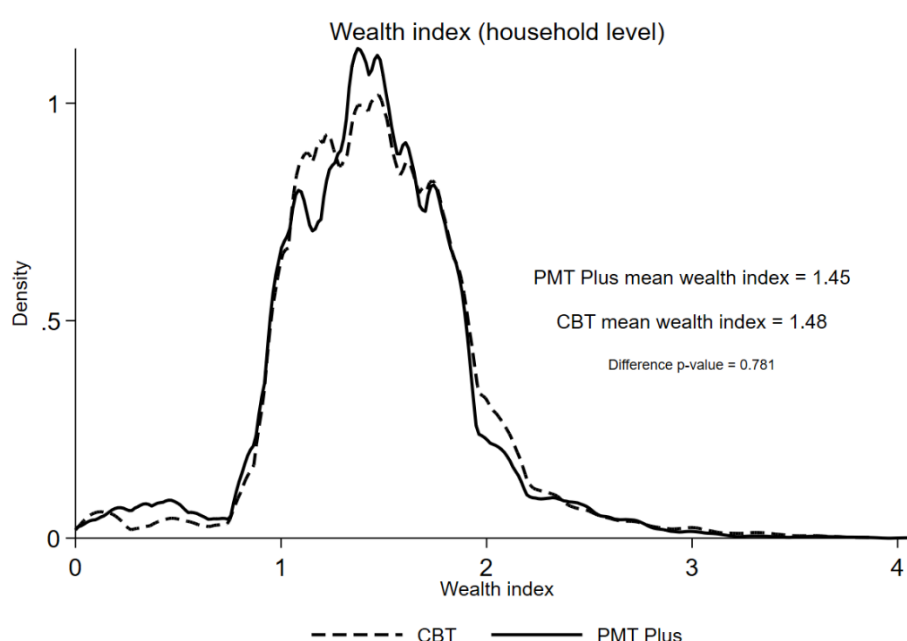
Literature	Country of Study	Targeting methods	Inclusion Error (Mean)	Exclusion Error (Mean)
Alatas et al. 2012	Indonesia	PMT	18%	
		Community	23%	
		Hybrid	22%	
Hillebrecht et al., 2020	Burkina Faso	Random	22%	78%
		PMT	15%	53%
		Assets	17%	62%
		PSI	18%	65%
		MPI	19%	69%
		Community	19%	67%
Stoeffler Q., Mills B., del Ninno C. (2016)	Cameroon	PMT	26%	47%
		Community	21%	17%
Sabates-Wheeler et al. (2015)	Kenya	Categorical by age	4%	17%
		Categorical by dependency ratio (DR)	30%	23%
Premand & Schnitzer (2020)	Niger	Community	42%	54%
		PMT	39%	54%
		FCS (food consumption score)	45%	59%
Brown et al. (2018)	Burkina Faso	PMT	41%	67%
	Ethiopia	PMT	45%	92%
	Ghana	PMT	32%	54%
	Malawi	PMT	41%	77%
	Mali	PMT	43%	81%
	Niger	PMT	51%	74%
	Nigeria	PMT	30%	46%
	Tanzania	PMT	36%	67%
	Uganda	PMT	40%	62%
	Mean	PMT	37%	72%

Altindag et al. (2021)	Lebanon	PMT	31%	27%
Basurto et al. (2020)	Malawi	PMT		14%
		Community		15%
Stoeffler and Schnitzer (2024)	Burkina Faso 1	PMT		27%
	Burkina Faso 2	PMT		54%
	Cameroon	PMT		24%
	Mali	PMT		27%
	Niger 1	PMT		48%
	Niger 2	PMT		36%
	Senegal 1	PMT		36%
	Senegal 2	PMT		41%
	Burkina Faso 1	CBT		36%
	Burkina Faso 2	CBT		69%
	Cameroon	CBT		34%
	Mali	CBT		32%
	Senegal 1	CBT		55%
	Senegal 2	CBT		50%
Average			40%	53%
			30%	53%
CI: lower bound			25%	45%
CI: upper bound			35%	63%

Targeting progressivity

198. The binary classification of households as either food secure or food insecure based on a fixed FCS threshold can obscure the severity of need and misrepresent targeting accuracy, for example, when households cluster just above the FCS threshold of 28.
199. To address this, the study uses the CGH and TD to assess the progressivity¹⁰⁵ of targeting, or the degree to which assistance is concentrated among the more food-insecure households. The CGH, proposed by Coady, Grosh, and Hoddinott (2004), and the TD, adapted from Galasso and Ravallion (2005), assess targeting performance using distinct but complementary approaches to evaluate how progressive each method is beyond a single cut-off.

Figure A6.1: Wealth index at baseline by targeting method



Note: The figure presents kernel density estimates of the household wealth index by targeting arm. The wealth index is constructed using principal component analysis (PCA) on 11 variables capturing assets, housing conditions, and access to basic services. Means are 1.45 for PMT Plus and 1.48 for CBT. The reported p-value corresponds to the difference in means from a regression with standard errors clustered at the block level.

200. In this context, “progressive” refers to the extent to which programme benefits are disproportionately directed toward more food-insecure households relative to random selection (CGH) or to better-off households (TD). The CGH evaluates whether the share of total transfers accruing to food-insecure households exceeds their population share, thereby capturing how concentrated benefits are among the vulnerable relative to a neutral allocation. In contrast, the TD measures the difference in participation rates between food-

¹⁰⁵ A targeting approach is considered progressive when it allocates a larger share of assistance to more food insecure households relative to a random allocation in the case of the CGH or when the proportion of food insecure households correctly included in the programme exceeds the proportion of non-food-insecure households incorrectly included in the case of the TD.

insecure and non-food-insecure households, capturing whether the vulnerable are more likely to be included in the program.

201. Additionally, the study uses a wealth index as an alternative benchmark to FCS to ensure robustness. The correlation between the standardized wealth index and the standardized FCS is just 19 percent, suggesting that these two measures capture distinct dimensions of vulnerability. Figure A6.1 shows the asset index distribution at baseline, which confirms similar distribution across the two study arms.
202. One issue with using FCS to assess targeting precision of the two methods is that, whereas the PMT Plus model (specifically the PMT component) explicitly targeted food insecurity based on the FCS, local committees in CBT blocs may have adopted a different definition of vulnerability to target assistance within their communities, potentially grounded on local understanding of food insecurity or need. Consequently, defining targeting precision based on FCS, may “mechanically” favour PMT Plus.
203. The index is constructed following standard practice by using Principal Component Analysis (PCA), a statistical technique that summarises information from a larger set of correlated variables into a smaller number of uncorrelated components that capture the greatest share of variance.¹⁰⁶
204. Eleven variables were selected after estimating the index and examining eigenvalues,¹⁰⁷ which measure the amount of variation in the data explained by each underlying component, for different combinations of variables. Variables associated with greater variation across households were retained. Specifically, the variables included were: ownership of agricultural tools, poultry, cooking pans, a mobile phone, a lamp, a radio, and a table, as well as dummy variables indicating access to sanitary facilities, access to an improved water source, use of wood as cooking fuel, and housing made of cement brick.
205. Table A6.3 presents the results of the CGH and TD analyses based on the FCS and the wealth index. The results show that looking at the asset poor rather than the food insecure yields broadly comparable conclusions. The differences between PMT-Plus and CBT remain statistically insignificant, again indicating similar ability to reach asset-poor households.
206. The CGH values are also similar across groups and indicate only a modest advantage in directing assistance toward worse-off households: at the 20- and 40-percent coverage thresholds, PMT-Plus records CGH values below one, meaning that it does not succeed in identifying asset-poor households above what would be expected from a random draw.

¹⁰⁶ Filmer, D., & Pritchett, L. H. (2001). Estimating Wealth Effects without Expenditure Data-or Tears: An Application to Educational Enrollments in States of India. *Demography*, 38(1), 115–132. <https://doi.org/10.2307/3088292>

¹⁰⁷ See for more details on PCA and eigenvalues <https://wiki.pathmind.com/eigenvector#principal>

Table A6.3: Measures of targeting progressivity

	FCS			Wealth Index		
	PMT Plus	CBT	Differential effect	PMT Plus	CBT	Differential effect
Panel A: Simulating coverage						
CGH Index	1.023	1.02	-0.003 (0.908)	1.002	1.116	0.114 (0.035)
CGH Index bottom 10%	1.102	1.027	-0.075 (0.584)	1.03	1.05	0.02 (0.808)
CGH Index bottom 20%	1.074	1.001	-0.073 (0.331)	0.959	1.062	0.104 (0.088)
CGH Index bottom 40%	1.056	1.014	-0.042 (0.406)	0.964	1.019	0.055 (0.273)
Targeting Differential	423.992	390.022	-33.97 - (0.629)	-63.707	-63.36	0.347 (0.993)
Panel B: Without simulation						
CGH Index	1.012	1.03	0.018 (0.500)	0.979	1.14	0.161 (0.003)
CGH Index bottom 10%	1.075	1.064	-0.011 (0.930)	0.991	1.088	0.098 (0.224)
CGH Index bottom 20%	1.069	1.018	-0.051 0.448	0.958	1.063	0.106 (0.062)
CGH Index bottom 40%	1.042	1.035	-0.006 (0.894)	0.958	1.025	0.067 (0.155)
Targeting Differential	388.471	425.543	37.072	-61.841	-65.226	-3.385
CGH Index			(0.592)			(0.927)

*Note: Panel A reports estimates from specifications that control for coverage, while Panel B reports estimates from specifications that do not control for coverage. The differential effect is calculated by regressing each targeting performance measure, namely the Coady, Grosh and Hoddinott (CGH) Index – its distributional variations (10 percent, 20 percent, and 40 percent) – and the Targeting Differential (TD), on a treatment indicator for blocs assigned to community-based targeting (CBT), while controlling for quadruplet fixed effects. Values in parentheses report p-values. On average more than 85 percent of households per bloc are food insecure, therefore, for the targeting method to be considered progressive (CGH>1), it must allocate more than 85 percent of resources to food-insecure households within this group. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

Heterogeneity analysis

207. A potential source of heterogeneity relates to differences in the distribution of food security across the population. To assess this, there are two metrics one could consider.
208. First is inequality in food security, measured by the coefficient of variation (CV)¹⁰⁸ of the FCS, which captures the extent of variation in FCS among households within each bloc. In the standard analysis (where coverage rates are not fixed), no meaningful differences in targeting performance are observed across blocs based on CV. The results of this analysis
209. Second is the overall level of vulnerability in the bloc, measured by the share of households classified as food insecure (FCS $\leq$ 28). When comparing differences based on bloc-level vulnerability, a modest difference in exclusion error emerges. Areas with a higher share of food-insecure households relative to PMT Plus have an exclusion error rate that is 0.70 percentage points higher.
210. Overall, neither analysis reveals meaningful variation in performance across these dimensions, when coverage rates are controlled for. Appendix Tables A6.4 and A6.5 report the results for exclusion and inclusion errors across blocs by food security inequality (CV) and overall vulnerability (share of food-insecure households), respectively.

Table A6.4: Performance heterogeneity by coefficient of variation

	PMT Plus (mean)	CBT	CV	CBT*CV
	(1)	(2)	(3)	(4)
Panel A: without controlling for coverage				
Exclusion Error Rate	0.189	0.310	-0.231	-0.456
		(0.222)	(0.377)	(0.468)
Inclusion Error Rate	0.11	-0.117	-0.532**	0.290
		(0.123)	(0.211)	(0.257)
Precision	0.89	0.117	0.532**	-0.290
		(0.123)	(0.211)	(0.257)
Panel B: controlling for coverage				
Exclusion Error Rate	0.189	0.087	0.013	-0.17
		(0.065)	(0.090)	(0.142)

¹⁰⁸ The CV is defined as the ratio of the standard deviation to the mean, $CV = \sigma/\mu$. It provides a unitless measure of relative variability, allowing comparison of dispersion across variables with different scales. It is usually expressed as a percentage ($CV \times 100$) (Everitt, B. S., & Skrondal, A. (2010). The Cambridge dictionary of statistics (4th ed.). Cambridge University Press.).

Inclusion Error Rate	0.11	-0.114	-0.535**	0.287
		(0.134)	(0.211)	(0.272)
Precision	0.89	0.114	0.535**	-0.287
		(0.134)	(0.211)	(0.272)

*Note: Estimates come from regressions at the bloc level, which corresponds to the level at which the precision indicators are estimated. Regressions include quadruplet fixed effects, with standard errors clustered at the bloc level and reported in parentheses. Panel A reports specifications in which coverage rates are allowed to vary across targeting approaches, while Panel B reports specifications that control for differences in coverage. The inclusion error rate is the complementary measure that captures the share of selected households that are not food insecure. The exclusion error rate measures instead the share of food insecure households that were not selected, calculated as the ratio of excluded food insecure households over the total number of food insecure households. The coefficient of variation (CV) is computed from the census data as the standard deviation of the baseline FCS divided by its mean within each bloc. Column 1 reports control means at census. Column 2 shows the coefficient on the indicator for selection through community-based targeting. Column 3 reports the coefficient on the CV of the FCS, calculated at the bloc level using census data. Column 4 reports the coefficient on the interaction term, which captures how the relative performance of CBT changes with this continuous measure of CV. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

Table A6.5: Performance heterogeneity by share of food-insecure households

	PMT Plus (mean)	CBT	Share of Food Insecure HHs	CBT*Share of Food-Insecure HHs
	(1)	(2)	(3)	(4)
Panel A: without controlling for coverage				
Exclusion Error Rate	0.189	-0.507	-0.469**	0.703*
		(0.334)	(0.199)	(0.388)
Inclusion Error Rate	0.11	-0.149	-0.779***	0.174
		(0.240)	(0.082)	(0.271)
Precision	0.89	0.149	0.779***	-0.174
		(0.240)	(0.082)	(0.271)
Panel B: controlling for coverage				
Exclusion Error Rate	0.189	-0.189	0.167**	0.231
		(0.237)	(0.071)	(0.272)
Inclusion Error Rate	0.11	-0.182	-0.846***	0.224
		(0.238)	(0.076)	(0.274)
Precision	0.89	0.182	0.846***	-0.224

(0.238)

(0.076)

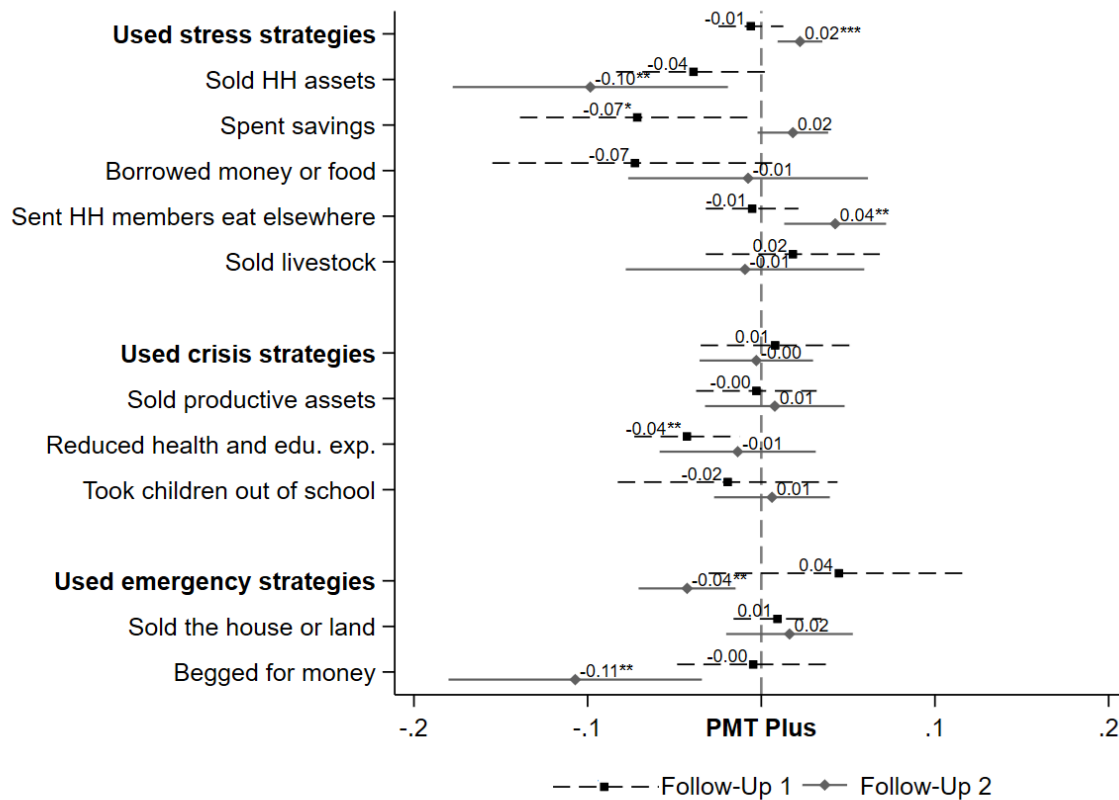
(0.274)

*Note: Estimates come from regressions at the bloc level, which corresponds to the level at which the precision indicators are estimated. Regressions include quadruplet fixed effects, with standard errors clustered at the bloc level and reported in parentheses. Panel A reports specifications in which coverage rates are allowed to vary across targeting approaches, while Panel B reports specifications that control for differences in coverage. The inclusion error rate is the complementary measure that captures the share of selected households that are not food insecure. The exclusion error rate measures instead the share of food insecure households that were not selected, calculated as the ratio of excluded food insecure households over the total number of food insecure households. The share of food-insecure households in each bloc is computed from the census data as the number of households with FCS below 28 divided by the total number of households in the bloc. Column 1 reports control means at census. Column 2 shows the coefficient on the indicator for selection through community-based targeting. Column 3 reports the coefficient on the share of food-insecure households in the bloc. Column 4 reports the coefficient on the interaction term, which captures how the relative performance of CBT changes with this continuous measure of baseline food insecurity. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

A7. Additional details and analyses on welfare impacts

Additional figures and tables on the main welfare analysis

Figure A7.1: rCSI components



Note: The figure shows the estimated impact of being assigned to CBT (ITT) blocs for both beneficiaries and non-beneficiaries on the likelihood of adopting livelihood-related coping strategies. Each point reflects the change in probability for households in CBT blocs, with values to the right indicating increased use and values to the left reduced use relative to those in PMT Plus blocs. Control variables were selected using post-double selection Lasso, and always include quadruplet fixed effects, with analytical weights applied to account for sampling and non-response. Standard errors are clustered at the bloc (treatment) level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7.1: Impact of CBT on primary welfare outcomes

	PMT Plus (mean)	CBT	PMT Plus (mean)	CBT
	FU 1	FU1	FU2	FU2
	(1)	(2)	(3)	(4)
Panel A: Main results				
Food Consumption Score	18.118	-1.956	14.561	0.394
		-1.221		-0.493
reduced Coping Strategies Index	21.182	-0.72	25.198	0.327
		-0.89		-0.721
Used stress coping strategy	0.966	-0.006	0.952	0.022***
		-0.011		-0.008
Used emergency coping strategy	0.618	0.045	0.711	-0.043**
		-0.046		-0.017
Used crisis coping strategy	0.754	0.008	0.774	-0.003
		-0.026		-0.02
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.561	0.703	10.825	-0.164
		-0.435		-0.337
Women's Agency (Index standard deviations)	-0.031	0.068	-0.015	0.076*
		-0.067		-0.042
Panel B: Controlling for coverage	FU 1	FU1	FU2	FU2
Food Consumption Score	18.118	-3.172***	14.561	0.085
		-0.905		-0.467
reduced Coping Strategies Index	21.182	-1.192	25.198	0.698
		-0.911		-0.757
Used stress coping strategy	0.966	-0.005	0.952	0.022**
		-0.013		-0.009
Used crisis coping strategy	0.618	0.081*	0.711	-0.046***
		-0.043		-0.016
Used emergency coping strategy	0.754	0.028	0.774	-0.01
		-0.02		-0.019
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.561	0.473	10.825	-0.062
		-0.368		-0.258

Women's Agency (Index standard deviations)	-0.031	0.054 -0.071	-0.015	0.083* -0.043
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*Note: Columns (1) and (3) report PMT Plus control means at follow-up 1 and follow-up 2, respectively. Estimates include quadruplet fixed effects and design weights, with standard errors clustered at the block level reported in parentheses. In panel B, all specifications additionally control for the baseline cluster-level coverage rate. The Food Consumption Score (FCS) ranges from 0 to 112. The food groups used for the FCS are: cereals, grains, roots and tubers; legumes and nuts; milk and other dairy products; meat, fish and eggs; vegetables and leaves; fruits; oil, fat and butter; and sugar. For the FCS, we control for households that were randomly assigned a recall period other than seven days during the follow-up 2. The reduced Coping Strategies Index (rCSI) follows WFP guidelines to measure the frequency and severity of coping behaviours related to food consumption. Life Satisfaction was constructed using the Cantril Ladder approach as the row total of three subjective well-being (SWB) questions, each measured on a 1–10 ladder scale, yielding a score that ranges from 3 to 30. The Women's Agency Index follows Jayachandran et al. (2023) and is computed as the standardized (z-score) mean of its components' z-scores. *, ** and *** denote statistical significance at 10%, 5%, and 1%.*

Table A7.2: Impact of CBT on Life Satisfaction

	PMT Plus (mean)	CBT	PMT Plus (mean)	CBT
	FU 1	FU 1	FU 2	FU 2
	(1)	(2)	(3)	(4)
Life Satisfaction (Index standard deviations)	-0.037	0.157 (0.098)	0.001	-0.027 (0.069)
Where you feel your life stands (MacArthur Scale: 0–10, 10 = best)	3.019	0.185 (0.183)	2.655	-0.086 (0.110)
How respected you feel in community (MacArthur Scale: 0– least, 10–most)	4.827	0.275* (0.157)	4.165	-0.141 (0.158)
How widely your opinion is held (MacArthur Scale: 0–least, 10– most)	4.714	0.332 (0.203)	4.005	0.105 (0.130)

*Note: Columns (1) and (3) report control means. Estimates include control variables selected using post-double selection Lasso and quadruplet fixed effects. Analytical weights are applied to account for sampling design and non-response. Standard errors are clustered at the bloc (treatment) level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

Table A7.3: Impact of CBT on the women's agency index and its components

	PMT Plus (mean)	CBT	PMT Plus (mean)	CBT
	FU 1	FU 1	FU 2	FU 2
	(1)	(2)	(3)	(4)
Women's Agency (Index standard deviations)	-0.031	0.068 (0.067)	-0.015	0.076* (0.042)
Opinion heard when an expensive item is purchased (Scale:1=Never, 6=Yes, always)	4.431	-0.020 (0.072)	4.540	-0.068 (0.053)
Doesn't need permission to buy clothing	0.409	0.017 (0.024)	0.344	-0.027 (0.029)
Allowed to go to the market w/o asking partner (Scale:1=Never, 6=Yes, always)	3.975	0.060 (0.105)	3.627	0.035 (0.098)
Allowed to visit/talk to other women (Scale:1=Never, 6=Yes, always)	2.727	-0.027 (0.024)	2.849	0.085* (0.045)
Allowed to visit using public transport (Scale:1=Never, 6=Yes, always)	2.553	-0.042 (0.055)	2.713	0.158* (0.081)
Allowed to go alone to meet your friends	0.665	0.048 (0.042)	0.703	0.066** (0.027)
Decides by yourself to purchase emergency medicine for child	0.904	0.003 (0.009)	0.911	0.008 (0.012)
Allowed to go alone to a relative's house inside the village	0.869	0.015 (0.025)	0.833	-0.007 (0.011)

*Note: Columns (1) and (3) report control means. Estimates include control variables selected using post-double selection Lasso and quadruplet fixed effects. Analytical weights are applied to account for sampling design and non-response. Standard errors are clustered at the bloc (treatment) level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

Table A7.4: Robustness checks on welfare outcomes

	PMT Plus (mean)	CBT	PMT Plus (mean)	CBT
	FU 1	FU 1	FU 2	FU 2
	(1)	(2)	(3)	(4)
Panel A: Dropping blocs with security issues				
Food Consumption Score	18.438	-2.105 (1.290)	14.273	0.512 (0.470)
Reduce Coping Strategy Index	21.311	-0.86 (0.938)	25.704	0.292 (0.768)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.751	0.708 (0.465)	11.027	-0.174 (0.370)
Women's Agency (Index standard deviations)	-0.019	0.072 (0.072)	-0.02	0.084 (0.045)
Panel B: Controlling for imperfect programme adherence				
Food Consumption Score	18.118	-1.927 (1.219)	14.561	0.442 (0.444)
Reduce Coping Strategy Index	21.182	-0.751 (0.908)	25.198	0.323 (0.731)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.561	0.708 (0.441)	10.825	-0.162 (0.336)
Women's Agency (Index standard deviations)	-0.031	0.07 (0.069)	-0.015	0.075 (0.042)
Panel C: Dropping IDP blocs with perfect coverage				
Food Consumption Score	18.061	0.393 (0.639)	14.613	0.012 (0.389)
Reduce Coping Strategy Index	21.103	1.092 (0.521)	25.14	-0.164 (0.534)

Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.636	0.183 (0.207)	10.759	-0.149 (0.285)
Women's Agency (Index standard deviations)	-0.022	-0.01 (0.048)	-0.012	0.041 (0.050)

*Note: Columns (1) and (3) report control means. Estimates include quadruplet fixed effects and design weights, with standard errors clustered at the bloc level and reported in parentheses. For the Food Consumption Score, the model additionally controls for non-seven-day recall at the second follow-up. Regressions for Panel A drop blocs affected by security issues. Regressions for Panel B include a dummy variable that indicates whether the household received the planned transfer. Regressions from Panel C exclude the three IDP blocs with perfect coverage. The symbols *, ** and *** denote statistical significance at 10%, 5%, and 1% levels, respectively.*

Heterogeneity analysis

211. The two targeting methods are compared based on whether their relative impact on household welfare varies with the size of the bloc, the level of coverage, and the level of food insecurity (as captured through the census). Specifically, the analysis tests for heterogeneous effects across:

- Table A7.5: blocs that are larger or smaller than the median bloc size in the sample (60 households),
- Table A7.6: blocs where coverage is above or below the median coverage rate (79 percent), and
- Table A7.7: blocs where the share of food-insecure households recorded in the census is higher than the median (87 percent).

Table A7.5: Welfare heterogeneity by bloc size (less than median)

	PMT Plus (mean)	CBT	Cluster size (<median)	CBT x Cluster size (<median)
	(1)	(2)	(3)	(4)
Panel A: First follow-up	FU 1	FU1	FU1	FU1
Food Consumption Score	18.118	0.830***	1.829***	-3.593***
		(0.799)	(1.276)	(1.361)
Reduce Coping Strategy Index	21.182	-0.64	-0.302	-0.612
		(0.804)	(1.113)	(1.315)
Used stress coping strategy	0.966	-0.028	0.008	0.025
		(0.012)	(0.015)	(0.019)
Used crisis coping strategy	0.754	-0.075**	0.015**	0.106**
		(0.030)	(0.034)	(0.046)
Used emergency coping strategy	0.618	-0.091**	-0.079**	0.179**
		(0.055)	(0.074)	(0.078)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.561	0.452	-0.571	0.434
		(0.372)	(0.482)	(0.616)

Women's Agency (Index standard deviations)	-0.031	0.028	0.04	0.051
		(0.058)	(0.087)	(0.106)

Panel B: Second follow-up	FU 2	FU2	FU2	FU2
Food Consumption Score	14.561	0.983	0.635	-0.663
		(0.498)	(0.983)	(0.762)
Reduce Coping Strategy Index	25.198	-0.136	-1.756	0.545
		(0.901)	(1.269)	(1.291)
Used stress coping strategy	0.952	-0.017***	-0.003***	0.049***
		(0.009)	(0.014)	(0.012)
Used crisis coping strategy	0.774	0.012	0.047	-0.018
		(0.019)	(0.029)	(0.032)
Used emergency coping strategy	0.711	-0.013	0.093	-0.036
		(0.025)	(0.026)	(0.032)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	10.825	0.211	1.044	-0.448
		(0.319)	(0.472)	(0.543)
Women's Agency (Index standard deviations)	-0.015	-0.038*	0.093*	0.145*
		(0.059)	(0.086)	(0.080)

Note: Panel A reports results for the first follow up and Panel B for the second follow up. Sample sizes range from 3031 to 3031 households at FU1. Sample sizes range from 3875 to 3960 households at FU2. The Food Consumption Score (FCS) ranges from 0 to 112. The reduced Coping Strategies Index (rCSI) follows WFP guidelines to measure the frequency and severity of coping behaviours related to food consumption. Both the FCS and rCSI were standardized. Life Satisfaction was constructed using the Cantril Ladder approach, aggregating responses from three subjective well-being (SWB) questions. The Women's Agency Index follows Jayachandran et al. (2023). Index standard deviation is computed as the standardized (z-score) mean of its components' z-scores. Cluster size is a standardized binary variable equal to one for blocs with a population size above the median, i.e., 60 households. Regressions with both a binary and a continuous outcome variable use an ordinary least squares (OLS) regression model. All specifications include a dummy variable

indicating whether the household received fewer than four transfers. For the FCS, the model additionally controls for non-seven-day recall at follow-up 2. *, ** and *** denote statistical significance at 10%, 5%, and 1%.

Table A7.6: Welfare heterogeneity by coverage rate (less than median)

	PMT Plus (mean)	CBT	Share of food insecure HHs (<median)	CBT x Coverage rate simulated (<median)
	(1)	(2)	(3)	(4)
Panel A: First follow-up	FU 1	FU1	FU1	FU1
Food Consumption Score	18.118	1.019***	-2.162***	-4.188***
		(0.797)	(0.880)	(1.076)
Reduce Coping Strategy Index	21.182	0.637*	2.606*	-2.520*
		(1.040)	(1.137)	(1.487)
Used stress coping strategy	0.966	-0.026*	-0.017*	0.032*
		(0.018)	(0.017)	(0.018)
Used crisis coping strategy	0.754	-0.042**	0.043**	0.077**
		(0.033)	(0.028)	(0.039)
Used emergency coping strategy	0.618	0.034	0.129	0.011
		(0.070)	(0.076)	(0.099)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.561	0.276	-1.598	0.888
		(0.442)	(0.651)	(0.741)
Women's Agency (Index standard deviations)	-0.031	-0.056*	-0.253*	0.212*
		(0.065)	(0.068)	(0.110)
Panel B: Second follow-up	FU 2	FU2	FU2	FU2
Food Consumption Score	14.561	0.614	-1.077	-0.167
		(0.649)	(0.564)	(0.706)
Reduce Coping Strategy Index	25.198	-0.958*	-0.042*	2.040*

		(0.795)	(0.902)	(1.163)
Used stress coping strategy	0.952	0.005**	-0.019**	0.030**
		(0.009)	(0.011)	(0.013)
Used crisis coping strategy	0.774	-0.062**	-0.066**	0.100**
		(0.033)	(0.034)	(0.042)
Used emergency coping strategy	0.711	-0.031	0.004	-0.018
		(0.035)	(0.026)	(0.043)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	10.825	-1.098**	-0.601**	1.540**
		(0.538)	(0.339)	(0.606)
Women's Agency (Index standard deviations)	-0.015	0.071	-0.126	0.101
		(0.092)	(0.060)	(0.109)

Note: Panel A reports results for the first follow up and Panel B for the second follow up. Sample sizes range from 3031 to 3031 households at FU1. Sample sizes range from 3875 to 3960 households at FU2. The Food Consumption Score (FCS) ranges from 0 to 112. The reduced Coping Strategies Index (rCSI) follows WFP guidelines to measure the frequency and severity of coping behaviours related to food consumption. Both the FCS and rCSI were standardized. Life Satisfaction was constructed using the Cantril Ladder approach, aggregating responses from three subjective well-being (SWB) questions. The Women's Agency Index follows Jayachandran et al. (2023). Index standard deviation is computed as the standardized (z-score) mean of its components' z-scores. Coverage rate is a binary variable equal to one for clusters where the share of the population selected for assistance, based on a simulated application of the PMT PLUS formula across both arms, is above the median. Regressions with both a binary and a continuous outcome variable use an ordinary least squares (OLS) regression model. All specifications include a dummy variable indicating whether the household received fewer than four transfers. For the FCS, the model additionally controls for non-seven-day recall at follow-up 2. *, ** and *** denote statistical significance at 10%, 5%, and 1%.

Table A7.7: Welfare heterogeneity by share of food-insecure households (less than median)

	PMT Plus (mean)	CBT	Share of food insecure HHs (<median)	CBT x Share of food insecure HHs (<median)
	(1)	(2)	(3)	(4)
Panel A: First follow-up	FU 1	FU1	FU1	FU1
Food Consumption Score	18.118	-1.369	-1.42	-0.957
		(1.605)	(1.659)	(2.317)
Reduce Coping Strategy Index	21.182	-0.902	-1.307	-0.056

		(0.978)	(1.295)	(1.559)
Used stress coping strategy	0.966	-0.009	0.029	-0.002
		(0.019)	(0.020)	(0.029)
Used crisis coping strategy	0.754	-0.015	0.019	0.046
		(0.045)	(0.045)	(0.066)
Used emergency coping strategy	0.618	0.046	0.113	-0.016
		(0.046)	(0.057)	(0.068)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.561	1.250*	0.670*	-1.127*
		(0.473)	(0.505)	(0.581)
Women's Agency (Index standard deviations)	-0.031	0.088	0.104	-0.067
		(0.086)	(0.088)	(0.114)

Panel B: Second follow-up		FU 2	FU2	FU2	FU2
Food Consumption Score	14.561	0.811	-1.025	-0.496	
		(0.664)	(0.645)	(0.986)	
Reduce Coping Strategy Index	25.198	-1.035***	0.610***	2.651***	
		(0.595)	(0.748)	(1.028)	
Used stress coping strategy	0.952	0.022	0.003	0	
		(0.009)	(0.012)	(0.015)	
Used crisis coping strategy	0.774	-0.047***	-0.008***	0.092***	
		(0.022)	(0.027)	(0.030)	
Used emergency coping strategy	0.711	-0.043	0	0	

		(0.030)	(0.022)	(0.040)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	10.825	-1.208***	-1.045***	2.399***
		(0.435)	(0.357)	(0.686)
Women's Agency (Index standard deviations)	-0.015	0.116	-0.044	-0.078
		(0.050)	(0.047)	(0.060)

Note: Panel A reports results for the first follow up and Panel B for the second follow up. Sample sizes range from 3031 to 3031 households at FU1. Sample sizes range from 3875 to 3960 households at FU2. The Food Consumption Score (FCS) ranges from 0 to 112. The reduced Coping Strategies Index (rCSI) follows WFP guidelines to measure the frequency and severity of coping behaviours related to food consumption. Both the FCS and rCSI were standardized. Life Satisfaction was constructed using the Cantril Ladder approach, aggregating responses from three subjective well-being (SWB) questions. The Women's Agency Index follows Jayachandran et al. (2023). Index standard deviation is computed as the standardized (z-score) mean of its components' z-scores. The median share of food insecurity is 87%. Regressions with both a binary and a continuous outcome variable use an ordinary least squares (OLS) regression model. All specifications include a dummy variable indicating whether the household received fewer than four transfers. For the FCS, the model additionally controls for non-seven-day recall at follow-up 2. *, ** and *** denote statistical significance at 10%, 5%, and 1%.

Impacts on beneficiaries and non-beneficiaries

Table A7.8: Welfare outcomes – beneficiaries

	PMT Plus (mean)	CBT	PMT Plus (mean)	CBT
	FU 1	FU1	FU2	FU2
	(1)	(2)	(3)	(4)
Panel A: Main results				
Food Consumption Score	18.555	-0.086 (0.704)	14.572	0.442 (0.396)
Reduce Coping Strategy Index	20.254	0.195 (0.604)	25.323	-0.300 (0.640)
Used stress coping strategy	0.963	-0.016 (0.012)	0.947	-0.003 (0.008)
Used crisis coping strategy	0.615	-0.045 (0.039)	0.708	-0.033 (0.020)
Used emergency coping strategy	0.760	-0.043* (0.023)	0.775	0.009 (0.017)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.578	0.482* (0.277)	10.687	0.217 (0.211)
Women's Agency (Index standard deviations)	-0.020	0.037 (0.046)	0.006	0.043 (0.046)
Panel B: Controlling for coverage				
Food Consumption Score	18.555	-0.609 (0.802)	14.572	0.026 (0.355)
Reduce Coping Strategy Index	20.254	0.504 (0.660)	25.323	0.304 (0.600)
Used stress coping strategy	0.963	-0.017 (0.013)	0.947	-0.001 (0.008)
Used crisis coping strategy	0.615	-0.029 (0.042)	0.708	-0.040** (0.020)
Used emergency coping strategy	0.760	-0.037	0.775	0.014

		(0.025)		(0.018)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.578	0.330	10.687	0.115
		(0.286)		(0.216)
Women's Agency (Index standard deviations)	-0.020	0.047	0.006	0.029
		(0.050)		(0.044)

*Note: Estimations were conducted on the sample of beneficiaries (N = 2,519). Columns (1) and (3) report PMT Plus control means at follow-up 1 and follow-up 2, respectively. Estimates include quadruplet fixed effects and design weights, with standard errors clustered at the block level reported in parentheses. In panel B, all specifications additionally control for the baseline cluster-level coverage rate. The Food Consumption Score (FCS) ranges from 0 to 112. The food groups used for the FCS are: cereals, grains, roots and tubers; legumes and nuts; milk and other dairy products; meat, fish and eggs; vegetables and leaves; fruits; oil, fat and butter; and sugar. For the FCS, we control for households that were randomly assigned a recall period other than seven days during the follow-up 2. The reduced Coping Strategies Index (rCSI) follows WFP guidelines to measure the frequency and severity of coping behaviours related to food consumption. Life Satisfaction was constructed using the Cantril Ladder approach as the row total of three subjective well-being (SWB) questions, each measured on a 1–10 ladder scale, yielding a score that ranges from 3 to 30. The Women's Agency Index follows Jayachandran et al. (2023) and is computed as the standardized (z-score) mean of its components' z-scores. *, ** and *** denote statistical significance at 10%, 5%, and 1%.*

Table A7.9: Welfare outcomes – non-beneficiaries

	PMT Plus (mean)	CBT	PMT Plus (mean)	CBT
	FU 1	FU1	FU2	FU2
	(1)	(2)	(3)	(4)
Panel A: Main results				
Food Consumption Score	17.164	1.654*	14.536	0.274
		(0.867)		(0.420)
Reduce Coping Strategy Index	23.204	0.354	24.902	1.269**
		(0.788)		(0.582)
Used stress coping strategy	0.974	-0.006	0.965	-0.001
		(0.011)		(0.012)
Used crisis coping strategy	0.625	-0.044	0.718	-0.041
		(0.046)		(0.035)
Used emergency coping strategy	0.741	-0.028	0.773	0.019
		(0.035)		(0.026)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.523	0.087	11.153	-0.279
		(0.340)		(0.262)
Women's Agency (Index standard deviations)	-0.054	0.146**	-0.065	0.007
		(0.074)		(0.049)

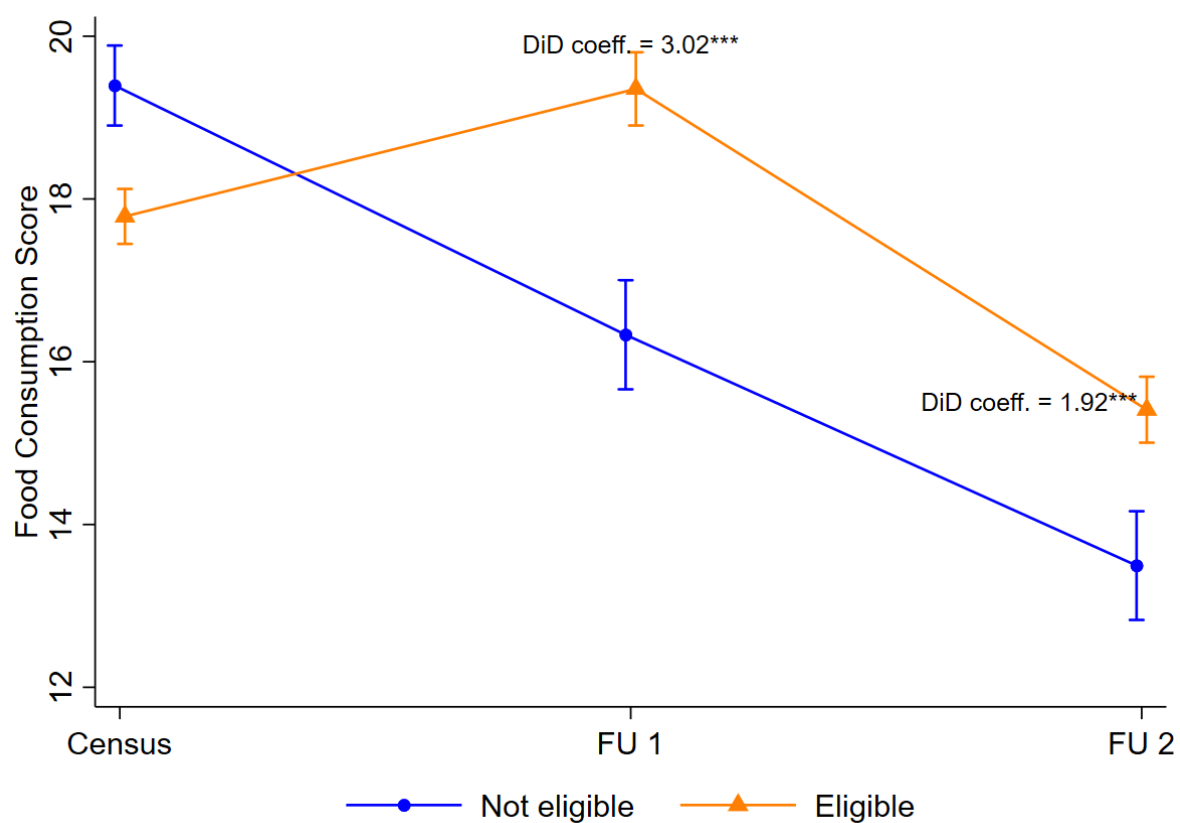
Panel B: Controlling for coverage

Food Consumption Score	17.164	1.295 (0.901)	14.536	0.175 (0.514)
Reduce Coping Strategy Index	23.204	0.898 (0.911)	24.902	1.503** (0.766)
Used stress coping strategy	0.974	-0.005 (0.012)	0.965	0.002 (0.013)
Used crisis coping strategy	0.625	-0.057 (0.056)	0.718	-0.075** (0.036)
Used emergency coping strategy	0.741	-0.068* (0.041)	0.773	0.004 (0.029)
Life Satisfaction (Cantril ladder sum, 3–30, higher=better)	12.523	-0.056 (0.351)	11.153	-0.546* (0.331)
Women's Agency (Index standard deviations)	-0.054	0.146* (0.084)	-0.065	0.048 (0.055)

*Note: Estimations were conducted on the sample of non-beneficiaries (N = 1,294). Columns (1) and (3) report PMT Plus control means at follow-up 1 and follow-up 2, respectively. Estimates include quadruplet fixed effects and design weights, with standard errors clustered at the block level reported in parentheses. In panel B, all specifications additionally control for the baseline cluster-level coverage rate. The Food Consumption Score (FCS) ranges from 0 to 112. The food groups used for the FCS are: cereals, grains, roots and tubers; legumes and nuts; milk and other dairy products; meat, fish and eggs; vegetables and leaves; fruits; oil, fat and butter; and sugar. For the FCS, we control for households that were randomly assigned a recall period other than seven days during the follow-up 2. The reduced Coping Strategies Index (rCSI) follows WFP guidelines to measure the frequency and severity of coping behaviours related to food consumption. Life Satisfaction was constructed using the Cantril Ladder approach as the row total of three subjective well-being (SWB) questions, each measured on a 1–10 ladder scale, yielding a score that ranges from 3 to 30. The Women's Agency Index follows Jayachandran et al. (2023) and is computed as the standardized (z-score) mean of its components' z-scores. *, ** and *** denote statistical significance at 10%, 5%, and 1%.*

Impact of cash assistance on food security

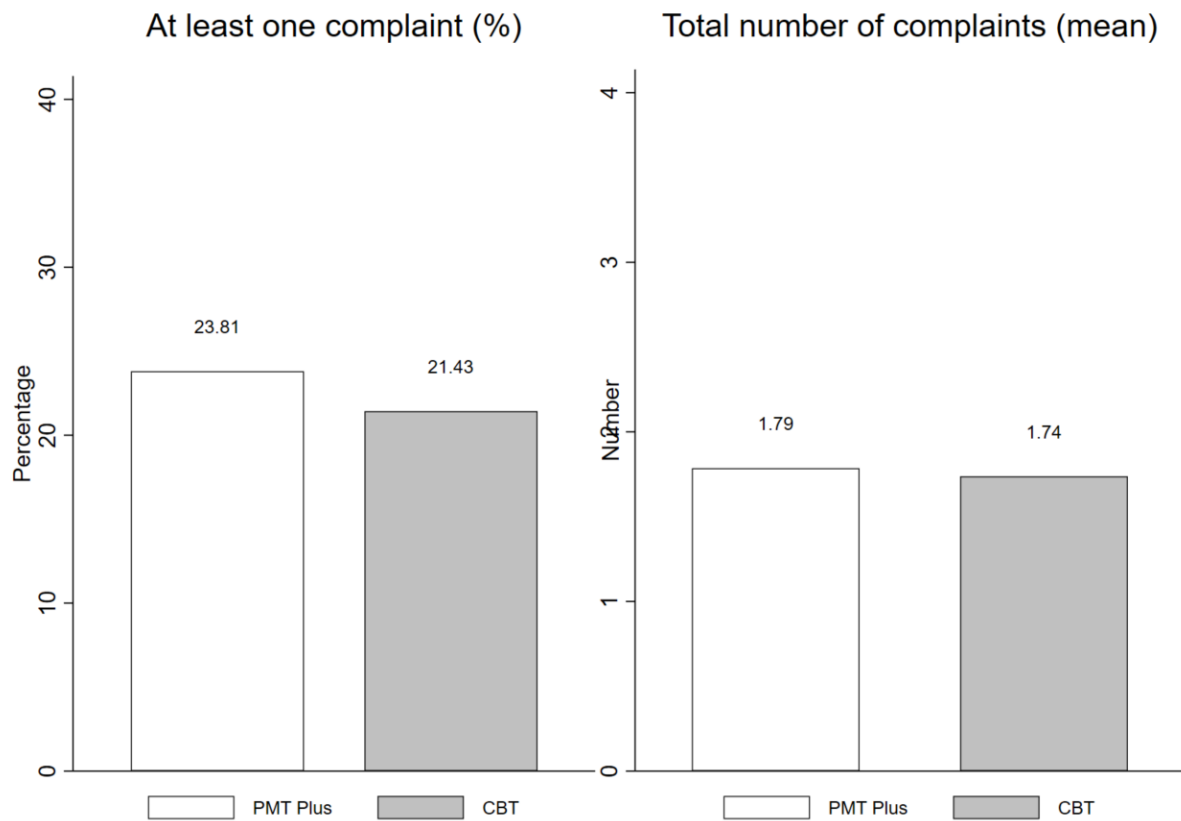
Figure A7.2: Impact of the programme on beneficiaries



Note: The figure presents estimates from a difference in differences design using the household panel across the census (baseline), first (FU1), and second follow-up (FU2). The outcome is the Food Consumption Score (FCS), the only indicator observed in all three waves. Points show mean FCS by baseline eligibility and survey round. Baseline values correspond to observed means, while FU1 and FU2 values correspond to regression adjusted predicted means from the DiD specification including household and time fixed effects. Standard errors are clustered at the household level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A8. Additional details and analyses on community acceptance and cohesion

Figure A8.1: Complaints by targeting method



Note: The figure presents the percentage of blocs with at least one complaint (left panel) and the average number of complaints per bloc (right panel), disaggregated by targeting arm. The reported estimates represent simple means. Coefficients shown on the left (Diff.) represent the difference between CBT and PMT Plus. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8.1: Impact of CBT on satisfaction indicators

	PMT Plus (mean)	CBT
	FU 2	FU 2
	(1)	(2)
Panel A: Main results		
Satisfaction	0.649	-0.00451 (0.0430)
Views targeting as understandable	0.681	0.0711** (0.0335)
Views targeting as fair	0.397	0.0190 (0.0670)
Views targeting as slow	0.171	0.0913*** (0.0271)
Views targeting as transparent	0.729	0.00931 (0.0133)
Views targeting as corrupt	0.0119	-0.00840** (0.00345)
Panel B: Controlling for coverage		
Satisfaction	0.649	0.0314 (0.0387)
Views targeting as understandable	0.681	0.0839** (0.0365)
Views targeting as fair	0.397	0.0228 (0.0722)
Views targeting as slow	0.171	0.0952*** (0.0307)
Views targeting as transparent	0.729	0.00752 (0.0154)
Views targeting as corrupt	0.0119	-0.00954*** (0.00355)

Note: The table report results from the second follow-up. Columns (1) report control means. Estimates include control variables selected using post-double selection Lasso and quadruplet fixed effects. Analytical weights are applied to account for sampling design and non-response. Standard errors are clustered at the bloc (treatment) level. The variables measuring satisfaction and views on targeting are binary indicators. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8.2: Robustness checks on satisfaction outcomes

	PMT Plus (mean)	CBT
	FU 2	FU 2
	(3)	(4)
Panel A: Dropping blocs with security issues		
Satisfaction	0.614	-0.002 (0.046)
Views selection process as understandable	0.736	0.073 (0.035)
Views selection process as fair	0.417	0.02 (0.071)
Views selection process as slow	0.197	0.092 (0.029)
Views selection process as transparent	0.743	0.009 (0.014)
Views selection process as corrupt	0.006	-0.01 (0.003)
Panel B: Controlling for imperfect programme adherence		
Satisfaction	0.607	-0.004 (0.043)
Views selection process as understandable	0.737	0.071 (0.033)
Views selection process as fair	0.411	0.019 (0.067)
Views selection process as slow	0.199	0.091 (0.027)
Views selection process as transparent	0.749	0.009 (0.013)
Views selection process as corrupt	0.005	-0.009 (0.003)

Panel C: Dropping IDP blocs with perfect coverage

Satisfaction	0.602	-0.012 (0.028)
Views selection process as understandable	0.742	-0.004 (0.020)
Views selection process as fair	0.415	-0.092 (0.030)
Views selection process as slow	0.202	0.064 (0.018)
Views selection process as transparent	0.751	-0.013 (0.016)
Views selection process as corrupt	0.005	-0.005 (0.004)

*Note: Columns (1) and (3) report control means. Estimates include quadruplet fixed effects and design weights, with standard errors clustered at the bloc level and reported in parentheses. For the Food Consumption Score, the model additionally controls for non-seven-day recall at the second follow-up. Regressions for Panel A drop blocs affected by security issues. Regressions for Panel B include a dummy variable that indicates whether the household received the planned transfer. Regressions from Panel C exclude the three IDP blocs with perfect coverage. The total number of clusters is 70, and the sample is 3,423 households. The symbols *, ** and *** denote statistical significance at 10%, 5%, and 1% levels, respectively.*

Table A8.3: Impact of CBT on additional satisfaction indicators

	PMT Plus (mean)	CBT
	FU 2	FU 2
	(1)	(2)
Targeting process was clear to respondent	0.786	0.016 (0.042)
Respondent knows criteria for vulnerability	0.318	-0.012 (0.056)
Respondent understands how targeting works	0.715	-0.025 (0.02)
Respondent believes targeting errors occurred	0.557	0.053 (0.057)
Respondent believes ineligible households were excluded	0.445	0.049 (0.058)
Respondent believes eligible households were included	0.211	0.018 (0.049)

Note: The table report results from the second follow-up. Columns (1) report control means. Estimates include control variables selected using post-double selection Lasso and quadruplet fixed effects. Analytical weights are applied to account for sampling design and non-response. Standard errors are clustered at the bloc (treatment) level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8.4: Impact of CBT on trust and social cohesion outcomes

	PMT Plus (mean)	CBT
	FU 2	FU 2
	(1)	(2)
Panel A: Main results		
Trust index	-0.015	0.0322 (0.0574)
Collective action index	0.032	-0.194** (0.0845)
Social cohesion index	-0.029	-0.0365 (0.0596)
Felt unsafe at home or bloc	0.627	0.0870** (0.0411)
Panel B: Controlling for coverage		
Trust index	-0.015	0.0485 (0.0545)
Collective action index	0.032	-0.255*** (0.0762)
Social cohesion index	-0.029	-0.0199 (0.0615)
Felt unsafe at home or bloc	0.627	0.0514 (0.0449)
<i>Observations</i>		3,956

Note: The table report results from the second follow-up. Columns (1) report control means. Estimates include control variables selected using post-double selection Lasso and quadruplet fixed effects. Analytical weights are applied to account for sampling design and non-response. The Social Cohesion, Trust, and Collective Action indices are standardized composite indices constructed from multiple survey items and re standardized to have mean zero and standard deviation one. The Social Cohesion index follows Bossuoy et al. (2022) and captures perceived support and community connectedness. The Trust index aggregates trust in different groups and institutions. The Collective Action index captures participation and engagement in community groups and activities. Standard errors are clustered at the bloc (treatment) level. Statistical significance is indicated by asterisks: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8.5: Robustness checks on trust and social cohesion outcomes

	PMT Plus (mean)	CBT
	FU 2 (3)	FU 2 (4)
Panel A: Dropping blocs with security issues		
Trust index	-0.048	0.03 (0.061)
Collective action index	0.074	-0.208 (0.088)
Social cohesion index	-0.02	-0.046 (0.062)
Felt unsafe at home or bloc	0.644	0.091 (0.043)
Panel B: Controlling for imperfect programme adherence		
Trust index	-0.015	0.032 (0.058)
Collective action index	0.032	-0.194 (0.084)
Social cohesion index	-0.029	-0.038 (0.058)
Felt unsafe at home or bloc	0.627	0.087 (0.041)
Panel C: Dropping IDP blocs with perfect coverage		
Trust index	-0.010	0.023 (0.050)
Collective action index	0.020	0.023 (0.050)
Social cohesion index	-0.045	0.049 (0.051)
Felt unsafe at home or bloc	0.629	0.087 (0.029)

*Note: Columns (1) and (3) report control means. Estimates include quadruplet fixed effects and design weights, with standard errors clustered at the bloc level and reported in parentheses. For the Food Consumption Score, the model additionally controls for non-seven-day recall at the second follow-up. Regressions for Panel A drop blocs affected by security issues. Regressions for Panel B include a dummy variable that indicates whether the household received the planned transfer. Regressions from Panel C exclude the three IDP blocs with perfect coverage. The total number of clusters is 70, and the sample is 3,423 households. The symbols *, ** and *** denote statistical significance at 10%, 5%, and 1% levels, respectively.*

A9. Ethical considerations

212. The study strictly adheres to ethical guidelines, including the 2020 United Nations Evaluation Group (UNEG) standards. The OEV and DI teams provided diligent oversight and enforcement of ethical protocols at all stages.
213. Key ethical principles and practices were rigorously implemented:
- Programme Exclusion: Every community in the evaluation received WFP assistance in accordance with the targeting method used. Survey respondents excluded from assistance were deemed non-vulnerable during the targeting process.
 - Institutional Review Board (IRB): The impact evaluation design underwent an ethical review by the University of Milan in Bicocca (#759), a public university affiliated with a principal investigator for the evaluation.
 - Informed consent: Households participating in the study initially consented to WFP programme involvement, followed by separate consent for participation in baseline, first and second follow-up surveys. Refusing to partake in the survey had no bearing on eligibility for WFP support.
 - Privacy during interviews: To ensure respondent privacy and comfort, to the extent possible, enumerator groups were gender balanced, and interviews took place in privacy at the respondent's household.
 - Training and protocols: Before each wave of data collection, enumerators underwent extensive one-week training followed by field piloting. The training ensured uniform and contextually appropriate questioning, and that enumerators follow survey best practices in terms of protocols and ethics.
 - Ethical oversight: Ongoing monitoring and management of ethical issues occurred during the study, with additional concerns addressed in line with established guidelines.
214. In summary, the study prioritizes ethical conduct, covering informed consent, privacy, cultural sensitivity and vulnerable participant protection. Ethical integrity was consistently upheld and monitored to safeguard participants throughout the research process.

A10. Limitations

215. **External validity:** The evaluation was conducted in a context characterized by relatively homogeneous vulnerability levels: on average, 86 percent of households per bloc met the vulnerability criteria. As such, findings may not generalize to settings with greater heterogeneity in household vulnerability. Additionally, the sample was limited to blocs with more than 40 households, excluding smaller blocs from the analysis.
216. **Internal validity:** The evaluation aims to estimate the impact of the programme on welfare indicators by comparing eligible and non-eligible households within each treatment arm. However, these two groups differ systematically.
- **Evaluation-driven effects:** Households may have altered their behavior or responses based on their understanding of the evaluation or their treatment status. To mitigate this, enumerators were not informed of the treatment assignments, and standardized survey protocols were used across all groups.
 - **Spillover effects:** There is a risk that programme benefits or information may have spilled over from treated to non-treated blocs, potentially diluting observed differences. While the design aimed to minimize such interactions, complete isolation between blocs could not be guaranteed.
 - **Excluded blocs:** A potential limitation of the study relates to the exclusion of 12 blocs due to security constraints prior to data collection. As randomization was conducted before the census, no baseline data are available to formally assess balance between treatment arms after these exclusions. However, the attrition was evenly distributed across arms, and the stratified design by quadruplets helps mitigate concerns about systematic imbalance. To further address this issue, robustness checks were conducted by excluding all quadruplets affected by these dropped blocs. The results remain consistent, suggesting that the main findings are not driven by the sample reduction.
 - **Survey Attrition:** Attrition at the first follow-up was substantial and primarily driven by conflict-related security constraints that were unrelated to programme implementation. These constraints were largely absent by the second follow-up, resulting in much lower attrition at that stage. Balance tests show no systematic differences in attrition between PMT Plus and CBT blocs. The quadruplet randomization design and the inclusion of quadruplet fixed effects further mitigate concerns related to attrition bias.
 - **Partial compliance:** A small number of households (470 HHs, 12 percent of the sample) did not receive all transfers as intended. These deviations are limited in magnitude and appear to be balanced across treatment arms, suggesting they are unlikely to affect the estimated impacts.
 - **Perfect coverage blocs:** A potential limitation arises from the presence of three blocs with perfect coverage, corresponding to IDP-only blocs in which all households were selected for assistance. This mechanically eliminates variation in targeting outcomes within these blocs and may affect the estimation of precision measures. However, these blocs represent a small share of the sample and are not systematically related to treatment assignment. To address this concern, robustness checks were conducted by

excluding these blocs from the analysis. The results remain consistent, indicating that the main findings are not driven by the inclusion of blocs with perfect coverage.

- **Cash Transfer Impact Analysis:** While the randomized design supports the plausibility of the common trends assumption underlying the difference-in-differences approach, the absence of pre-intervention panel data precludes a formal test. As such, the results should be interpreted with this limitation in mind, although the experimental setting reduces the likelihood of meaningful deviations from parallel trends.

217. **Short implementation cycle:** The programme consisted of only 4 transfers over a 5-month period. As endline data collection occurred shortly after the final transfer, the evaluation could not assess long-term impacts. The results therefore reflect only short-term effects.

A11. Stakeholder analysis

218. The stakeholder analysis for this evaluation identifies those who may influence or be influenced by the evaluation's outcomes. Stakeholders encompass both internal and external parties, including programme beneficiaries. While the WFP CO in the DRC is the primary user, the evaluation seeks broader utilization of its findings. Stakeholders were categorized as follows:

- Internal DRC-based stakeholders, including key personnel CO personnel;
- Internal stakeholders outside the DRC, such as OEV, the WFP regional bureau, and headquarters divisions;
- External stakeholders, including international NGOs, donors, United Nations agencies, and local forums;
- National stakeholders, encompassing government entities at national and subnational levels, and local NGOs.

219. This study actively contributes to the existing literature by addressing the scarcity of rigorous evidence on targeting methods in humanitarian settings. To ensure diverse perspectives were considered and that the evaluation's results are effectively used, the evaluation actively engaged with various stakeholder groups. The evaluation team designed the results as public goods, intending them to provide a foundation for further analysis on this critical topic.